Math 16: Mathematics for Life Sciences
Guide for teaching teams

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Part I

About Math 16

Introduction

Welcome to Math 16! As part of the teaching team for Math 16, you are part of a broad movement which reimagines math education for biologists. It is dangerous to say exactly when this movement began (someone always tried it earlier!), but our inspiration is a class called LS30 which began at UCLA in Fall 2014. The UCLA team has written an article about their experience [1], where they describe their motivation as follows.

> There is a growing realization that traditional "Calculus for Life Sciences" courses do not show their applicability to the Life Sciences and discourage student interest. There have been calls from the AAAS, the Howard Hughes Medical Institute, the NSF, and the American Association of Medical Colleges for a new kind of math course for biology students, that would focus on dynamics and modeling, to understand positive and negative feedback relations, in the context of important biological applications, not incidental "examples".

Our motivation is the same, and we are encouraged by the fact that the UCLA class has demonstrated more than 10 years of success at scale. Other institutions, including Harvard, Arizona, Texas State, Utah State, Cornell, Johns Hopkins, UC Santa Barbara – private and public, R1 and undergraduate-teaching-focused, large and small – have been inspired by UCLA's course. We have formed a loose consortium, meeting to develop our courses and share resources. Our collective efforts have attracted grant funding, and can now be called a broad transformative movement.

Math 16, like LS30 at UCLA and others, is hard to explain because very few mathematicians and scientists have taken such courses. Contrast this with calculus. Everyone who has taken a calculus course knows what a calculus class is. You start with a bit of functions and limits, do the definition of the derivative, a whole lot of derivative rules, and some applications to mins and maxes. Linear approximation and some Taylor series are worked in. Students learn antiderivatives, all sorts of rules and subtle techniques for integrals, areas under curves, some solids of revolution and stuff. Calculus is an Olde Institution, and you can compare a modern textbook to one used 100 years ago and find the same broad structure. Modern textbooks cost upwards of 200 dollars, presumably for the large number of vetted application exercises and integrated technology. Also, because the large publishing companies are driven by profit. The core calculus is largely the same, with some changes in fashion.

What is Math 16

So what is Math 16? It is a course which *replaces* traditional calculus, while having *some conceptual overlap* with calculus. But the structure is completely different, which is why a student cannot jump from Math 16A into Math 11B (second-quarter calculus), or from Math 11A into Math 16B at UCSC.

So what is Math 16? Math 16 is a *modeling-first* approach to *mathematics for life sciences*, which focuses on identifying quantities of interest as **state variables** and other quantities as **parameters**, and modeling biological phenomena by **algebraic relationships** and **change equations**. The term "change equation" is our friendly way of saying "system of ordinary differential equations." Change equations are really the main characters in the story of Math 16, and we start off with discussions related to predator-prey, quickly developing the Lotka-Volterra equations. These would never be found in an introductory calculus sequence, and probably would not appear until near the end of a first course in differential equations. And yet... it works.

The best way to understand what Math 16 is to read the lab manual and see what students do. By the end of the course, you will probably develop your own way of understanding what the course is. And please share with your colleagues, so more people know about it.

Mindset

Math 16 is a course in mathematics for the life sciences, a.k.a., biology. If you are a mathematician, of the "pure" (non-applied) sort, then teaching this class can be challenging and ultimately rewarding. Success in the course requires a certain **mindset** of both students and instructors.

Most of the students in this class are just entering college (mostly frosh), and taking this course because their biology major requires such a course. They probably have a choice between this class and a more traditional calculus sequence. The students have a huge variety of mathematical experiences, and many of them do not like math or do not see themselves as math people. Most of the students are not interested in math for its own sake. Most of the students *are* very interested in science though. Their proposed majors include everything from molecular/cell biology (e.g., MCDB BS = molecular, cell, developmental bio) to health (GCH BS= global and community health), to big pretty organisms (marine biology, plant science) to broad paths of biology (BIO BS) and ecology (EEB BS = ecology and evolutionary biology).

As a member of the teaching team, you are probably trained in mathematics. Perhaps you have experience with some science and applications. But if you are a graduate student at UCSC in mathematics, then you are probably most interested in math for its own sake.

This disconnect, between your priorities and your students' priorities, is a challenge. This course can turn the disconnect into an opportunity... if you can start with the appropriate mindset. Read the challenges and opportunities below.

Challenge I am not the expert who knows everything about what we are teaching. I don't know any biology, modeling techniques, etc.

Opportunity I have experience reading carefully, working through problems methodically, and responding honestly even about uncertainty.

Challenge Students will not respect me if I cannot do every problem.

Opportunity When I am faced with a problem I can't solve, I know how to identify the sticking point, invent a simpler problem, write about what we do and don't know, and ultimately how to ask others and do my own research to answer questions in time. I know that authentic problems cannot be solved in a minute or an hour or sometimes even a day, and students need to learn this too.

Challenge Students want to learn by imitation. They are going to revolt. I'm scared.

Opportunity A bit of learning by imitation is possible as a first step. But they will recognize the value of deeper engagement, even if it takes a bit longer. I can be patient and get students to work towards a common goal, as long as they know **I am on their side**.

Challenge I just don't care about biology. Cells, frogs, whatever... not interesting to me.

Opportunity Biology is a big world, and there are probably 100 different reasons that people do get interested in biology. I am open-minded and possess a scientifically curious mind, so I will find something interesting in there!

Challenge I don't have the time for this class. I have other classes, jobs, research, family, etc.

Opportunity I recognize that doing a job well requires commitment. And with practice, things become easier. When I can, I'll push myself to take on a worthwhile challenge. Especially one with a clear beginning and end like teaching a class.

Consider the above *opportunities*. And think of how they may help you in a future career. Careful analysis and problem-solving strategies, honesty in the face of uncertainty, advocating for others, open-mindedness and curiosity about the world, commitment to goals. These are worth striving for. And the teaching team will be there to support you.

Students will benefit when they see you take these opportunities. It is valuable for them to see your process when you are really struggling with a problem. It is valuable when you bring certain skills (mathematics, problem-solving maturity) and students bring certain skills (recent experience across chemistry and biology). It is valuable when they recognize that you are struggling alongside them for a common goal of learning. It is valuable when students see you exhibiting curiosity and a growth mindset about a field where you have very little prior experience or interest. Because they will have to do the same!

Pedagogy

Math pedagogy is a deep well of politics and fear and excitement and research and "research" and it is hard to know where to start. For this course, we are trying to create discussion sections in which students are **deeply engaged**. We are, to put it briefly, trying to build a discussion section where **students think**.

Without endorsing every word in the book, we can recommend the recent book *Building Thinking Clasrooms in Mathematics*, by Peter Liljedahl [2]. This is quite a popular book (the BTC Teachers Facebook group has about 80000 members), and has its own share of controversies. While written in the K-12 context, the ideas are relevant for our discussion sections where we too hope to build a thinking classroom! The motivation of Liljedahl can be found in his introduction:

> Over 15 years ago I reached out to my connections in the teaching community and asked them to recommend to me teachers that they had heard were good mathematics teachers—teachers who were respected within their schools and within the school division and were known to have students who performed well in mathematics. Based on these recommendations, I visited 40 classrooms in 40 different schools. I visited classrooms of every grade from kindergarten to Grade 12. I was in low socioeconomic settings and high socioeconomic settings, English-speaking classrooms and French-speaking classrooms, and I was in public schools and private schools. And in every classroom I visited, I saw the same thing – students not thinking in ways that went beyond mimicking the

> teacher. Closer investigations revealed that within a 60-minute lesson, 20% of students spent 8–12 minutes thinking, while 80% spent zero minutes thinking. This is a problem. This is what has been missing. Thinking is a necessary precursor to learning, and if students are not thinking, they are not learning.

For 15 years, Liljedahl experimented with teachers to identify "optimal practices for thinking." From this work, he identified 14 such factors that can be tuned to build a thinking classroom. These factors are:

1. Give thinking tasks.
2. Frequently form visibly random groups.
3. Use vertical non-permanent surfaces.
4. Defront the classroom.
5. Answer only keep thinking questions.
6. Give thinking task early, standing, and verbally.
7. Give check-your-understanding questions.
8. Mobilize knowledge.
9. Asynchronously use hints and extensions to maintain flow.
10. Consolidate from the bottom.
11. Have students write meaningful notes.
12. Evaluate what you value.
13. Help students see where they are and where they are going.
14. Grade based on data (not points).

You don't have to go "all-in-BTC" like some school districts are trying. (Education trends can be a bit culty.) But you can think about the above optimal practices (according to Liljedahl) and how and whether they can be implemented in your discussion section. For example:

1. Give thinking tasks: does your opening "warm-up" task get students really thinking? Is it highly engaging, and not just getting to the next step of the curriculum? Does it require students to just find an answer, or does it get students to go deep? How do **you** construct a good thinking task?
2. Frequently form visibly random groups: When we start a new lab (every 3 weeks), we form new random groups. How can we make it so that students enter their new groups with the mindset that they are going to be active contributors? How does randomness break down social barriers? Active contribution and inclusivity are the goals Liljedahl seeks here.
3. Use vertical non-permanent surfaces: Most of our rooms have ample blackboard space. Are your students standing in their groups next to a designated section of board? Liljedahl writes

> The fact that it was non-permanent promoted more risk taking, and the fact that it was vertical prevented students from disengaging. Taken together, having students work, in their random groups, on VNPSs had a massive impact on transforming previously passive learning spaces into active thinking spaces where students think, and keep thinking, for upwards of 60 minutes.

Can you create longer (20+ minute) tasks where students are standing most of the time?

4. Defront the classroom. Can you rearrange the furniture? Can you make sure that students are facing every which way?

5. Answer only keep thinking questions. Liljedahl describes three categories of student questions:

   > (1) proximity questions – the questions students ask because you happen to be close by, (2) stop-thinking questions – the questions students ask so they can reduce their effort, the most common of which is, "Is this right?" and (3) keep-thinking questions – the questions students ask so they can keep working, keep trying, and keep thinking.

   How can you redirect type (1) and (2) appropriately, to keep the students thinking? (I wouldn't be so strict as to only answer type (3) questions... but it's a nice ambition.)

6. Give thinking task early, standing, and verbally. Where are you, and where are your students, during the warm-up? Do you just write a problem on the board? How is your posture? How do you phrase your question?

7. Give check-your-understanding questions. This may be within your exit ticket. How will students know if they understand? And how will you know if students understand? Independent of grading, you need to have prepared check-questions like this.

8. Mobilize knowledge. To make group work and discussion effective, the vast variety of knowledge your students bring must be mobilized. How are your students learning about what their peers are thinking? Do they take time to look at the work of others within the group, and within other groups? How can their diverse experience in math, biology, and chemistry be useful to the classroom?

9. Asynchronously use hints and extensions to maintain flow. Different groups will work at different paces, and get stuck at different times. How can you "float" among the groups, and provide the groups with that little hint or push, to keep them in the thinking flow?

10. Consolidate from the bottom. Consolidation is where we try to bring everyone together and build a cohesive concept. Referencing Schoenfeld (1985), Liljedahl writes "For over 100 years, this has involved teachers showing, telling, or explaining the learning that the teachers desired for the students to have achieved. The problem is that it doesn't work." Instead, Liljedahl recommends the following bottom-up approach: "working upwards from the basic foundation of a concept and drawing on student work produced during their thinking on a common set of tasks." What does that look like, at the end of an activity, or at the end of a discussion section?

11. Have students write meaningful notes. The lab manual is meant to enforce this, in a sort of traditional way which contrasts with the ephemeral current norms. Instead of clicking buttons on a changing screen, or watching videos that end, or typing in a text box that vanishes, get

the students to **write on paper**. And even better, get the students to write their notes in a sort of scratch notebook, and then final drafts in their lab manual. Liljedahl writes, "In a thinking classroom, on the other hand, notes are a mindful activity involving students deciding for themselves what notes their future selves will need. The research showed that this way of taking notes kept students thinking while they wrote the notes and that the majority of students referred back to these self-created notes in both the near and far future."

12. Evaluate what you value. Do we do this in the way we grade? How can we improve?

13. Help students see where they are and where they are going. How can your students receive formative assessment, not for a grade, but to know where they are and where they have to work to learn?

14. Grade based on data (not points). Liljedahl advocates for standards-based or evidence-based grading. I don't understand it well... but am willing to dive into it if it is feasible!

Outcomes

Math 16A does not destroy student confidence! In fact, in the first two years of the course, it increased student confidence (see Fig.1)! This itself is an achievement for the students we have, and it contrasts with the usual effects of precalculus/calculus.

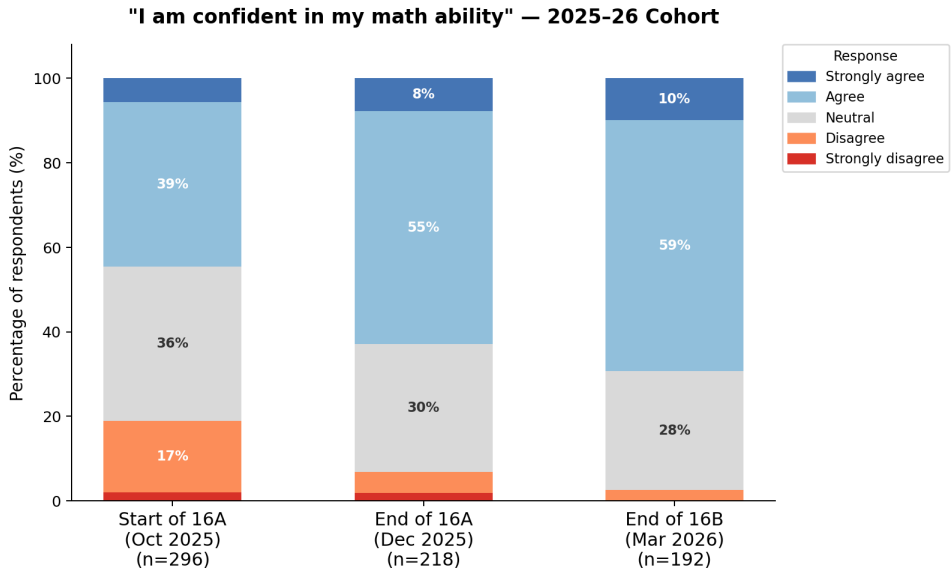


Figure 1: Math confidence distribution among students at three timepoints. One might be concerned that we are simply losing the not-confident students along the way (survival bias), and we do lose track of the students that withdraw or do not respond to the survey. But among the students who stuck with the class and responded to the survey, 39% of individuals increased their mathematical confidence and 15% of students decreased their mathematical confidence.

We asked the Math 16A students, "Did this class change your perception of mathematics?" At the end of the quarter, here is what a few of them wrote.

> The connection between math and science became much clearer during this class. Before this class I viewed math as a set of challenging formulas and complicated problems, but have now come to see the connection math has to science and the real world.
>
> I have previously taken math courses that were strictly about memorizing formulas and applying them. This class not only uses formulas in a way that can be applied to various situations but also presents them in a way that is relevant to more than just the math itself. Science based math like the math in this class has resparked my love for math!
>
> This class helped me understand the importance of, as well as greatly improved my relationship with math. Because of the lab notebook style of teaching, as well as the helpful lectures and sections I no longer really dread math class.

We asked the Math 16A students, "If someone asks you what this class is about, what would you tell them?" At the end of the quarter, here is what a few of them wrote.

> This class is unlike any other math class it focuses on relatively easy concepts but more so teaches you how to think and apply mathematical concepts to science in a way that easily explains and represents scientific phenomena. It does delve into slightly more complex ideas as well, using Greek symbols etc. but nothing unmanageable.
>
> It's about relationships, more specifically, change relationships. It's about seeing a relationship and being able to make a mathematical representation of it with reasoning, algebra, and a touch of calc.
>
> I would tell them that this class is about looking at math in a literal, more practical sense. This class is more real-world preparation than any other math course I have taken.
>
> It's different than any math classes that you had took, if you like problem solving or word problems you should definitely take it, but if you are looking for straight forward math I recommend you look for something else. This class is about understanding math, chemistry, physic, and biology at the same time. If you know some of it, it would be easy for you, but if you don't then you have to figure it out like a puzzle.

TA Feedback

We conducted a survey of TAs from our first offering of Math 16A and 16B, and here are some things they said.

For student success, the most important roles of the TA are to rephrase and clarify the questions from the lab manual, facilitate group work, and provide effective feedback to students.

Most TAs found it rewarding to teach Math 16, because the connections between math and science led to greater student engagement and learning. TAs did report that they were learning material alongside students, but it did not pose a problem.

The most challenging aspects were the time commitment, particularly the additional section times. TAs reported a typical 10-12 hour workload per week, with 16-18 hours in a heavy week. These hours included prepping for two sections per week, running 4 sections per week, having 2 office hours, 1 teaching team meeting, attending/viewing lecture, designing rubrics and point distributions for problems, grading, and Canvas/Gradescope management.

Some TAs found it challenging to teach in a course that merged math and biology; the basic biology content for the problems was not so difficult, but they had difficulty answering follow-up questions from students.

TAs (and students) wished for more connection between lecture and the lab manual, which has been a continual work in progress. TAs also wanted a bit more structure provided for sections and assessments, besides the large labs due every three weeks. This has also been done now, with lesson plans and rubric. TAs wished for an answer key to the manual, and this is mostly complete in Year 3 with our Instructor Edition of the Lab Manual, thanks to Hadrian Quan!

TAs recommend that future TAs make sure to work through the lab manual on their own, and to attend lectures to understand the science and its connection to the mathematics. This course may require more time to prepare for section than other classes. TAs also recommend separate meetings among the TAs and other members of the teaching team, to enhance learning and consistency.

Running the machine

Personnel

If you are reading this manual, you are probably one of the following:

Lead instructor Good luck! Talk to Marty Weissman, visit a previous class, and plan ahead.

TA Each graduate student Teaching Assistant typically has two discussion sections of about 40 students each, meeting for two hours weekly. That is four hours of discussion section per week, and two lesson plans.

ALF Our ALFs are "active learning facilitators". They primarily assist the TA in making the discussion section run smoothly. ALFs can easily float around the room, making sure groups stay on task, answering some questions but more often directing the groups to try different things – explain things to each other, write solutions on the board, etc. ALFs can also assist with passing out worksheets, taking attendance, etc.

ACE ACE may run supplemental sections, and be embedded within our teaching team. ACE brings pedagogical experience and often experience with Math 16 specifically.

LSS This course is often supported by one or more tutors from Learning Support Services.

The whole teaching team should meet once each week to debrief from the previous week and prepare for the next!

Canvas and Gradescope

Canvas is the UCSC learning management system, and the primary way that students receive information, track their grades, know the syllabus and schedule, etc. We also use Canvas for a few surveys/quizzes throughout the class.

Gradescope is used by students to submit their lab assignments. Each lab has one or two "check-ins" to keep students on track, and a final submission. They complete the lab by scanning the pages (preferably using the Gradescope app, to keep size reasonable and integrated with Gradescope), uploading to Gradescope, and clicking a "submit" button.

There is probably too much in the lab manual to read all the work by every student. We need to figure out a way to give honest and useful feedback, and identify a way for students to correct/improve their work.

Grading

Attendance and names

Attendance and participation in discussion section is **required** for all students. And participation means showing up on time, staying until the end, and working and thinking with others while there. You may need to remind students of this if they bend these rules. Please note that students are allowed to miss two discussion sections per quarter without grade penalty. Late/incomplete participation can be marked down by 50%. Please update the Gradebook at least once per week to keep the discussion records up to date.

Students should bring in one piece of paper to all discussion sections. On Tuesdays, students should bring in their (printed and completed) worksheets. These are exercises to make sure students practice basic techniques before section starts. On Thursdays, students should bring in their lecture digests. These are brief reflections to make sure students know the basic biological (or scientific) system being studied, and the mathematical models we apply to the system. These documents should be checked for completion at the beginning of each section.

Using students' names is fundamental to having participation, and convincing students that you are on their side. I find it very hard to remember lots of names. Thankfully, an old-fashioned approach works well for both record-keeping, group management, and name-reference.

The table-tent name-card method

The TA and ALF can share the responsibility for the following tasks during discussion section.

1. Before sections begin, purchase a bunch of blank white cards (4 inch by 8 inch works well). Folding them lengthwise yields a "table-tent" that is about 2 inches high and 8 inches long.

2. On the first day of discussion section (and when new students occasionally join), give everyone a card, and ask them to neatly fold it lengthwise, and write their name on *both sides* of of the resulting table-tent, large enough that it can be read from across the room. Markers are nice.

3. All the students hand their table tents back to you. You then shuffle the tents, and deal them out on tables to form random groups of 4-6 students. Just place 4-6 cards on each table, and students will find their card! By leaving these on the tables throughout discussion section, you can call on students by name easily, and students will learn each other's names quickly. DO refer to students by name, e.g., "I saw something interesting in Jianna's approach on that board..."

4. At the end of class, gather up the name cards, and use a clip or rubber band to keep them in the same groups for the next class. At the next class, you just deal them out in the same groups!

5. To take attendance, look for any cards that do not have a person next to them. You will be able to tell who is there or not, and record attendance accordingly.

6. At the beginning of each new lab (every three weeks or so), you can use the same name cards, shuffle them, and repeat the previous steps.

The shape of a discussion section

Every discussion section lasts 65 minutes. Each lesson plan in this manual budgets its activities to that total. Students meet for five or six discussion sections per lab, on Tuesdays and Thursdays, over roughly three weeks. You will not get through every page of the lab manual in section, and you are not meant to. Section is for getting students a strong start on the hard parts together; the rest is homework. The BEFORE SECTION / DURING SECTION / AFTER SECTION block at the top of each plan is where that split is spelled out — read it first, and tell students at the end of each section exactly what lands in the AFTER column.

Groups of 4-6 students stay together for a whole lab, and are re-shuffled at the start of each new lab (see Attendance, above, for the name-card method that does both jobs at once).

Pedagogy and technology: no-tech and high-tech

Most of our discussion section rooms are low-tech, with no projectors or document cameras. If you're lucky, there may be some working outlets. But there are some great ancient tools: blackboards and tables and chairs. These lesson plans are written with such a room in mind, and students will bring their own technology with them. Our approach is that every activity should be deliberately one of two modes, and you should say which one out loud when you start it.

No-tech mode. Laptops closed. Even better if phones and laptops are put away. Work happens on scratch paper, in the lab manual, or **at the blackboard**. This is the right mode for sketching time-series and trajectories by hand, for setting up change equations, for translating between English and equations, and for any discussion where a screen would just be somewhere to hide. There are amazing things we can do *without* technology, like (1) walk around the room looking at group solutions on boards, (2) have longer conversations without screen interference, (3) build stronger algebraic and graphing skills by putting a pen/pencil on a piece of paper.

High-tech mode. At least two laptops open per group, with students looking over each other's shoulders at a shared screen. This is the right mode for the Modeling Synthesizer tools, for spreadsheets, and for Desmos. Technology is an amazing tool, and when it's time to use it... we use it! Aim for two laptops per group — the point is that nobody is watching alone, and nobody has to crane their neck around the table to see the screen. For spreadsheeting, it may be better to have *everyone* with their laptop open, so that *everyone* learns how to work with spreadsheets.

Announce the mode. Practice saying "It is time to put all your computers and phones away, because we are about to work on paper and the boards." And say "Ok class – it's time to pull out your laptop computers! Please make sure there are two laptops open on each table. Keep some paper and pens for notes too."

One consequence for TAs: because you cannot project, anything you want the whole class to look at either has to go up on the blackboard, or already be printed in the lab manual (a great deal of it is — check the page before assuming you need to reproduce a figure), or be loaded by the students themselves on their own machines. When a plan below says to demonstrate a tool, it means talking the room through the clicks while they follow along on their own screens, not showing them yours.

Part II

Lesson Plans

Chapter 1

Flow

About Lab 1

The two big systems in this lab are (1) sharks and tuna, and (2) insulin and glucose. The sharks and tuna example will recur throughout Math 16, because it is such an intuitive one for students. Sharks and tuna swim around, and sharks eat tuna. What could be easier, right? The shark-tuna system and insulin-glucose system exemplify negative feedback loops.

- When the tuna population is high, they feed the sharks, leading to growth in the shark population, which leads to more predation of tuna, which leads to shrinking of the tuna population.
- When glucose in the bloodstream is high (e.g., you eat a candy bar), specialized cells in your body release insulin into your bloodstream, which leads to more glucose uptake by other cells, effectively removing glucose from your bloodstream.

The new instructor might ask: why do we begin the class with a two-variable system? Wouldn't it be simpler to start with something basic like exponential growth, then do logistic, then think about two species? That is traditional after all!

UCLA folks convinced me that it is better to think about sharks and tuna first. The reason is that we really don't want students to start by *solving* equations. Instead, we want students to start by *declaring variables* and *representing systems* with *trajectories in state space* and *time series plots*. When it comes to representations in state space, the 2-dimensional case is *more intuitive* than the 1-dimensional case (phase portraits come in Lab #3). Having two variables really primes the conceptual approach. If you begin with single-variable exponential growth, then (1) some students will know it already and be bored on day one, and (2) you will condition students to think that math is about finding exact solutions to equations. That's just not true in biology.

In fact, for mathematicians, finding exact solutions to differential equations has not really been the focus for the past century. We often teach ODE (Ordinary Differential Equations) as a course in finding exact or series solutions through a cornucopia of techniques. But most systems of ODEs do not have nice closed-form solutions. Therefore, since the work of Poincaré, V.I. Arnold, and others, the 20th century view of ODEs has been highly visual and geometric. In the 21st century, the labor of making such pictures has vanished thanks to computers. So our students, in the 21st century, may study systems of ODEs in the modern visual/geometric manner with the assistance of computers. That is one way to describe this course to mathematicians. We introduce systems of ODEs in a modern way with visual, geometric, and computational tools throughout.

General advice

Two parts of this lab are essential, and should be emphasized even if they are not sharks and tuna nor insulin and glucose. These are the *bathtub problem* and the pair of pages on *interpreting/expressing with change equations*. The bathtub problem is crucial so that students can break down change equations and identify inputs and outputs. The pages on interpreting/expressing are core to the class, and students will struggle for a while. But they'll get it by week 10! Keep coming back to these pages, even weeks and months later.

Note that we do sharks and tuna first, and then insulin-glucose a bit later. We do both in Lab 1 to set a standard for the class. It's not an ecology class nor a molecular biology class. A lesson for everyone is that modeling applies to biology at all scales, from molecular to micro-organisms to physiology within an organism to whole populations in their environment. We hit examples at all scales because modeling is relevant, and because our students will study biology at various scales.

Instructors should use the following words all the time in this class: **model** and **modeling**, **state variable**, **parameter**, **time-series plot**, **state space**, **trajectory**. Avoid vague references to "draw a graph" and be more specific to distinguish different kinds of graphical representations: time-series plots and trajectories in state space for now. Later there will be semi-log time series, phase portraits, and more. Here are some canned questions that you can try out in section.

- What does that state variable/parameter **mean** in this context?
- What are the possible and plausible values for that state variable/parameter? What does your state space look like?
- What are the appropriate units for that state variable?
- Is that state variable going to increase? Decrease? For how long?
- What are the assumptions underlying this model?
- How did you choose that approach?
- Did anyone else choose a different approach around the classroom?
- What does that graph tell you about the system?
- Can you draw the time-series on the board so everyone can see?
- Does this model predict anything that is unrealistic? Under what circumstances?

Do you have other questions to ask students? Things to keep them in flow? Write some down to remember them for later. Also, do not be afraid to make requests of your students, such as:

- Please hand in your worksheet.
- Please put away your phone/tablet.
- Please write that up on the blackboard to share.
- Please write your solutions on scratch paper. Rewrite the solutions in the lab manual after class, when you are confident and have time to reflect and edit.
- Everyone needs to take a seat and open your computer if you have one with you.

Do you have tricks for classroom management? Getting everyone to quickly reassemble or shush? Share with your teaching team!

Lab 1, Section 1

Lesson summary: Students explore shark-tuna dynamics and visualize population changes using time-series plots and state space.

Key terms: Change equation, State variable, Parameter, Rate of change, Time series plots, State Space, Trajectory.

Learning objectives: After this lesson students will be able to:

- Identify the factors that affect two populations in a system.
- Connect the rate of change of a variable to the slope of a time-series plot.
- Identify "change" in a trajectory in state space, and in a time-series plot.

Before section: Complete lecture digest, and bring into section.

During section: ST1, ST2, ST3, ST4.

After section: Complete above. Also complete BT1, BT2, N1, N2, and worksheet for the section ticket.

Plan

Getting settled (5 minutes): It takes a little while on the first day!

- Everyone sits down, but don't get comfortable or unpack bags.
- TA and ALF distribute 4x8 blank cards. Instruct students to fold in half to make 2x8 tents. Then students write name on both sides in dark pen/marker.
- ALF collects all the cards, shuffles them, and distributes the cards (4-6 per table) during introductions below.
- TA and ALF collect *lecture digests*.

TA and ALF Introduction (5 minutes): Introduce yourself! Don't forget...

- (TA) My email address is...
- (TA) Office Hours (what they are, where they are, when they are, why to go there)
- (TA and ALF) What I like about math / science is.

Expectations (5 minutes): It is *crucial* to set expectations on the first day of section.

- Students are expected to bring lab manual (official hard-copy!), laptop, scratch-paper and pen/pencils to **every section**.
- Expect students to work with each other, and make the classroom a positive place for learning, asking questions. Expect students to write their ideas on the blackboard, especially incomplete and possibly wrong ideas! Expect students to address each other by name and with respect, even when there are mistakes.

- Students should have expectations of the TA and ALF. State these! E.g., "You can expect us to come to class prepared, and our first priority will be helping you to learn. We won't give you "the answers" but we will try to help you understand and productively struggle."

ST1: (20 minutes) Individual and group work through ST1, focused on time series plots.

- Ask everyone to move to the table which has their name card! Grab a seat and meet your Lab 1 group. Open up Lab Manual. **Write name in lab manual!!** No computers or phones on the table! TA and ALF float to keep things moving. Make sure students are NOT writing in their lab manual at the beginning – they need to discuss, and use the lab manual for *final drafts*. Scratch paper is important here.
- Ask students to work through ST1 on their own for 5 minutes. Each table should choose a representative to draw the time series on the board.
- As you wander around, encourage the groups to get the important elements of a time-series plot: axes with clear labels and ticks, a "legend" or other visual device to distinguish the two plots, good choice of scale.
- The **key discussion point** is the last question on ST1, because this is where a conceptual calculus idea begins. When one variable is high, the other has a high rate of change! A high rate of change is not reflected by vertical position of the time-series... it is reflected in the slope. Students will have lots of ways of answering this question, and see if you can tease out the big ideas, having them write sentences on the board to help. It is important to develop *precise language* to qualitatively describe the graph.
- If students are done, they can carry out an icebreaker while TA is floating. Icebreaker: Take turns answering the following: "State your name (what you like to be called), what college you are in, and the best thing you've found on campus or in Santa Cruz so far."

ST2: (15 minutes) The page ST2 is good for a short lecture. Focus on language, declaring variables with sentences ("Let it be" sentences!), and the Delta notation, as in the written introduction on ST2. Ask students to draw diagrams to answer the exercise on that page on the board. Answers can vary! A diagram is just a way of visually communicating something. Don't worry about "giving away the answer" because their answers can be their own.

ST3-4: (15 minutes) Don't expect to get through all of this! They will finish at home. Students will have seen trajectories in state space during lecture.

- Demonstrate how you would start the problem on ST3.
- Let them work with their same groups through ST3 and as much of ST4 as they can get through.

Conclusion Remind students of the time of the next section. Remind students to bring the printed and completed worksheet to the next section. Note students should complete ST1, ST2, ST3, ST4 (final drafts only in the lab manual!), and then complete BT1, BT2, N1, N2 on their own or with classmates, before the next section. **Don't forget** to collect the name cards in batches to keep groups for next section. And update gradebook based on attendance and lecture digest.

Lab 1, Section 2

Lesson summary: Students explore the general form of change equations with a single state variable, input and output terms, and parameters.

Key terms: Change equation, State variable, Parameter, Rate of change, Time series plots, Birth/Death rate. Semilog plot.

Learning objectives: After this lesson students will be able to:

- Build a change equation from input/output terms.
- Connect the rate of change of a variable to the slope of a time-series plot.
- Identify and interpret relative change, in the context of per capita birth/death rates.
- Utilize a semilog plot for time series, in the context of exponential growth.

Before section: Complete ST1, ST2, ST3, ST4 from last time. Compete BT1, BT2, N1, N2, and Worksheet before Section.

During section: BT1, BT2 discussion. Then Pop1, Pop2. Start SL1, SL2.

After section: Complete above. Also complete ST5 by the next section.

Plan

Distribute name-tents: Before students arrive, put the students name-cards back out on the tables, in the same groups.

Students find their groups (5 minutes): As students enter, they find their groups... same groups as previous section. Review expectations.

- Bring out lab manual, scratch-paper, and pen/pencils.
- Today is a no-tech day. No laptops or phones please.
- We will collect your worksheets soon! Get this out.
- Lab Manual Work. **Complete all work on scratch paper first!**. First lab check-in coming up soon.
- Please ask questions. We will try to keep you in "thinking flow" – might not answer questions completely.

BT1-BT2 Review: (20 minutes)

- Write the equation $\Delta W = iT - jD$ on the board. Going from group to group, see if someone can tell you what the letters stand for, i.e., "Group 1, what does W mean?" and "Group 2, what does T mean?"
- Ask each group to select one person to draw their time-series plot on the board (BT2), and other group members to fill in the axis labels, stages, and rates of change.
- **Collect the worksheets**, and while doing this, take 5 minutes to walk around the room and look at the other plots. What is similar? What is different?
- Question for the class: of the five letters (W, i, j, T, D), which ones change and which ones are constant? Which ones do we control, and which are set for us? Which one is the key "state variable" that we care about?

Pop1-Pop2: (25 minutes) Lecture on Pop1. The key point is how to go from a familiar situation (doubling bacteria) to the "change equation" $\Delta B/\Delta t = B$. What does it mean that the state variable is on the right side of the equation? How would you describe the change equation $\Delta B/\Delta t = B$ in words? How is this different from $\Delta B/\Delta t = 1000$?

- Give the students about 5 minutes to answer the exercise on Pop1, by setting up a *table* on the blackboard with columns: t, B, Δt, ΔB, and $\Delta B/\Delta t$. They've done an exercise like this on the worksheet.
- Then move to Pop2, with birth and death rate. Ask the groups to work on this by *first reading aloud* the top of the page. Then ask them to work together on the exercise there.
- Stop them at the end of the 25 minutes. They will finish this for homework. The challenge problem (EX* with a star!) is *required*, but do not give it away. Students can think about it more on their own.

SL1-2: (15 minutes) Don't expect to get through all of this! Use this time to

- Ask the class a few questions about the world population. Give them 1 minute to look at their Lab Manuals (SL1) and try to answer the questions. Good questions are: what was the population of the world in the year 1000AD? In 2000BC? How much did the population grow between 1AD and 1000AD? In absolute terms? In relative terms? Each group can choose a representative to speak, or you can cold call! The punch line comes afterwards: which graph (standard y-scale or log y-scale) was most useful to answer these questions?
- A key concept here is that semilog plots allow the reader to quickly see "fold-change". For example, a $\log_2$ scale allows one to quickly see when a quantity doubles or halves (doubling-time, half-life, etc.). A $\log_{10}$ scale allows one to quickly see when a quantity is multiplied by 10, a "10-fold change". Try to drive students towards this concept and descriptive language, e.g., getting them to write a sentence like "the electricity usage doubled from about 1960 to 1970."
- Then the groups can work through the exercises in SL1-2 until the end of class. If you have time to carry out a few steps to create the semilog plot on SL2, it would help students to get a good start!

Conclusion (5 minutes) If you haven't collected the worksheets, collect them! Record attendance and worksheets. Ask students to complete Pop1, Pop2, SL1, SL2, and ST5 by the next section, which will be all Sharks and Tuna. Remind them about the upcoming Lab 1 Checkpoint on Gradescope.

Lab 1, Section 3

Lesson summary: This is a key section in which students use computational tools (simulations) to study the Lotka-Volterra predator-prey model.

Key terms: Interaction term. Empirical law. Proportionality. Vector field. Trajectory

Learning objectives: After this lesson students will be able to:

- Provide empirical evidence that the number of interactions is proportional to the two interacting quantities.
- Identify and interpret birth, death, and interaction terms in Lotka-Volterra equations.
- Carry out Euler's method, to make predictions from a change equation.
- Connect a vector field to how the state of a system changes.
- Gain insight on a system by running a model within a computer simulation.

Before section: Students have turned in their Lab 1 Checkpoint, through ST1-4, BT1, BT2, N1, N2. They have also completed ST5 and SL1-2.

During section: ST6,7,8.

After section: ST9 and ST10: Reflections and model evaluation. Lang1, Lang2 before next section. Worksheet 2 due on Tuesday.

Plan

Students sit in their same groups, name-cards, etc. That machine should run pretty fast now!

Introductory lecture (10 minutes): The goal of today's section is to understand the Lotka-Volterra equations – a "change equation" model of predator and prey populations. These were introduced in lecture on the previous day. The first assumption is that we have two species – the sharks and the tuna – and that tuna would reproduce and thrive in the absence of sharks, while sharks would slowly die off in the absence of tuna. How would we model shark and tuna populations if they were kept separate? The hope is that students will (from their previous Pop1 and Pop2 exercises) guess something of the form $\Delta S/\Delta t = -\delta S$ and $\Delta T/\Delta t = \beta T$.

From there, it is reasonable to ask: how should this be modified in the presence of shark-tuna interactions? The goal is to get some sort of answer like "a positive influence on the number of sharks, and a negative influence on the number of tuna." Then, you might note that the *number of shark-tuna interactions* (chomp!) is what gives the influence, so we need to figure out *how many interactions* to expect. That is the point of ST6.

Shark-Tuna Interactions (25 minutes): Provide the following instructions to students. They will work at their tables, in the same groups as before. They will use our online tools at `https://martyweissman.github.io/modeling-synthesizer/`. Everyone should open their laptop! **Collect the lecture digests!**

- One person will be the data analyzer, using the "Linear Regression with Log Scaling" tool. They should press the "Clear Data" button to clear out the table. Today, X will represent the number of tuna placed, and Y the number of tuna eaten (interactions).
- Each group is given a fixed number of sharks (one group 10 sharks, one group 20 sharks, etc.). The rest of the group (not the data analyzer) should run the *Shark-Tuna Interactions* tool. Each person should choose an arbitrary number of tuna, and click the "Place" button. The output will show the number of tuna eaten. They report that number to the data analyzer, who enters that data into the table. (Remembering X is the number of tuna, and Y the number of tuna eaten).
- The data analyzer should record about 15 such observations, with a variety of X-values (and some repeated are ok!). After that's finished, they can click "Fit Linear Model" to get a best fit line. A reasonable question is: how good is this linear model? Some might know about R^2, which is reported in the tool.
- If time permits, they can try the same experiment, but fix the number of tuna, and let X be the variable number of sharks. Is there a linear trend there too?
- To wrap up, note the linear dependence of I (the number of interactions = tuna eaten) on both S and T. You can discuss the challenging exercise at the end of ST6, just in broad terms like "when I depends linearly on both S and T, the appropriate formula is $I = kST$. It will be challenging to estimate k, but some will have good ideas.

Lotka-Volterra: Tabulating (15 minutes): It is useful to display the Lotka-Volterra equations in ST7 to begin, with the constants given in the exercises. Demonstrate for students how you complete Rows 0 and 1 from the given information in the problem. Then ask the groups to figure out the rest. Leave the time series plot for homework.

Lotka-Volterra: Simulation (10 minutes): Here in ST8, students will use a simulator. For your knowledge, this simulator uses Runge-Kutta to solve differential equations.

- Point the students towards the *Shark-Tuna Trajectories* tool. You may want to float around, showing the functionality. Clicking on the vector field to start a trajectory (multiple trajectories can be displayed). Slow the animation speed to make it easier to see. The tool displays the differential equations, but they can think of S' as being like $\Delta S/\Delta t$ for now. Point out the trajectory in state space (top plot) and time series (bottom plot) with legend. Stop the trajectory for a moment, and ask the group "how many tuna and how many sharks are there right now?"
- Ask the students to load the simulation themselves and work through the exercises in ST8. They can hand-sketch the trajectory later. Float among the groups, to discuss the arrows (vector field) and what they mean.
- An important lesson, and the point of the "kill sharks" and "kill tuna" buttons is this: depending on when you click these buttons, it is possible to kill sharks (for example), and end up on a wilder trajectory that will yield even more sharks in the future. Messing with a dynamical system can put you on a different trajectory, and possibly yield results that are opposite of what you want! A mathematical model can provide insight into this (see ST9: insightful models!). Lotka-Volterra isn't really an accurate model for description or prediction, but is does provide insight.

Conclusion (5 minutes) Remind students to complete ST10 and Lang1, Lang2 by next section, as well as the worksheet (to bring in the hard copy). Insulin and Glucose starts next week. Record attendance and lecture digests.

Lab 1, Section 4

Lesson summary: Students explore a simple physiological model of the insulin-glucose system, analogous to predator/prey, but with a new phenomenon: saturation.

Key terms: Insulin, glucose, sugar, hormone, saturation, Hill function.

Learning objectives: After this lesson students will be able to:

- Provide examples of quantities that occur in physiology: concentrations of molecules in the bloodstream.
- Identify parameters of physiological importance, and how they affect a system.
- Study a two-variable system by first looking at each variable in isolation.
- Identify interaction terms, and positive and negative effects of one quantity on another.
- Use Desmos to explore parameter changes on a family of functions.
- Explain saturation graphically, and with a sigmoid function (Hill function).

Before section: Finished sharks and tuna. Lang1, Lang2.

During section: IG1, IG2, IG3, IG4.

After section: Final drafts through ST10 for checkpoint on Wednesday. Complete IG1-4 as needed by Thursday.

Plan

IG1 Warmup (20 minutes):

- Quickly have groups assemble at their tables together. This warmup is a **no-tech** time. Take 5 minutes to have students read out loud the paragraphs in IG1. One person can read the first paragraph, one person can read the next, etc., reading aloud to the whole class. You can write up the formulas on the board. Note the bracketed expressions, e.g.,

$$\Delta G = [\text{meals + liver production}] - [\text{cell usage and cell storage}].$$

  Please use this sort of notation. The point is to scaffold the creation of change equations, and refer back to simple systems like the Bathtub (BT1 and BT2). We understand change by understanding sources of positive and negative change.

- Then ask the groups to work on just the first exercise from IG1, and draw their graphs. Note here $\Delta G/\Delta t = m$, so their graphs should look like a line. You can ask what the slope of the line is (it is the parameter m!). This should take just about 5 minutes too. It's a quicky.
- Then students can work on writing up their solutions to the first exercise, and work on their second exercise in their groups. I think it's good to float around the groups, to discuss the meaning of the parameter s. Note that a high value of s means that insulin is "doing a better job" at clearing glucose from the bloodstream. And a low value of s is called "insulin resistance" – it means that insulin is not doing a good job, and glucose rises as a result. This is dangerous and can lead to Type II diabetes.

IG2-3 (25 minutes) These are exercises on understanding the "Hill function" $H(G) = G^2/(1 + G^2)$ and saturation.

- Start by having the groups read aloud from IG2, about how insulin down-regulates glucose, and write the change equation for $\Delta I/\Delta t$. There is a complicated term in there, and the goal is for them to understand it!
- Then ask the groups bring out their laptops and go to the graphing calculator at Desmos.com. They can start by plotting the basic function

$$H(G) = \frac{G^2}{1 + G^2}.$$

  This will let them sketch the graph for the first IG2 exercise.
- Now turn to the change equation for insulin. We will have discussed this in lecture too.

$$\frac{\Delta I}{\Delta t} = qb\frac{G^2}{1 + G^2} - \gamma I.$$

  Tell them that we're going to start understanding this with the parameters q, b equal to 1. So the equation simplifies to

$$\frac{\Delta I}{\Delta t} = \frac{G^2}{1 + G^2} - \gamma I = H(G) - \gamma I.$$

  Leading questions: What if there were no $H(G)$ term? What does this describe about the change in insulin when there is no glucose to consider? Answer: Insulin slowly degrades, e.g., if $\gamma = 0.1$, this would represent a 10% decrease in insulin during each unit of time.
- Have the students work on the exercise at the bottom of IG2 in groups, and then move onto IG3. The IG3 exercise requires the students to modify their $H(G)$ function in Desmos by adding parameters. Help the students to add the parameters (k and n), and appropriate sliders to answer that exercise.

IG4 (10 minutes) Now it is time to conclude this introduction to Insulin and Glucose, by dissecting all the terms. I think the groups should each take one term (like sIG or γI or $qbG^2/(1+G^2)$, and *write on the board* their sentence describing what it represents. Then they can present their sentence and/or walk around and see what the other groups wrote.

Conclusion (10 minutes) Remind students that...

- Lab 1 Checkpoint 2 is due tomorrow (Wednesday)
- They should finish IG1-4 by Thursday. Note: IG5-9 on Thursday, and then wrap-up next week with synthesis problems, and looking back at Lang1, Lang2 and others.

Lab 1, Section 5

Lesson summary: Students do a deep dive into the insulin-glucose model, with implications for health. They look at homeostasis, as it appears in trajectories in state space.

Key terms: Equilibrium, beta cell, concentration, sensitivity, diabetes, trajectory, state space, homeostasis.

Learning objectives: After this lesson students will be able to:

- Find the physiological implications of a mathematical model.
- Apply Euler's method, to tabulate values of two state variables governed by a change equation.
- Visualize two changing state variables with a trajectory in state space.
- Compare a system with closed trajectories to a system with a stable attractive equilibrium point.

Before section: Students have completed IG1-4. They have gone to two lectures related to insulin and glucose, physiology and health.

During section: IG 5-8.

After section: Complete IG 5-8. Draft Flow1 (model reflection). Start practice problems online, related to discrete time modeling, and trajectories and time series.

Plan

Students assemble into their Lab 1 groups. Today is a high-tech day: ask for at least two laptops open per group, so students can look over each other's shoulders.

Introduction to the simulator (10 minutes): Have everyone load up the simulator before you start talking, and give the tour by describing what to click while they follow along on their own screens. The tool is the *Minimal Insulin-Glucose Simulator*. Note this is *not* the same as the *Insulin-Glucose Regulation* tool, which is the fuller model used in Lab 4. Note there are light and dark and unicorn modes, for students to enjoy.

Explain how you can change the parameters with the sliders. The meals button should be self-explanatory. The "challenge" button refers to a Glucose challenge. This is a common test for diabetes (or pre-diabetes). Basically, the doctors make you drink a big bottle of glucose solution, and they study (via blood tests) how your glucose/insulin levels recover to a stable state. Note from the challenge mode, the insulin response looks a bit delayed – there is not a "delay equation" being used here... this is just what happens when you have the sort of ODE we study.

Note the units on the two y-axes. Note insulin concentrations are always much much lower than glucose concentrations (picomoles vs millimoles per liter)!

Students can reset with the Baseline button, or totally reset with the reload button (top-right). The help button (?) is also at top-right. Have students mess around with these!

Group study IG5-6 (25 minutes): Collect the Lecture Digests!

Ask the students to work through IG5-6 in their groups, and use this time to float among the tables. In addition to "start work" instructions, please ask each group to select one person to **read aloud the exercise** to the others. There should be ample time to float, and discuss the open-ended questions on these pages.

Insulin-Glucose trajectories IG7 (20 minutes): I would take a little intermission to recap what you heard as you moved among the groups. And then, go into IG7. This is a more traditional math section. I would suggest demonstrating Euler's Method to complete rows 0 and 1 of the table shown. Then students can work with their groups to complete the table and plot the trajectory in state space.

IG8 (10 minutes): The tool is the *Dynamical Systems Calculator*. Enter the system of equations. You might note that Xmin and Ymin should be zero, to be physically relevant! (State space does not include negative concentrations of glucose or insulin!) You can ask the questions of these exercises to the class, give them one minute to think, and then take answers from raised hands around the room. It can be a nice discussion, especially towards the end of IG8. Note: to see 5000 trajectories at once, just click the "Start" button. It's a nice way to see what physiologists call "homeostasis." But important note: this is a very simple mathematical model. In reality, the insulin-glucose system exhibits oscillation and other complicated behavior. We will return to it a bit in Lab 4.

Conclusion (5 minutes) Remind students that...

- Big due dates next week. End of Lab 1! Order and pick up Lab 2!
- Next Tuesday will be the last discussion section for Lab 1. Time to recap. Complete a draft of Flow1 and Flow2 before next section, but not final draft yet.

Lab 1, Section 6

Lesson summary: Students complete Lab 1, reflecting on two systems they have modeled over the past weeks.

Key terms: State variables, parameters, change equations, absolute and relative change, interaction terms, saturation.

Learning objectives: After this lesson students will be able to:

- Read a complicated change equation and interpret the terms and parameters.
- Compare two change equations, and explain how their differences reflect different situations and modeling strategies.
- Evaluate the quality of a model in multiple dimensions.

Before section: Students will have completed Lab 1, through IG8, and drafted Flow1.

During section: IG8 if necessary, Flow2 for discussion. Then TA choice!

After section: Complete Lab 1 and submit to Gradescope! Then read Page 1 of Lab 2 before Thursday.

Plan

Welcome (10 minutes): This is a more reflective day, when students may be stressed about the final Lab 1 deadline on the next day. It is a day to solidify concepts from the last few weeks. You may wish to survey the class, to see if there is one page from the lab manual that many students would like to revisit. Lang1 and Lang2 may be good ones, or Pop1 and Pop2. Note that page Flow2 is already reflective, and should probably be done during discussion section.

Flow2 (20 minutes) I think these are good questions for structured individual, group, and all-class sharing. For example, you can start by having students write the answer to each question on paper, with the class silent. (Just 1 minute). Then ask each group to designate a speaker, and discuss as a group for 3 minutes. When the group is done with discussion, and confident of their answer, the speaker can raise their hand or hold up a "ready" card. When all groups are ready, some of the speakers can speak their answer (2 minutes for 4 speakers).

Check Worksheets (2 minutes) Don't forget to check the worksheets!

Open: Your choice (30 minutes)! I think it is worth going back to something challenging from this first lab, even from the beginning. Ideas: Lang1 and Lang2 may get interesting responses, and they have core concepts. You could return to ST7 and IG7, and work on creating the tables. The Bathtub, and Pop1/Pop2, are also good and lead nicely into the next lab.

Conclusion (5 minutes) Remind students how to turn in their Lab 1. Lab 2 begins at the next section! They will be fine with just the PDF for that one, but should have their printed copy in hand for the section after.

This may be a good time for comment cards to get feedback on section before. Here are a few suggested questions:

- In what ways does discussion section help you learn?
- Is there anything we can do to make discussion section more helpful for comfortable for you?
- Other constructive comments?

Chapter 2

Growth

About Lab 2

Lab 2 is about *growth*, and that refers to a deep study of three different models of growth. These are: Linear growth, Power law growth, Exponential growth

Note that, unlike a traditional calculus course, we think of these as *models* rather than as *abstract functions*. We don't have to completely avoid the word "function" but in this lab the independent variable will always be t (time), and we are considering a quantity that is growing over time. We *use* these functions to model that growth. So the functions exist to describe something real.

This lab may be very challenging for students, because it has a touch of calculus. This touch of calculus can cause problems because it exacerbates the differences among our students. Students who have taken calculus might think they know the stuff, and will start invoking calculus formulae that their peers haven't seen. And students who have not taken calculus might then think they have to learn all of differential calculus in a few weeks.

So I think the best approach is to *avoid* teaching this material in the style of calculus. Maybe just avoid the whole word calculus, even though we're doing some derivatives. One should definitely not teach this by falling too deeply into old calculus habits, because our focus is very different. Please do not fall back onto using x for the independent variable and y for the dependent, and f for the function. We're trying to get students in the habit of thinking concretely of variables as representing quantities, and time (t) is a sort of special quantity throughout.

Here is the approach we take. We consider three families of models, and require students to understand each from multiple angles.

Linear growth First and foremost, X satisfies the change equation $\Delta X/\Delta t = m$ (a constant). We will transition to $dX/dt = m$ later in the lab. Geometric perspective: the time-series plot is a straight line. In words: X undergoes the same absolute change each unit of time. Not so important perspective for our class: $X = mt + b$ for some number b. I call this the "predictive model" and we also see it through linear regression.

Power law growth These we see first as functions, $X = Ct^p$ for parameters C (usually positive) and p (always positive). We discuss the power rule from calculus, $dX/dt = pCt^{p-1}$. Another important angle: if you plot this on a log/log plot, i.e., graph $\log(X)$ against $\log(t)$, the plot is a straight line with slope p. Power laws used to be a sort of hot topic in biology... these days, much less so, and power law models are often challenged.

Exponential growth This is absolutely core to biological systems. First and foremost, X satisfies the change equation $\Delta X/\Delta t = kX$, or eventually $dX/dt = kX$. The constant k is called the

first order growth rate. The time-series plot is a straight line, on a semi-log plot! In words: X undergoes the same relative change each unit of time. Lastly, $X = Ce^{kt}$ in "natural form" or $X = C \cdot b^t$ for some positive b.

As we proceed sequentially through these three models of growth, we hit two big mathematical concepts. First is the derivative, which we introduce primarily through estimation (choose a very small Δt), and drawing (the usual secant line approaches the tangent line). Second is the logarithm, which haunts biology students terribly until they get used to it.

To introduce the derivative, we start with a "tabular" sort of approach and a qualitative "slope" approach. In the tabular approach, they might know the value of a quantity Q at some time, like $t_{now} = 1$. If they also know the value soon after, e.g., at $t_{later} = 1.01$, then they can estimate the derivative:

$$Q' \approx \frac{\Delta Q}{\Delta t} = \frac{Q_{later} - Q_{now}}{t_{later} - t_{now}}.$$

On the qualitative side, students should be able to estimate the slope of a printed graph, and *very importantly* apply the appropriate units. For this, they can just sketch a tangent line at that moment in time with a ruler. And then compute the slope via rise over run. But units and the graph scale are crucial! In a math class, students often assume axes are evenly scaled, so "slope 1" looks perfectly diagonal. But in science, for the time series we consider, the slope will be in units "quantity units" divided by "time units" and it's meaningless to have "evenly scaled" axes when they measure completely different things in different units!

For logarithms, please *do not* introduce logarithms as inverse functions to exponential functions. They've probably seen that in precalculus, and it won't help them. They need to know what logarithms mean, e.g., $\log_{10}(1000) = 3$ because $10^3 = 1000$. And they need to log-scale axes to transform exponential change to linear change and estimate parameters. They will use calculators to compute logs to solve some equations. But again, it is not about the function concept.

Important pacing note: This is a difficult lab, not just because of the two challenging concepts (the derivative and logarithm). Also, it is difficult because there is a midterm right in the middle. Students have three sections, then the midterm, then three more sections. They have midterms in their other class(es) too. So it can be kind of stressful. Note that the midterm takes place during our lecture time, so this means that discussion section does a bit more work to introduce concepts too.

Lab 2, Section 1

Lesson summary: Students revisit a familiar topic from algebra – equations and graphs of lines – in the mathematical modeling framework.

Key terms: Linear growth, observation, proxy measurement, line of best fit, slope, intercept,

Learning objectives: After this lesson students will be able to:

- Characterize linear growth by change equation, plot, and predictive equation.
- Relate the slope of a time-series plot to the rate of change.
- Use appropriate units to describe rates of change.
- Identify elements of published time-series graphs, including lines of best fit and observations.
- Apply linear regression with a computational tool, in order to fit a linear model to data.

Before section: Get some rest after Lab 1! Read first page of Lab 2.

During section: LG1, LG3, LG4.

After section: LG2, F1, F2, N3, Log1.

Plan

Welcome back (10 minutes): Today you should create new groups randomly: shuffle the whole pile of name cards and deal 4-5 onto each table, as at the start of Lab 1. Remind students of this. They will need a little time to introduce themselves to their group-mates, in a similar fashion to Lab 1. This is a good time to tell students that the main goal of today is to understand lines in a different way. They will all have seen things like $y = mx + b$ in their prior algebra experience. But here we are thinking about lines as *models of growth* – how to fit a model, and what sorts of change are (or are not!) linear.

Before sending the students into the groups, I think you can pose the first three questions on LG1 to the whole class. This part is no-tech — laptops closed. The main graph (Fig 6 of Monod) is printed right there on page LG1, so just have everyone open their lab manual to it. Then ask them to write down (don't share yet!) their answers to the first three questions on LG1. You may wish to put these questions up on the board too. Then send them off to their new tables.

New group start (15 minutes) Allow 5 minutes for students to introduce themselves, majors, what they found most interesting from Lab 1. Then their goal is to finish LG1 by discussing their written answers and working on the last question. The last question asks them to estimate the slope and intercept of the given graph, and there is no "right" answer. The intercept looks like it is between 5 and 10. But note the "best fit" line has a higher intercept than the observed data point at $t = 0$. So this is a subjective question which can spur discussion. How *should* one estimate the slope and intercept? This is debatable, and leads directly into the idea of linear regression (LG4).

LG3 and LG4 (35 minutes) I think that it is safe to skip LG2 in discussion section, and leave that one for homework. The page LG3 contains three characterizations of linear growth, and these are crucial concepts! All three characterizations are crucial – the "predictive equation" like $X = mt+b$ and the change equation $\Delta X/\Delta t = m$, and the graphical characterization that the time-series plot is a line. Students are accustomed to learning *one* way to undertstand something, and here we are asking them to learn all three ways of understanding linear growth. I think that's worthy of a mini-lecture, before sending them into their groups to complete LG3. The exercises are meant to have the students shift among the three characterizations of linear growth, in particular contexts.

Once you have walked the room and checked LG3 is complete, the rest of the group time can be used for LG4. This is technology-time, and students will need laptops (one or two per group). Make sure that one person from each group is assigned to read the introductory paragraphs for LG4 before the groups begin! One person can also be assigned the role of data entry, running the Linear Regression tool on their laptop where the rest of the group can see it.

Conclusion (5 minutes) Next Tuesday, a worksheet will be due as usual, for the in-class section. Complete LG2, F1, F2, N3, Log1 for homework by Tuesday as well as the worksheet! It's a big batch of homework, but much of it is precalculus review/drill.

Lab 2, Section 2

Lesson summary: Students learn the derivative concept, and estimate the derivative, through making tables and drawing graphs. The derivative is identified as the instantaneous rate of change. This is then applied to study the rate of growth of a redwood tree.

Key terms: Derivative, time interval, tangent, instantaneous rate of change.

Learning objectives: After this lesson students will be able to:

- Estimate a derivative dX/dt at a given time t from an appropriate computation of $\Delta X/\Delta t$.
- Interpret dX/dt as an instantaneous rate of change.
- Use appropriate units and geometry to describe areas and volumes of objects and their densities and masses.

Before section: LG1-4, F1, F2, N3, Log1.

During section: D1, D2, RT1, RT2.

After section: Finish D2, RT2.

Plan

Introduction and D1 (15 minutes): Find groups, to match new random groups created during last section. Ask everyone to open the lab manual to D1. Each group should have *one* person with a computer open, and the rest with electronic devices put away (unless they have accommodations).

Draw the table from D1 on the board, with values of the function $P(t) = 100t^{1/2}$, and ask each group to use their computer (a calculator app is helpful) to compute one value of the function. This is a good time for a reminder that the one-half power of a number is the same as the square root of the number. Use the group's responses to complete the table.

While they're working, copy the sentence from the lab manual on the board: "On the time interval $[1, 1.1]$, $\Delta t =$ _____ and the rate of change is _____, and demonstrate how you can compute this from the data in the table. Ask the groups to compute the next two values of $\Delta P/\Delta t$ (using their assigned calculator-person).

Then it is time to open up the next question to the groups for discussion. One minute of discussion within groups then sharing out should be sufficient. Guide their responses towards what we are looking for: that these values of $\Delta P/\Delta t$ seem to approach the whole number 50. And that is the thing called dP/dt. The *derivative* dP/dt at a time t (here $t = 1$) is the rate of change $\Delta p/\Delta t$, for time intervals that start at t, and end just the tiniest moment after. How tiny? Well, what we really mean is the limit. As graduate students who have taught calculus, you probably have a good explanation in your pocket :)

Visualizing the derivative (10 minutes) Ask the groups to finish D1 with Desmos, with one person running the computer and the rest discussing to answer the questions. They may need a bit of help adjusting the axes, but that is important practice! They should find that a slope of 50 makes the line tangent. And this is a good time to share that the derivative dP/dt captures the slope of the *tangent line*.

Estimating the derivative (10 minutes) Lead the groups through completion of the first exercise of D2. (They may finish another one). For the first one, the function $P(t) = 100t^{1/2}$ is the same as the previous page. But they are starting at $t = 4$ instead of $t = 1$. What value of Δt should they use? That is *their choice*. The point is they should try a value that is small enough that they can make a good estimate. They might even try multiple values of Δt, which is good!

Redwood tree (25 minutes) This is an extended sequence of exercises which is going to be challenging for students. The first exercise requires some judgment from the students. How are they going to use a little image to estimate the radial growth rate of a tree? I would hope they "count the rings" as they go across the tree cross section, realizing that each ring represents a year of time. It is fine, and expected, that each group uses a slightly different estimate. Try to limit them to 5 minutes for that first exercise – if they have a somewhat realistic number for $\Delta R/\Delta t$, then they are good. I think it is good to then use this 5-minute point as a checkpoint, and use *your own* estimate of $\Delta R/\Delta t$ (e.g., 0.8 cm per year) to give your own linear function to model. For example, $R(t) = 0.8t$. Make sure that students **do not use a y-intercept** like $R(t) = 0.8t + 1$, or else the rest of the problem gets hard! This can be justified by saying that the initial radius of a redwood sapling is so small that it is negligible in our estimates.

Provide about 15 minutes for the rest of RT1. They will want to use a calculator for these basic science sort of exercises. These require unit conversions, area/volume formulas, and using density to convert a volume into a mass. While it is "basic science," it will be hard for many students. Make sure to practice, and look for the more experienced scientists in each group to play a bigger role. E.g., if anyone took AP chemistry, or high-school physics, they might be more active in explaining the process to others.

Finish the section with RT2. You can do the "1-year-old" problems at the board, using functions $H(t)$ and $M(t)$ that you have found in the room. Then the groups can try the 10 and 100 year old examples. The finishing exercises of RT2 are fine for homework – they require a bit more reflection.

Conclusion (5 minutes) Don't forget to check worksheets! Ask students to finish D1, D2, RT1, RT2 for homework, particularly D2 and RT2.

Also note that D3 and D4, which ask students to use the power rule, are saved for the next section. The power rule will be taught with examples in lecture on Wednesday, which will help them complete these pages on Thursday. I don't want to cloud the conceptual foundations of the derivative in this section with power rule drills – the power rule here is much more of a side-note in this lab rather than a central concept.

Lab 2, Section 3

Lesson summary: Students meet the power rule, and practice finding derivatives of linear and power functions. They connect this rule to what they found by approximation in the previous lesson. Then they review for the midterm.

Key terms: Power rule, derivative, slope, power function.

Learning objectives: After this lesson students will be able to:

- Apply the power rule to give the exact value of dX/dt in easy circumstances.
- Find dP/dt for a linear function, and interpret it as the slope.
- Describe the meaning of the derivative in at least two ways (e.g., slope of tangent, instantaneous rate of change, limit of $\Delta X/\Delta t$, etc.)

Before section: D1, D2, RT1, RT2.

During section: D3, D4.

After section: PK1, PK2.

Plan

Introduction and D3 (25 minutes): The page D3 introduces the "power rule," which is the first *real* rule of calculus – the first time that slopes of a curve were found by some symbolic method. For historical accuracy, note that Archimedes was able to figure out the slope of a parabola by more purely geometric means (this is important to understand how parabolic lenses focus light!), but here we have rules for all power functions. Even powers like 1/2 (square roots)!

Students struggle with "rules" in purely symbolic form, so it is useful to practice lots of examples at the board. Students are strong in pattern-finding, so you can write up many examples and ask students lecture-style how to fill in the gaps. E.g.,

$$\text{If } P = t^5, \text{ then } dP/dt = ...?$$

$$\text{If } P = t^{13}, \text{ then } dP/dt = ...?$$

Don't forget some odd ones like $P = t^{1/2}$, asking what the "power 1/2" means. Similarly for $P = t^{-1}$. The same "drop the exponent" sort of process works for all these, even $P = t^1$ which is just linear.

Please call on students rather than just getting volunteers for this one. Many (half?) of our students have taken some sort of calculus. And the power rule is just the sort of thing they remember. I want all the students to do the pattern-finding.

There are two key ideas in D3 worth stressing, once pattern-matching practice is done. One is that (miraculously) the derivative they find via the power rule matches the derivative they estimated in D1/D2. That's the reason for the first exercise.

The second idea is that derivatives are *geometric*. Vertically stretching a graph (e.g., stretching from $P = t^3$ to $P = 100t^3$ just stretches the derivative... it is a rise/run, and the rise gets multiplied by 100 here. Also geometrically, if you *shift* a graph upwards/downwards, e.g., shifting from $P = t^3$ to $P = 100 + t^3$, then the derivative doesn't change at all. The slopes don't change if you just move the graph!

Derivative drills, D4 (20 minutes) Page D4 contains numerous derivative rules. It may be easy for students who have taken calculus in their past, and hard for students who are new to calculus. I think it is worth doing 6-8 of these (3-4 from each batch) in class, at the board. Students at the board should be asked about two things: first, the *method* they used to compute the derivative; second, *the conceptual basis* for that method. They don't need to explain why the power rule works. But if a function is linear, they should be able to identify that the derivative just gives the slope. Or if the function is stretched or shifted, geometry can be part of the conceptual basis. We want students who understand derivatives graphically rather than just applying the power rule blindly.

Midterm review (15 minutes) This is a good time to go through some of the midterm review materials circulated over Canvas and in announcements. Since derivatives are fresh in their minds, you might have the groups work through the derivative-related problems from the previous midterms. This can help answer the question "what do I have to know about derivatives for the test?"

Conclusion (5 minutes) Remind students that the midterm is on Monday, during lecture. The Lab 2 checkpoint is due on Friday. Ask students to complete PK1, PK2 for homework (no rush). Note that PK1,PK2 is the first time they have a "situation" to understand on their own. The point is partly to see if they can apply what they've learned in an unfamiliar context, with a bit less support. That includes reading a bit of technical background. They can definitely work together and go to office hours, but PK1 and PK2 are really for them to work on outside of section.

Lab 2, Section 4

Lesson summary: Students develop an exponential function from first principles about bacterial reproduction (binary fission). And then they explore the general exponential function in the $F(t) = Cb^t$ form, estimating its derivative to find the key principle: that exponential functions are proportional to their rates of change (derivatives).

Key terms: Doubling time. Exponential function. Derivative. Proportional.

Learning objectives: After this lesson students will be able to:

- Describe a "doubling" situation with an exponential function, in a way that conveys doubling time.
- Interpret exponential functions as the results of repeated multiplication.
- Estimate the rate of change of an exponential function.
- Identify the relationship between an exponential function and its derivative.

Before section: D3, D4. Also PK1, PK2, though these may be safely delayed if students need more time.

During section: Exp1, Exp2, Exp3, Exp4.

After section: Log2, Log3.

Plan

Introduction (10 minutes): You can introduce the situation of Exp1 (bacteria undergoing binary fission) in a little lecture, and set up the table of values of $P(t)$ when $t = 0, 20, 40, 60$. You can lead the class through these values (first exercise on Exp1), taking responses from around the room. And they can copy these down on their lab manual. Then I would pose the questions to the class from the second exercise: How is $P(20)$ related to $P(0)$. How is $P(40)$ related to $P(20)$. How is $P(60)$ related to $P(40)$? If you pose these questions separately, and write the responses up on the board, you should get some like "$P(20)$ is 1.5 times $P(0)$." The challenge for students is how to express these three answers in just one formula. What I have in mind is students writing something like $P(t + 20) = 1.5P(t)$. Or even $P(t + \Delta t) = 1.5P(t)$. The purpose of asking such questions is that we use *variables* to describe a general situation instead of a specific situation. So it is one thing to be able to write $P(60) = 1.5P(40)$. It is a more difficult abstract task to write $P(t + 20) = 1.5P(t)$. I want students to gain experience in basic algebraic abstraction.

Bacteria and doubling (25 minutes) To finish Exp1, I want to lead students towards an exponential function like $P(t) = 1000 \cdot (1.5)^{t/20}$. First, they should recognize that $P(120)$ requires multiplying by 1.5 six times. So $P(120) = 1000 \cdot (1.5)^6$. That you might even lead them towards with lecture. But send them to work in their groups pretty soon. Can they generalize that formula to $P(t) = 1000 \cdot (1.5)^{t/20}$? Key leading questions are "what if instead of $P(120)$, I had asked you about $P(200)$?" and "What is the relationship between the time t in $P(t)$ and how many times you have to multiply by 1.5?"

After 10 minutes, ask all the groups to send you their "answer" – their formula for $P(t)$. If it is not easily equivalent to $P(t) = 1000 \cdot (1.5)^{t/20}$ give advice. After 15 minutes, wrap it up and get to Exp2. The groups should be able to complete Exp2 on their own. The toughest is probably the doubling-time. Students love to look up these formulas on Khan academy, Google, etc. The problem is that half-life and doubling formulas all use slightly different notation and are valid in slightly different settings. Here I really want students to understand how they arise from first principles. The goal is to get a formula like $P(t) = 1000 \cdot 2^{t/d}$, where d is the doubling time. Students should think of this as being repeated multiplication by 2. How many multiplications? The answer is t/d (the amount of time elapsed divided by the amount of time to double).

The general exponential function (25 minutes) Now that students have seen exponential functions in two forms (with base 1.5 and 2), introduce the general exponential function $F(t) = Cb^t$. Note that C and b can always be positive here. You can run the Desmos show as a lecture, varying the parameters C and b (set your sliders to avoid negative values), and asking students to describe what they see. They can write in their lab manual as the show goes on. That should take 5-10 minutes.

Then it's time for group work. The derivative estimation of Exp3 and Exp4 should be very familiar from the previous section, except here there are exponential functions and previously there were power functions. They may appreciate a bit of help with calculating these quantities, how to use calculators on their computer. Desmos also has a nice free scientific calculator. Students will probably not finish Exp3 and Exp4, since there are a few similar exercises. But if they can do a few together, they can finish on their own.

Conclusion (5-10 minutes) To wrap up, some students may have graphed the derivatives at the end of Exp4. You can illustrate this on Desmos too. Discuss the meaning of the derivative graph. It captures not the time-series of the quantity, but the time-series of its rate of change. Desmos has a nice feature students might like: if you graph a function by using a letter, e.g., graph $L(t) = 2t + 1$, then you can type $y = L'(t)$ to graph the derivative. That's a nice one to show students to help them finish Exp4. The concluding principle from Exp3 and Exp4, worth repeating, is that every exponential function is proportional to its rate of change. This is meant to get students to associate a change equation like $dP/dt = kP$ with an exponential function!

Note for the TA: In fact, this proportionality works even if you use a discrete version $\Delta P/\Delta t$ with some fixed time interval Δt; the proportionality constant is just a bit different. E.g., if $P(t) = 2^t$ and $\Delta t = 1$, then $\Delta P/\Delta t = P$.

Lab 2, Section 5

Lesson summary: Students become comfortable with multiple representations of exponential functions, including the number e and the "natural" form Ce^{kt}.

Key terms: The constant e. Derivative (as a function). Natural logarithm ln. First order growth rate.

Learning objectives: After this lesson students will be able to:

- Connect a change equation of the form $dP/dt = kP$ to an appropriate exponential function.
- Describe a process of doubling or halving using an exponential function with base 2.
- Using reference materials, change between forms of an exponential function (different bases).

Before section: Log2, Log3.

During section: Exp5, Exp6, Exp7, Exp8.

After section: Finish Exp7, Exp8.

Plan

Derivative warmup (Exp5, 15 minutes) The first exercise on Exp5 asks students to write down the derivatives of $F(t) = t^3$ and $L(t) = 5 - 4t$ and $P(t) = 10^t$. Ask each group to write the derivatives on the board by their table. Then ask one group to present each, and explain what their answer means and how they got it. The goal is to understand the very different methods. For $F(t) = t^3$, they will probably use the power rule (some students know it or remember it from lecture). For $L(t) = 5 - 4t$, see if they can figure it out by noticing this is a line of slope -4. For $P(t) = 10^t$, they have previously estimated the derivative numerically. This leads into the last question, where they should find this first order growth rate k, as the constant relating $P'(t) = kP(t)$.

Note: this may be the first time they are using the notation $P'(t)$ instead of dP/dt. This takes some time to adjust to. The reason for using $P'(t)$ is that we are thinking about the derivative as a function here, and not just computing rates of change.

This warmup should lead to a brief lecture about first-order growth rate. Exponential functions $P(t)$ are characterized by the fact that $P'(t)$ is *proportional to* $P(t)$. This phrase "proportional to" is not in their everyday vocabulary, so it is worth emphasis: this means that $P'(t) = kP(t)$, for some number k. This number is called the first-order growth rate of the quantity P. This sets up the goal of Exp6: to explore the first order growth rate.

The emergence of e, Exp6 (20 minutes) Ask one or two people in each group to be Desmos operators. As a quick warmup, ask the students to load up Desmos, and graph $F(t) = t^2$ (in one line) and $y = F'(t)$ on the next line. They should see a parabola and a straight line ($y = 2t$) on their graphs. Walk around until they all have these two graphs. It is a good time to ask what the line represents – in lecture, I discussed this fact: the derivative of $F(t) = t^2$ is the function $F'(t) = 2t$. So the linear graph represents the rate of change in the parabolic graph. Once everyone is on the same page, they can work on Exp6 for 15 minutes. This page can be *very* quick, if students know what they need to do. They are meant to twiddle the graphs on Desmos to estimate e. Here e is the unique real number for which $P(t) = e^t$ is its own derivative. In other words, this is the unique real number for which the exponential function $P(t) = e^t$ has first order growth rate 1.

Natural logarithms, natural form, Exp7-8 (25 minutes) Students have already done some drill involving logarithms base 10 and 2. The page Exp7 contains similar problems about the logarithm base e, which is called ln. The should be able to compute logarithms by rephrasing logarithm questions as exponent questions – that is the first batch of exercises on Exp7. The estimation of $\ln'(100)$ is a very nice (I hope) problem which reinforces prior knowledge of estimating derivatives with their new natural logarithm concept. Note that Desmos also has a scientific calculator to assist them in computations of ln. But hopefully they compute $\ln(e^2)$ without a calculator!

It is a good idea to finish Exp7, and do the first half of Exp8 during section. This involves the mechanical task of converting bases to base e, using the natural logarithm: the formula is given in Exp7:

$$b^t = e^{\ln(b)t}.$$

The point of this is that it lets you put all exponential functions into a "natural form," where you can "read off" the first-order growth rate k. Note in the above that $k = \ln(b)$. You might demonstrate this with

$$100^t = e^{\ln(100)t} = e^{4.61t}.$$

It is a good habit to put in concrete numbers and simplify/round. Then note that the first-order rate constant is 4.61. In other words, if $P(t) = 100^t$, then $P'(t) = 4.61P(t)$. When you get caught up in mechanics like this, it's good to bring it back to the basics as often as time allows.

The last exercise here is the Carbon-14 "word problem." This is a good thinking question, and may be best assigned for homework and review during the next section.

Conclusion (5 minutes) Remind students that homework is easing up again (after the stress of midterm). Their homework is to finish the Exp5-8 stuff, particularly Exp7 and Exp8. This *really* should be finished in time, because Lab 2 is due next week!

Lab 2, Section 6

Lesson summary: This is the final discussion of Lab 2, which focuses on the idea of *fitting* a model to given data. Three independent scientific contexts relate to the three models of growth (linear, power law, exponential growth)

Key terms: Fitting a model, R^2, regression, log-scaling, base-pairs, metabolic rate, degradation.

Learning objectives: After this lesson students will be able to:

- Apply linear regression with a computational tool, to find the line that "best fits" given data.
- Apply log-scaling to the horizontal and/or vertical axis, to assist with fitting a power law or exponential model.
- Appropriately select/filter data, based on situational awareness and reasoning.
- Interpret the R^2 statistic, in the context of simple linear regression.

Before section: Exp5-8.

During section: Fit1, Fit2, Fit3, Fit4.

After section: MG1, MG2.

Plan

Linear regression demonstration (10 minutes) To begin, it is a good idea to complete the first question on Fit1 as a demonstration. Students have already used the linear regression tool, but this is a chance to show how to efficiently use the tool to enter data, fit the model, and find the regression coefficients. The tool will report R^2, and also will help visualize the RSS (residual sum of squares). Note that fitting the model means minimizing the RSS, so this is an important concept for understanding what people mean when they say the line is the "best fit". The quality of the fit is assessed with the RSS.

GC-content linear regression (10 minutes) The rest of Fit1 is very easy to complete, by just imitating the process you have demonstrated. It is just another linear fit. What makes it complicated is the molecular biology setup. Fundamentally, there are two variables: the "GC-content" of a chunk of DNA and the "melting point" of the chunk of DNA. These are roughly linearly related. Recall that the "letters" (nucleotides) of DNA are A,T,G, and C. DNA is shaped like a ladder, and each rung of the ladder is either an A-T rung or a C-G rung. The GC-content captures what percentage of the rungs are C-G rungs.

The importance of this is that it takes more heat to separate the C-G rungs, when DNA is "unzipped." When DNA replicates, e.g., when a biology lab replicates DNA via the process known as PCR, there will be uneven amounts of replication depending on GC content. Some genes (chunks of DNA with 100s or 1000s of rungs) will have a higher GC content than others. It is sort of amazing to me how GC content has diverse impacts across biology. For an example on the ecology and evolution side, see `https://doi.org/10.1073/pnas.1321152111`. On the genomic side, see, e.g. `https://pmc.ncbi.nlm.nih.gov/articles/PMC3315510/`.

The *only* thing students have to do in Fit1 is to recognize there are two (perhaps mysterious!) variables, and fit the linear relationship even with the distraction and complexity of the underlying science.

Log-scaling and fitting (15 minutes) The page Fit2 is meant to show how log-scaling can be used to apply linear regression to fit power-law and exponential models of growth. They should notice that appropriate log-scaling will improve the fit (measured by R^2), and hopefully will use linear regression to find a good model of $\ln(Y)$ as a linear function of X (in the first example) or as a linear function of $\ln(X)$ (in the second example). Where students will probably need some help is using the given formulas to transform a relationship into a power-law or exponential model. E.g., how to convert $\ln(Y) = mX + B$ into $Y = Ce^{mX}$. You may wish to demonstrate this with an example or two before students get started. For example, you could convert $\ln(Y) = 3X + 2$ into $Y \approx 7.4e^{3X}$, to help them with that key last step.

Note the power of this log-scaling technique. This is used in practice, particularly with semilog plots and exponential models. It can be used to estimate growth rates for populations, in epidemiology, etc.

Big examples (25 minutes) The pages Fit3 and Fit4 are two very real examples. The first involves a metabolic power law (one of the few power laws in biology that seems to survive scrutiny... maybe). The first question there asks students to select only 10 of the 13 given examples. This is important, I think, as one should never blindly apply statistics. Students need some training in thinking about the meaning of the given data, what may be outliers for what reasons, etc. There is not a "right" answer to what animals to keep in the data set, but there are certainly better and worse answers! (I would exclude the birds, since the rest are mammals!)

The last example is Fit4, which relates to pharmacokinetics. It is a nice example of exponential decay (and not radioactive decay!). It is ok not to finish this one, but I do hope it brings together many of the techniques from Exp1-8 in an important biological context.

Conclusion (5 minutes) This is the last section of Lab 2. Remind students that there is no lecture tomorrow (Nov 11), due to the Veteran's Day holiday. Complete MG1 and MG2, and submit Lab 2 on Gradescope by tomorrow. Lab 3 begins this Thursday!

Chapter 3

Equilibrium

About Lab 3

Lab 3 is about equilibrium, and it brings together some of the general dynamical systems ("change equations") ideas from the first lab with some of the mathematical ideas of the second lab. It begins with a thorough study of the logistic growth model. To understand it, we can begin with a typical exponential growth model:

$$P' = kP \text{ with exact solution } P(t) = Ce^{kt}$$

If "crowding" causes a problem, then we find a negative "self-interaction" term proportional to P^2. The equation becomes:

$$P' = kP - \ell P^2.$$

This is the logistic change equation. One often factors to get something like

$$P' = \beta P\left(1 - \frac{P}{C}\right),$$

where β is the birth rate parameter, and C is the carrying capacity. One can exactly solve this equation, but the more important thing is understanding the qualitative behavior through a deeper study of equilibria.

The logistic growth model leads into a treatment of phase portraits for change equations in one variable. Phase portraits are visual devices, in which stable equilibria are marked with a filled dot, and unstable with an open dot. With one state variable, the state space is just the real line (or perhaps the positive real line). Trajectories are movements along this line, and they "flow" away from unstable equilibria and towards stable equilibria. Students will first see these for the logistic equation above, and then for more general 1-dimensional dynamical systems. Students will be able to go through a step-by-step analysis:

1. Looking at a change equation, like $P' = kP - \ell P^2$, and when asked to find equilibria, they will know to solve the equation $P' = 0$ and why (equilibrium = no change, i.e., $P' = 0$).

2. Students will use a variety of algebraic techniques to solve the equation. E.g., solving $kP - \ell P^2 = 0$ by factoring.

3. Students will draw the phase portrait, with equilibrium points at the solutions to the equation above.

4. Students will identify whether equilibria are stable or unstable by one or two methods. E.g., looking *near* equilibria and seeing whether P' is positive or negative. Or, looking at the slope of the graph of P' vs P. The latter method "works" but I find it less intuitive.

Once students have a functional understanding of equilibria in a single state variable, we think about how equilibria change with parameter changes. This is a **really important point**. A parameter shift can make equilibria appear or disappear: an unstable and stable equilibrium point can collide and annihilate, or a pair (unstable and stable) of equilibrium points can appear from nowhere!

This phenomenon is incredibly important in biology, e.g., in biological switches and tipping points. For example, an ecosystem may be at or near an equilibrium point. Then, some intervention (pollution? warming? etc.) may shift a parameter. You might think that the equilibrium will just shift continuously with the parameter. But sometimes the shift can make the equilibrium point dissappear, at which point the system may suddenly shift to a faraway equilibrium point (e.g., extinction). Bringing the parameter back to "normal" may not be enough to bring the system back to its old equilibrium point, even if it reappeas. It may be stuck in its new equilibrium state.

We see an important example in the lactose system, which introduces the "Lac Operon". This is a textbook example in biology of central historical importance in the understanding of gene regulation, and it's a complicated system! It will take up a solid day of discussion section. We go on to study more 1-dimensional systems. One is related to gene expression, and two are related to chemistry. That will finish our study of phase portraits and dynamical systems in one state variable. There are some key principles:

1. One can almost entirely understand 1-dimensional dynamical systems through identifying the equilibrium points and their stability (looking "near" the equlibrium points).

2. Parameter changes can create/destroy equilibrium points in pairs.

3. 1-dimensional dynamical systems can never oscillate on their own. At least, not until we introduce delay equations in Lab 4.

The rest of this lab relates to 2-dimensional systems and the shapes of their equilibria. Our systems will be mostly ecological, as a sort of advanced return to shark-tuna type systems. But instead of just predators and prey, we study pairs of species in competition (so a negative interaction term for both!) or cooperation, etc. These lead to generalized Lotka-Volterra equations. On the visual dynamical systems side, we see how much more complicated equilibria are when there are two state variables instead of one.

With just one state variable, the possibilities are really just stable or unstable equilibria. (Semistable are not so important). With two state variables, one finds spiral nodes, saddle points, etc... as shown in the opening figure of the Chaper (from Lotka, 1925). We work with nullclines just a bit in these 2-dimensional systems, since they can be useful tools for understanding dynamics in what follows.

Note that this is the last Lab of Math 16A. After this, students will move onto Math 16B, which starts with oscillation!

Lab 3, Section 1

Lesson summary: An introduction to growth with carrying capacity: the logistic model. This bridges the previous lab (Growth) to the current one (Equilibrium) in a foundational theoretical model.

Key terms: Logistic model, equilibrium, stable and unstable, phase portrait.

Learning objectives: After this lesson students will be able to:

- Interpret the terms of the logistic change equation $P' = \beta P - \gamma P^2$ in terms of birth rate and self-interaction (crowding) effect.
- Identify and interpret the carrying capacity, given a logistic change equation.
- Interpret the phase portrait associated with the logistic model.
- Use the parameters in the logistic change equation to predict the shape of the time-series plot.

Before section: Students are coming out of Lab #2. Nothing due ahead of time.

During section: LO1, LO2, LO3, LO4.

After section: Finish LO1-4. Ph1. Worksheet.

Plan

Introduction (10 minutes) The introduction can cover the first two exercises (discrete vs continuous time models) from LO1, in a brief lecture based on the *Vibrio natriegens* bacterium. One way to do this is to give a brief lecture with the background on these bacteria, followed by a mostly-complete solution to the two exercises. For the first, you can create a table from the change equation $\Delta P/\Delta t = 0.07P$,with $\Delta t = 1$ minute. Then ask the students to fill in the blank to the sentence "Each minute, the population increases by _____ percent," and take a few answers from around the room. Then have the students open Desmos on their laptops (at least two per group) and graph $P(t) = Ce^{0.07t}$ themselves, walking them through it step by step. Then you can ask the class again (once everyone has the graph nicely zoomed), to fill in the blank "Each minute, the population increases by _____ percent." There should be a slight difference between these answers, due to the discrete vs continuous time difference. But $\Delta t = 1$ minute is pretty small, so the difference is not too great.

Self-interactions (30 minutes) Begin by making new groups, shuffling and dealing out the name cards as usual at the start of a lab. Give them a 5-minute ice-breaker to meet their new groups.

This should be introduced as on the bottom of LO1. The point is to count "self-interactions" when there is a population in a contained space. The punch line is that the number of self-interactions is proportional to P^2, when the population is P. Students will discover this empirically using the *Self-Interaction Explorer*. Namely, they will analyze the relationship between H (the number of "hits" or self-interactions) and P (the population). This is a quadratic relationship, $H = kP^2$, and this should be revealed by a log-log plot: $\log(H) = 2\log(P) + \log(k)$. Linear regression on a log-log plot should yield a slope near 2. There are also more theoretical arguments, if you care to provide them!

Logistic model exploration (LO3, 20 minutes) This is where the exponential growth model is complicated by putting in the self-interaction (crowding) term. So the equation becomes $\Delta P/\Delta t = 0.07P - 0.000035P^2$. The first two exercises can be done at the board, and students should observe that the population may increase or decrease when it starts at 1000 or 3000, respectively. The techniques should be review from Lab 1, with basic table-making. The key new idea is the last exercise, where students need to find the *equilibrium point* for the first time, by solving the equation $0.07P - 0.000035P^2 = 0$. It is absolutely crucial for students to understand *why* this is an important equation to solve: it is because equilibrium happens when rates of change are zero, i.e., when things are not changing.

The rest of LO3 can be a lecture, where you explain how the phase portrait is a tool for visualizing the equlibria, and thinking about the long-term behavior of the system qualitatively.

Note that while a single state variable (1-dim state space) seems like it should be simpler than two state variables (2-dim state space), and it is mathematically simpler, the visual devices are tricky for students. If you are confined to a number line, it is sort of weird to draw vectors along the line, and trajectories look like paths along the line. Everything sort of overlaps and is tricky to see at first. But students will get it!

Conclusion (5 minutes) Remind students to complete LO1-4, Ph1, and the worksheet before the next section. If you have extra time, you might load up the logistic growth explorer and give a demonstration to help students with LO4. The style of LO4 should be similar to previous tasks students have done.

Lab 3, Section 2

Lesson summary: A close study of equilibria and phase portraits, in the somewhat abstract setting of a differential equation of the form $X' = F(X)$.

Key terms:

Learning objectives: After this lesson students will be able to:

- Solve equations of the form $F(X) = 0$, using multiple techniques (factorization, reasoning about fractions, reasoning about positivity).
- Relate solutions to equations to equilibrium points, given an equation of the form $X' = F(X)$.
- Perform stability analysis via a test point method.
- Draw and interpret phase portraits for differential equations of the form $X' = F(X)$.

Before section: Students have studied the logistic model, and practiced solving equations (Ph1).

During section: A deep study of phase portraits in 1-dimension: Ph1, 2, 3, 4.

After section: Students read background on the Lac Operon (Lac1) and watch a video on the topic, e.g. Khan Academy. Students complete Ph1,2,3,4.

Plan

Group setup (15 minutes) New groups assemble at tables for the first time, allow each table to go around for introductions. While the groups are going through introductions, draw two horizontal lines on the board next to each table. These will become phase portraits later on.

Assign each group two equations from Ph1, so that all 8 equations are covered (most covered by two groups). Each group must write their solutions to these equations neatly at the board (two writers from each group). These solutions should be above or below the horizontal lines.

Equilibrium points (15 minutes) Ask each group to explain one of their equation solutions, to get through all eight equations. They need to explain the technique they used. If another group wrote the same equation on their board, ask them if they solved it the same way. The point here is: there can be many correct ways to solve an equation, there can also be some definitely incorrect ways to solve an equation (if you lose solutions!), and there are better and worse ways to solve an equation. We are trying to build up everyone's equation-solving technique so they can solve equations with confidence and efficiency.

Once all 8 equations on Ph1 are solved and discussed, ask each group to simply put (empty for now!) circles on the horizontal lines corresponding to each solution, and to replace their equation-solving work with the change equation. E.g., a group would erase $(X-3)(X-2)(X-1) = 0$ and replace it by $X' = (X-3)(X-2)(X-1)$. At this stage, the groups have gone through the first step of drawing the phase portrait. What they have drawn is the *state space* (the horizontal line) and the equilibrium points (circles). You might note that in many examples, the state space will only consist of non-negative values of the quantity, e.g., populations, concentrations, heights, etc., cannot be negative. But other quantities can be negative, e.g., temperature in Celsius, sea levels relative to a 0-marker, etc.

Stability analysis (10 minutes) Ask each group to choose "nice" points on their state space which are between their drawn equilibria. Include points that are to the left and right of all equilibria as well. At this point, they should take each of these "test points" as X, and plug it into the formula for X' to determine whether X' is postiive or negative at the point. Then draw the arrow right or left accordingly. You may wish to demonstrate this for one equation and one test point just to get started. For example, you could demonstrate for the test point $X = 0$ in the change equation $X' = (X - 3)(X - 2)(X - 1)$.

Once their stability analysis is complete, the room should be able to draw the complete phase portrait for all of the equations on Ph2. They can take some time to walk around the room and copy things down – this is a good time to look for mistakes, discrepancies between groups, etc. If time permits, you can reflect on easy and hard parts of this process.

Allee effect (20 minutes) The "Allee effect" is used to describe systems that are just a bit more complicated than logistic growth. If you wanted to put things in order of complexity, the simplest population growth model is exponential, with change equation $P' = \beta P$ (with β the per capita growth rate). The next simplest accounts for a negative effect of crowding, with change equation $P' = \beta P(1 - P/C)$ (with C the "carrying capacity" marking the upper equilibrium). The next model accounts for a population that has trouble growing without a critical mass, i.e., the population dies off if it is too small. There can be lots of biological reasons for this. Maybe the organisms need to work together for protection or food gathering. We might also be tracking a subpopulation with a trait (e.g. a genetic trait). Imagine we had a tub fill of goldfish, and a few of them had a weird mutation making them blue. Over time, they all reproduce. If there are just a few blue ones, they might be bred out by chance, but if there is a critical mass, the blue ones might thrive and become a larger chunk of the population.

In the Allee effect, the change equation looks like $P' = \beta P(1 - P/C)(P/D - 1)$ with equilibria at $0 < D < C$. Here C (the largest equilibrium point) is the carrying capacity, and D is that critical mass... if the population is bigger than D, it will rise towards the carrying capacity at C. If the population is less than D, it will die off towards extinction (stable equilibrium at 0).

With a bit of an introduction, the groups should be able to work together to complete the exercises on Ph3. This is a good time for them to put their noses in their papers, use Desmos, etc., to work through the page while you float around.

Collect worksheets for attendance!

Conclusion (5 minutes) The conclusion is a good time to discuss the last two exercises on Ph3. This is where students will be *interpreting* the phase portrait. What is nice here is that the phase portrait can be used to make real predictions about system behavior. That is exactly what they need for Ph4, which can be left for homework.

Please ask students to complete Ph1,2,3,4, and read Lac1 by Thursday. They may also wish to watch a video about the Lac Operon – Khan academy has one, and there are many others – before Thursday. I'll discuss this in lecture as well. The tricky thing is that the Lac Operon is very complicated from a gene regulation standpoint. Students will not need to know all about the intricacies (as they did not need to know all about how β cells produce insulin). Instead, we will focus on a few critical *quantities* and how they influence each other to make a system with a biological switch. So they can watch videos... but they should not worry if they don't understand all the technical aspects of gene regulation!

Lab 3, Section 3

Lesson summary: Lac Operon! How a shift in parameter can yield a dramatic change in equilibrium points, and the role this plays in a biological switch.

Key terms: Parameter, switch, sugar (glucose and lactose), all or nothing, first-order,

Learning objectives: After this lesson students will be able to:

- Create a model of change based on input input and output terms.
- Identify equilibrium points as those where input and output terms are in balance.
- Identify a bistable system, and express implications for its behavior.
- Analyze the effect of a parameter change on equilibrium points.

Before section: Ph1 - 4, particularly the creation and interpretation of phase portraits for general one-variable change equations. Reading Lac1 for background.

During section: Lac 1-4 (exercises start at Lac 2).

After section: GE1 and GE2 and a worksheet.

Plan

Constructing the change equation (15 minutes): This can be done mostly as a lecture. The point here is that the science is really very complicated, but we are just tracking a single state variable: L is the amount of lactose within the *E. Coli* bacteria, called *intracellular* lactose. This is where every word matters, when declaring a state variable. We cannot say L represents lactose, because we are trying to understand how much lactose there is in a certain place!

Once you accept this state variable, you need to identify the input and output terms. How does L increase? How does L decrease? Students can read about this in Lac2. Input is due to the cells (bacteria are single cells!) bringing lactose inside, through their cell membrane, with the assistance of lactose permease. Lactose permease (coded by the LacY gene) is a protein that forms a transporter to bring lactose molecules from outside the cell into the cell. The gene regulation complexity shows up here, where lactose permease production (the gene being turned on/off) depends on things like the available amount of glucose.

Output is due to metabolism: the lactose molecules inside the E.Coli are metabolized, which means they are broken down into other molecules. Lactose is a disaccharide, which means it is basically two little sugars (glucose and galactose) stuck together. When lactose is metabolized, it is broken into these two little sugars, which the bacterium can use for energy.

Cutting through all this, L increases due to "lactose permeation," i.e., lactose coming in through the cell membrane, assisted by the big molecule lactose permease. And L decreases because lactose is digested and used for food (metabolized).

System analysis (25 minutes) With the equation set up, Lac2 provides a brief opportunity to analyze a Hill type function. This is a slightly different Hill function from what they saw before, but still has the characteristic sigmoid shape. It may be interesting for students to compare... what does the extra 0.01 term do, for example? Why is it there? (It is because there is always a *little* bit of lactose permease present.

In Lac3, students are asked to plot the functions $P(L)$ (lactose input term) and $M(L)$ (lactose output term) on the same plot, with L on the horizontal axis. The point is that where the graphs cross is where input equals output, and hence where there is equilibrium. So instead of solving an equation like "input - output = 0", students look at "input = output". It is a mathematically trivial shift, but can be a real helpful shift in perspective!

The equation in this instance cannot be easily solved with algebra, so students will need to use a tool like Desmos (assign one Desmos operator per group!) to find when input = output. They can create the phase portrait, and discuss implications to finish Lac3.

Parameter shift (20 minutes) Things get very interesting when glucose is involved. Scientifically, the presence of glucose changes the gene regulation, altering the presence/absence of lactose permease. In this way, if g is the amount of glucose (viewed as a parameter), then g will shift the lactose input term $P(L)$. Now students can take the same old equations in Desmos, and put in the g parameter. So in Desmos, students are looking at the graph of

$$P(L) = \frac{0.01 + L^2}{1 + (1 + g)L^2} \text{ and } M(L) = 0.4L.$$

See `https://www.desmos.com/calculator/vdtewmhqh2` for this! (Students should make this themselves though.)

When g is high (above 0.6 or so), the graphs of $P(L)$ and $M(L)$ intersect just once, at a very small value of L. This is a stable equilbrium point, so when there is a lot of glucose, there will be very little lactose permeation. In other words, the bacteria will not eat the lactose, even if it is around. But when g is low (below 0.6 or so), the graphs of $P(L)$ and $M(L)$ intersect three times. These three intersection points correspond to three equilibria in the classic stable-unstable-stable pattern which makes a "switch". There is a stable state with very low lactose. And there is a stable state with very high lactose. So low glucose creates the opporunity for the bacterium to "switch" and become a lactose eater. How it actually flips the switch from off to on is more complicated... and I don't think that biologists fully understand. This is a real subject of research... see the Nature paper "Multistability in the lactose utilization network of Escherichia coli" at `https://www.nature.com/articles/nature02298` for more on bistability.

I think it is challenging to explain what Monod found using just the model we are using. Observant students might notice that as glucose decreases, the system may naturally be "stuck" in the off position of the lactose switch. But Monod found the switch is turned on after a short delay. The model is insufficient to fully explain how the lactose switch is turned on. But it does show that the switch is *possible* after glucose runs out.

Conclusion (5 minutes) Remind students to complete their homework (worksheet and GE1, GE2) by the next section. A Lab Checkpoint is also coming up. Note that GE1 and GE2 are a completely new system. Like the pharmacokinetics, students will have to read about a new biological system, and synthesize what they have learned in an unfamiliar setting.

Lab 3, Section 4

Lesson summary: Students apply their knowledge of change equations to study two examples from chemical kinetics.

Key terms: Concentration, mole, molar, reactants, products, kinetics, rate constant, ion, dissociation.

Learning objectives: After this lesson students will be able to:

- Work with concentrations as state variables, with appropriate units and chemistry notation.
- Relate a change equation $C' = -kC$ to exponential decay.
- Reason about quantities which are linked, e.g., an increase in one is always accompanied by a decrease in another.
- Work effectively with scientific notation and units, e.g., constants like $1.23 \cdot 10^{45}$.

Before section: Students have finished their work on the lac operon, and completed two pages on a gene expression system. Also they have completed a worksheet on phase portraits.

During section: Chem1 and Chem2.

After section: Thanksgiving! Students should complete a worksheet and thoroughly review Sharks and Tuna (Lotka-Volterra system) from Lab 1.

Plan

A brief introduction to the scientific setup (10 minutes): In this chemistry-focused day, we are going to work with substances that are dissolved in water (and water itself!). Chem1 is all about hydrogen peroxide, which is the molecule H_2O_2. Structurally, it looks like $HOOH$. You might find it in your medicine cabinet (or your parents), since it is a cheap disinfectant at low concentrations. A fun activity (safe? I don't know) is the "Elephant's Toothpaste Geyser" – add hydrogen peroxide with some dish soap and maybe a catalyst to make a bubbly fountain.

The chemical reaction is often written as:

$$2H_2O_2 \to 2H_2O + O_2.$$

But this hides a lot of complexity. Really, what is happening is something like this: the HOOH (hydrogen peroxide) has a week O-O bond, so it breaks apart into two HO molecules. These are *very* reactive. From there, one could find an $HO + H_2O_2 \to H_2O + HO_2$ reaction, and then an $HO_2 + OH \to H_2O + O_2$, and probably a bunch of others. These occur rapidly, and chemists have to work very hard to untangle the chains of possibilities. The *net result* of these reactions is the main one on Chem1,

$$2H_2O_2 \to 2H_2O + O_2.$$

The introduction can say a bit about the scientific setup, and the goal is to declare the state variable (concentration of hydrogen peroxide) and the chemistry notation $[H_2O_2]$ that is used to represent it. For a deeper dive into the science, you might look at "Decomposition of hydrogen peroxide in aqueous solutions at elevated temperatures," by Lin et al, 1991, `https://doi.org/10.1002%2Fkin.550231103`. Particularly, the pages pp.974–975 show the sort of first-order (exponential) decay that you find in the lab manual, with the techniques of semilog plots to identify first order reaction rate. Those pages could probably be integrated into the discussion section, in fact!

Exponential decay (20 minutes) Chem1 is really a review of exponential decay, connecting it to the phase portraits we have been drawing. The only equilibrium point is at 0, and it is a stable equilibrium point, so the concentration will tend towards zero. This is a good time to review the connection between the change equation $C' = -kC$ and the predictive equation $C(t) = C(0)e^{-kt}$. Groups can use this opportunity to practice exponential/logarithm skills to solve the exercises.

Chem2, dissociation of water (15 minutes) This is a scientifically important page, all about the "dissociation" of water. What this means is that water molecules naturally break apart sometimes: an HOH (water molecule) breaks apart into an H^+ (hydrogen ion) and an OH^- (hydroxide ion) sometimes. In this breakup, the OH gets an extra electron, which is why it is an "ion" OH^- while the hydrogen H loses that electron to become a hydrogen ion H^+. Note that a hydrogen ion H^+ is just a proton!

With pure water, this dissociation happens frequently, but the ions also come back together to form water. In this way one finds two sources of change in the amount of water: the formation of water from ions meeting (the $\beta[H^+][OH^-]$ interaction term) and the loss of water due to dissociation (the $-\delta[H_2O]$ term). The main job of the first two exercises of Chem2 is to really understand all the terms in the change equations. This can be done in a more lecture-like format, giving time for students to discuss and write their answers.

Probably the biggest challenge in Chem2 is that there are three variables: the concentrations $[H_2O]$, $[H^+]$, and $[OH^-]$. We haven't worked with three variable systems before! The crucial insight is that these three variables are tied together. The hydrogen and hydroxide ions always appear/disappear in pairs. And each loss of an H_2O molecule corresponds to a gain of an H^+ and OH^- ion. So really if you track one variable carefully, the other two are completely determined. This is what I mean with the learning objective: "reason about quantities that are linked."

Chem2, pH (15 minutes) The last exercises in Chem2 relate to the fact that the pH of water is 7. The pH is a really weird scale, at least for mathematicians. What pH means is $-\log_{10}([H^+])$, the negative logarithm of the concentration of hydrogen ions. In chemistry, there are a lot of similar scales, e.g., pK would be the negative logarithm of the concentration of potassium ions. All these things are critically important in cell biology.

In water, what happens is that the dissociation of water and reformation of water end up in equilibrium, in a state where the concentrations of $H+$ and OH^- are stable. The exercises ask the student to find this equilibrium, walking through a few tricky steps. Scientific notation and units are crucial here, so students will need a bit of time.

Conclusion (5 minutes) Remind students to have a great Thanksgiving! And... complete a worksheet for next Tuesday. And... to thoroughly reread the shark and tuna Lotka-Volterra stuff from Lab 1. That will be generalized in our last week (Week 10) of class.

Lab 3, Section 5

Lesson summary: Students examine 2-variable equilibria and the generalized Lotka-Volterra equations in the context of two ecological systems with competing species.

Key terms: Competition, phase portrait, equilibrium, nullcline, time series, state space, trajectory.

Learning objectives: After this lesson students will be able to:

- Interpret all terms in a Generalized Lotka-Volterra system of equations.
- Analyze a system with two species by first analyzing the system with no interactions.
- Distinguish competing species from predator-prey relationships by the form of the equations.
- Plausibly model a two-species system with state variables and change equations.
- Sketch nullclines when they occur as linear equations in state variables.

Before section: Students review Sharks and Tuna from Lab 1, particularly the Lotka-Volterra equations.

During section: GLV 1, 2, 3, 4.

After section: GLV 5,6.

Plan

Moose and Squirrel Intro (10 minutes): Welcome to the last week of class! The beginning of class is a good time for a quick (interactive) lecture, to review the Lotka-Volterra equation for sharks and tuna. Hopefully students remember this well enough to interpret all the terms. Then you can present the (fictional) system of moose and squirrel, governed by the generalized Lotka-Volterra equations:

$$M' = 2M - M^2 - 0.5MS.$$

$$S' = 3S - S^2 - MS.$$

Having both of these equations on the board, students should be able to gather in their groups and work out the exercises on GLV1. In particular, they can draw the two phase portraits (squirrels if $M = 0$ and moose if $S = 0$). One person from each group can present an example of a difference between Moose-Squirrel and Shark-Tuna, to collectively solve the first exercise on GLV1. It is true that the species are different, but the key differences to observe include:

- We used discrete-time modeling for sharks and tuna. Now we're using continuous-time modeling with derivatives instead of Δ.
- The non-interacting terms are based on a logistic model with Moose and Squirrels, rather than an exponential model with Sharks and Tuna.
- The interaction terms are both negative, reflecting a competetive relationship of Moose and Squirrels. For sharks and tuna, the interactions (chomp!) were good for sharks and bad for tuna.

Nullclines and stability (30 minutes) Nullclines are shown in lecture, but this is the first time for students. The way I explain it is that the "X-nullcline" represents the states where X does not change, i.e., $X' = 0$. When there are two variables, like S and M, you have two nullclines: the S-nullcline and the M-nullcline. The word "nullcline" sounds like "line" but nullclines are not always lines, and they are not always connected. (That's information for the TA!) In the GLV = Generalized Lotka-Volterra setting, each nullcline can be a *pair* of lines. Note that one pair of lines often intersects another pair of lines in 4 intersection points... those will be the equilibrium points. But sometimes those intersections can be degenerate, or lie outside the state space (i.e., if $S < 0$), etc. And self-intersections (the S-nullcline intersects itself!) are not equilibrium points! The math here is fundamentally algebra, with tricky conceptual and notational aspects.

To run GLV2 in Section, I recommend doing one example (one nullcline) at the board, following the marginal note, and then letting the groups draw the plot and the second nullcline at their boards. That will allow you to float around checking boardwork and equilibrium points. They can photograph their boards and fill out their manual later. Then for test points (for the vector field), each member of the group can get a different test point. You could suggest 5 test points, and ask each group to complete all 5 – each member of the group completing two test points to get a double-check.

While students are working, it would be good to write up the parameters for the Generalized Lotka-Volterra simulator. This can be tricky due to sign conventions, variable names, etc. The correct parameters (if P corresponds to Moose and Q to Squirrels) are:

$$\alpha = 2.0, \quad \beta = 3.0, \quad \gamma = 1.0, \quad \delta = 1.0, \quad u = -0.5, \quad v = -1.0.$$

The tool should allow them to show nullclines to check their work, identify the equilibrium point type (Stable Node). Clicking a few points on the vector field should tell thm about long-term behavior, or clicking the Start button will show thousands of trajectories. Note how they get swept in towards the central equilibrium point at $(1, 2)$. Students can experiment with the premade examples (predator-prey, like sharks and tuna, or competetion with mutual exclusion) too.

Gerbils (20 minutes) The rest of the class can be devoted to GLV3 and GLV4, which asks students to model a pair of Gerbil species. There is academic study of this pair of species, e.g., in `https://doi.org/10.1046/j.1365-2656.2001.00520.x`. These two species of gerbils coexist by taking turns: one active early in the night and the other active later in the night. This is all very cool... but in the exercise students are just asked to plausibly model the pair of species. The hope is that they cook up reasonable parameters for the GLV equations, and perhaps incorporate the fact that one gerbil is twice as large as the other. (Maybe that makes them better competitors? Maybe worse since they need more food? Just look for plausibility!) Students can use the GLV tool in the Modeling Synthesizer to help answer the questions in GLV3 and GLV4, and even to improve their choice of parameters. Note that the tool will provide time-series plots and trajectories in state space, which can help with the exercises too. Suggest using the tool, especially if time is tight!

Conclusion (5 minutes) Second-to-last section of the quarter! Please remind students to complete their SETS, and tell them when the last section meets. Students should complete GLV5 and GLV6 before then... these are again exercises which require some creativity and the goal is plausible modeling to reflect the interactions between species.

Lab 3, Section 6

Lesson summary: Students complete their study of the Generalized Lotka-Volterra Equations with exploration of parameters, and Gause's analysis of two species of Paramecia. Then they consider shapes of equilibria and their interpretation for the interaction of species.

Key terms: Competition, Predator-Prey, Cooperation, Attractive Equilibrium, Spiral Trajectory,

Learning objectives: After this lesson students will be able to:

- Interpret the coefficients in the Generalized Lotka-Volterra Equations.
- Identify systems with competition, cooperation, and predator-prey relationships.
- Interpret trajectories in state space, as predictions for the survival of species.
- Find examples of GLV equations which exhibit various behaviors near an equilbrium point.

Before section: GLV 1-6.

During section: GLV 7-8 and Eq1,2.

After section: Study for final exam, and complete Lab 3!

Plan

Introduction (15 minutes): To get started you might write the GLV equations with parameters $\alpha, \beta, \gamma, \delta, u, v$ as shown in the Lab Manual. You can pose the first two questions of GLV7 (What does it mean if $u = 0$ and $v = 0$? What does it mean if $u > 0$ and $v > 0$ but $\alpha < 0$ and $\beta < 0$?) to the whole class, taking a show of hands or a few spoken answers. Answers: if $u = 0$ and $v = 0$, it means the species do not interact... or at least not in a way that affects their population growth. And if $u, v > 0$ while $\alpha, \beta < 0$, this means that the species would die out without each other, but their interactions may help them both to survive (codependence).

Once you're on the same page there, it is time to send students back to their groups to finish GLV7. I would explain what is meant by an "attractive equilibrium point" first – this is an equilibrium point (P, Q) where all nearby points will have a trajectory leading to (P, Q). They can test this by clicking on nearby test points to see if indeed the trajectory just leads into the equilibrium point. I would also assign one GLV-explorer within each group to run the simulation on their laptop, where the rest of the group can see it. By "describe the dynamics" I mean for students to describe what happens when they click on various points. It may be that faraway points are drawn into other equilibrium points, for example. They can use their own words!

Fitting parameters (30 minutes) The page GLV8 is a pair of exercises to lead students into the process of model-fitting, just as the end of Lab 2 had students fitting linear, power-law, and exponential models of growth. Finding parameters to best fit a system of differential equations is a central task of mathematical modeling, and we don't have time to study this rigorously. Instead, we take a conceptual approach assisted by computational tools.

The first exercise uses the Logistic Growth Explorer, where students should enter the given data table. Within each group, a few students should take population P (to study parameters α and γ), while the others should take population Q (to study parameters β and δ). The idea is that we can estimate those four parameters by studying paramecia living separately (no interaction terms!). Students should be able to guess the carrying capacity C from the data tables, and tweak the birth rate parameter and t_0 to fit the data as best as possible. They can try to minimize the RSS if they want to do a good job! Doing that, I get:

$$\alpha = 0.65, \quad \gamma = 0.0011,$$

$$\beta = 0.93, \quad \delta = 0.0046.$$

Note that the logistic growth explorer uses parameters b and C and t_0. The tool displays the equation in factored form, $P' = bP(1 - P/C)$, so the lab manual's α is just b — that one can be read straight off. But γ is not displayed anywhere, so students do have to convert: $\gamma = \alpha/C$. (The lab manual page now says this explicitly, since otherwise there is just enough arithmetic/algebra to confuse/delay lots of students.) The t_0 parameter does not appear in the change equation – it just affects the initial value, by sliding left/right. So students should adjust this parameter to get a good fit, but it does not appear in their solution in the lab manual.

The second exercise uses the GLV explorer to consider the two remaining parameters: the interactions coefficients u and v. Students should be able to enter their α, γ, β, and δ. And some experimentation with u and v should roughly fit the given data. It looks like the two paramecia species are competing, with the P outcompeting Q by a small margin.

It is not clear whether the populations in the second data table have reached a stable equilibrium, or whether one species may eventually "win" while the other goes extinct. Many answers are possible. For example, $u = -0.004$ and $v = -0.004$ is not a bad starting place, and there will be competitive exclusion (saddle equilibrium). And $u = -0.002$ and $v = -0.001$ will be close, with a stable equilibrium point. As a starting point, you might recommend choosing u and v between -0.001 and -0.009. Students can use the Show Nullclines button to try to give the trajectory the appropriate shape.

Spiral node (15 minutes) To finish the section, you can look at the spiral node on Eq1. Groups can experiment with parameters, and the canned examples on the GLV simulator, to find a spiral node. What do the nullclines look like? What are the parameters? Does this reflect species in competition? Cooperation? Predator-prey? There are multiple answers possible, but they should be backed up by concrete numbers and explorations. Students can work through Eq2 on their own.

Conclusion (5 minutes) Remind students about review sessions, wish them a happy and safe holiday, etc.

Thank you for teaching Math 16A this quarter!

Chapter 4

Oscillation

Welcome to Math 16B! This is a continuation of Math 16A, so there will be familiar things and new things. Students will be familiar with the rhythm of the class – the lab structure, the lectures and sections each week, the big and small deadlines, and exam format. They will be ready to jump in. Additionally, they will be accustomed to the active learning environment of section. BUT... different TAs run their sections differently, so that is not completely dependable. Students will be probably be new to their particular section group, and their TA.

The TAs on the other hand may be new to the course – in a way they will benefit from the students being ready for section, but they will need some time to adjust! New TAs should definitely read the introduction to this manual, and work through some of Lab 3, and the beginning of Lab 4 to get a hang of things. Some of the math may be new, and most of the science will probably be new to math graduate students.

About Lab 4

Lab 4 is about oscillation – state variables that exhibit a long-term repetetive behavior of increase and decrease. I use the term oscillation a bit loosely. The most strict definition might be *sinusoidal*, meaning that a variable X obeys a predictive formula of the form

$$X(t) = a\sin(bt + c) + d.$$

I use a looser definition, that X oscillates if exhibits an interval of strict increase, then strict decrease, then strict increase, then strict decrease, etc., with the lengths of those intervals remaining constant. E.g., I would include damped oscillation like $X(t) = e^{-t}\sin(t)$. The main thing is that oscillations have two associated quantities: the *period* of the oscillation (time between peaks) and the *frequency* of the oscillation (number of peaks per unit time). They also have an amplitude – the height of the peak or depth of the valley, but this can variable in cases of damping and other irregularities.

Oscillation in this class comes from three distinct and important sources.

- The simple harmonic oscillator (SHO). I use SHO to describe any pair of state variables X and P which obey equations of the form $X' = uP$ and $P' = -kX$. In classical mechanics, if X is the displacement of a mass on a spring and P is the momentum, the first equation states that P is the momentum and $u = 1/m$ (where m is the mass). The second equation is Hooke's law, that P' (the force) is $-kX$, where k is the spring constant. We use the SHO to introduce the sine and cosine functions, as solutions to such equations. And the SHO arises in physics... and physical oscillations can drive biological oscillations! The other physical oscillations biologists

should know about are *molecular vibrations*. These are well-modeled by the SHO, and give each molecule its own signature set of frequencies. These affect interactions with light, and allow us to distinguish molecules in a somewhat non-invasive way, by shining light at them! Knowing which molecules are present in a sample is crucial for modern biology.

- The next source of oscillation is the *limit cycle*, and these occur in biological systems. We will see examples in glycolysis (a fundamental chemical reaction that occurs in the cell), and in ecology – with the Holling-Tanner predator-prey model. The key fact about limit cycles is that they provide a source of robust sustained oscillation. Robust means that small parameter changes or model modifications will not destroy the oscillatory behavior. And sustained means that they don't just damp away to nothingness... the amplitude remains visible for long periods of time.

- The final source of oscillation is the combination of negative feedback (seen earlier) with a time delay (new in this lab!). Time delay is crucially important, and sort of obvious. Everything in the world takes some time to have an effect, from nerve signals to environmental changes. We will see how time delays can yield oscillations in a physiological context (muscle tremor) and elsewhere.

If students understand these sources of oscillation well, the "icing on the cake" would be for them to understand Hopf bifurcations a bit. This fits into a broader theme of how parameter changes can yield drastically different system behavior. We saw this in Lab 3, where a parameter change can make a system go from having 3 equilibrium points to having 1 equilibrium point. This means that a parameter change can activate or deactivate a "biological switch," as exhibited by the Lac Operon.

Here something similar can happen. A parameter change can activate or deactivate oscillations – and we can find critical thresholds for which sustained oscillations occur. This task – locating the emergence of oscillation – will occur towards the end of a few models we study.

Lab 4, Section 1

Lesson summary: After settling in, students dive into the simple harmonic oscillator, becoming familiar with the equations, eliptical trajectories, and sinusoid time series.

Key terms: Simple harmonic oscillator (SHO), period, frequency, elliptical, sinusoid.

Learning objectives: After this lesson students will be able to:

- Recognize when a system of change equations is an example of the SHO.
- Draw the trajectories in state space, and the time series, associated to an SHO.
- Identify the key features – period, frequency, amplitude – from a time series of an SHO.
- Describe a sinusoid by an appropriate variant of the sine function.
- Know the derivative of sine and cosine by heart.

Before section: Students complete a worksheet very similar to SHO1, and they complete the previous three pages (including Osc1, Osc2) as a lab startup.

During section: Students work through SHO1, SHO2, and some of SHO3,

After section: Students complete SHO3 and SHO4.

Plan

Group making (15 minutes): It is important that students get settled quickly into groups, while leaving a little time for introductions and ice-breakers. Here is my recommendation. Make a big bag of blank table-tents for each student (fold a 4.25" by 11" piece of paper in half like a tent-shape... so two table-tents per standard 8.5 by 11 paper). Pass these out to students as they enter, and ask them to write their name clearly on both sides of the tent (first name large, last name small). Collect all the completed table-tents, and scramble them in the bag. Then put them out in groups of 4-6 at tables to make the first round of groups. **Students should stay in the same group throughout Lab 4.**

Once the students are in their groups, each group member gets 1 minute to describe the most interesting thing they have learned at UCSC. Try another 5-minute icebreaker if you've got one! Please *use* the table tents for two purposes:

- Feel free to cold-call students based on the name you see on the table tent!
- Use the table tents to record attendance in the Canvas Roll-Call utility.

Don't forget to collect table-tents at the end of class, stacked in their groups, to pass out again the next time. (Note from Marty: If you want me to purchase some sturdier table-tents, let me know. It's on me)

Discrete-Time SHO (20 minutes) To begin with, ask each group to copy the table from the first exercise of SHO1 onto the blackboard, and start filling it out. One student should have the chalk, and the others should help them fill out the table. It should be pretty easy if they've done the worksheet! Note that $\Delta t = 1$ at first, so really $\Delta X = P$ and $\Delta P = -X$.

time	X	P	ΔX	ΔP
0	1	0	0	-1
1	1	-1	-1	-1
2	0	-2	-2	0
3	-2	-2	-2	2
4	-4	0	0	4

FYI, the discrete-time SHO exhibits oscillations that blow up in amplitude. One can show algebraically that the "total energy" $X^2 + P^2$ doubles at every time-step.

The next part of SHO1 asks students to use a time step of $\Delta t = 0.5$ instead. So there, $\Delta X/\Delta t = P$ means that $\Delta X = 0.5P$ during each time-step of 0.5. The oscillations will still "blow up" – but not as quickly. As $\Delta t \to 0$, we move towards the continuous-time SHO in which the oscillations as perfectly sinusoidal.

Continuous-time SHO (25 minutes) The page SHO2 begins with a very brief call-back to Lab 3 on equilibrium. I think one can simply ask the groups to answer that exercise, and cold-call or ask for a volunteer to explain. We want students to be able to find the equilibrium points of the system $X' = uP$ and $P' = -kX$ through a brief thought process:

- Equilibrium means things are not changing, so $X' = 0$ and $P' = 0$.
- In this system, this means that $uP = 0$ and $-kX = 0$.
- Since u and k are positive constants, this implies $P = 0$ and $X = 0$. That is the only equilibrium point.

With that quick conceptual review complete (in just a few minutes if possible), it is time for the groups to open up laptops (2 laptops per table should be sufficient) and load the SHO simulator in the Modeling Synthesizer. These tools are all linked from the Canvas page. There they can experiment with u and k and see what the trajectories look like. They should note that when $u = k = 1$, the trajectories are circular. But changing u and k stretches the circle into an ellipse. Floating around, see if you can get the students to write complete clear sentences, e.g., "When $u > k$, the trajectory is an ellipse that is stretched horizontally." The tool has a "detect period" button that they can use for the final exercise on SHO2.

That should go pretty quickly, so students can get into the harder exercise in SHO3. This asks them to collect data on the period they find for various values of k (holding u fixed) and various values of u (holding k fixed). The "answer" is that the period is $2\pi/\sqrt{uk}$, or equivalently, $2\pi u^{-1/2}k^{-1/2}$. If they are clever and remember Lab 2, they may try a log-log plot with the tool, and discover this. But this is a really tough one. I think it is good to have students tabulate their results at their boards, and share insights, e.g., if $u = k = 1$ then the period is 2π (given to them). If $u = k = 2$ then the period is π. But if $u = 2$ and $k = 1$, then the period is somewhere between π and 2π. There are opportunities for cleverness and interesting discussion here.

Conclusion (5 minutes) Once students have seen the trajectories, you can set up the rest for homework by mentioning that they are seeing sine and cosine graphs in their time-series plots. And the reason is that the derivative of sine is cosine, and the derivative of cosine is the negative sine. So $X = \sin(t)$ and $Y = \cos(t)$ satisfy the change equations $X' = P$ and $P' = -X$.

Don't forget to collect table tents and take attendance!

Lab 4, Section 2

Lesson summary: Students examine period, frequency, and wavelength (for light), in a variety of scientific contexts, to become familiar with how these are used in practice to describe and reason about oscillations.

Key terms: Period, frequency, wavelength, photon, speed of light.

Learning objectives: After this lesson students will be able to:

- Correctly use the words frequency and period to describe oscillations.
- Convert between period and frequency, in standard units.
- Use the $f\lambda = c$ to convert between frequency and wavelength of light.
- Have a rough sense of the periods, frequencies, and wavelengths that occur in nature.

Before section: Students have completed SHO 1-4, on the simple harmonic oscillator.

During section: Osc3 and Osc4, and then Gly1.

After section: There is no lecture between this section and the next, so it is important for students to review the glycolysis lecture and read the lab manual before going onto Gly2,3,4.

Plan

Periods and frequencies of oscillation (20 minutes): Page Osc3 brings up a bunch of real examples of oscillations, and asks the reader to convert between period (how much time between peaks) and frequency (how many peaks per unit time). Note that frequency units are "per" units, which can trip up students! For example, a standard unit of frequency is Hertz (Hz). Note that one Hz is one "per second" – in SI units, one would say 1 Hz equals $1s^{-1}$, where s stands for seconds. This is terribly confusing when students first learn it... which might be during this section! What do we mean by s^{-1}? We mean "per second". Just like "meters per second" (a unit of velocity) is m/s or ms^{-1}.

Until the last two problems on Osc3, all can and should be done without a calculator. These problems could be made more interesting if each group has to write up (**on the board!**) one more oscillation whose period/frequency they know. Surfers might think of the period of a swell (time between waves) for example. The menstrual cycle is about 28 days. Anything that is cyclical has a period... ask for students to be creative and use these words "period" and "frequency" to describe something they know. That is how they will incorporate these words into their vocabulary. Sound waves and their frequency may be familiar to string instrument players. Standard tuners will show the Hz of the note, e.g., an A is 440 Hz. Going up an octave doubles the frequency, and going down an octave halves the frequency.

The last two problems on Osc3 will cause some problems and require a calculator. For example, the students need to think to convert a period of 3.83 ms (milliseconds) into a frequency (how many per second). It is good to prompt before going for the calculator! Here are some prompts to scaffold:

- If the period were 1 ms, then what is the frequency in Hz? (Answer 1000, because 1 ms is 1 thousandth of a second!)
- If the period were 10 ms, then what is the frequency in Hz? (Answer 100, because 10 ms is 1 hundredth of a second!)
- If the period is 3.83 ms, then estimate the frequency in Hz. Between what and what? (Answer: Between 100 and 1000!)
- If the period is 5 ms, then what is the frequency in Hz? What division problem answered this? (Answer 200, which is $1000 \div 5$).

Light (20 minutes) There is some scientific content here, and I would start by asking a volunteer to read aloud the first paragraph, and another volunteer to read aloud the second paragraph. You can write the key formulas on the board: $c = 3 \cdot 10^8 \text{m/s}$ and $f\lambda = c$. These are physics foundations, and relevant across the sciences. We want all science students to know that light comes in photons of various wavelengths, and that all these differently named things (X-rays, microwaves, etc.) are all just light at different wavelengths and frequencies. The toughest part for students here will probably be units and working with scientific notation. But science majors need this practice!

To fill out the table, with five students in a group, each student can choose a "name of light" to look up its wavelength and compute the frequency. Students could work closely with their group at a table here, but then you can give time for the groups to wander and share their tables or answers with other groups.

Glycolysis (20 minutes) Glycolysis is a chemical reaction that every biologist learns about, because it is crucial for the survival of just about every cell (human and otherwise). It is how cells turn sugars into usable fuel (ATP and NADH) to do all the things cells do (reproduce, twirl their flagella, etc.). Glycolysis is also a really complicated 10-step reaction in organic chemistry, each step with its own particular complexity. We are sort of cutting through all that, to focus on the fact that in some settings glycolysis exhibits long-term sustained oscillations.

The page Gly1 is pretty easy, but (1) there is a lot of jargon, introduced in lecture, but probably not recalled by students. Moreover it is good to dwell on the stuff that seems easy, and share methods of answering those "easy" questions. For example, how do you estimate the period of oscillation from a (printed) graph? From the data points? From the sinusoids the author has drawn? There are choices to make in such estimations, and no particularly right answer for this class!

Students may be able to get through Gly1 and Gly2 during this section, but do not go beyond Gly2. The further Gly3 and Gly4 really require different techniques as they get into limit cycles in a dynamical system! So timing is important... use time wisely, and stretch things out, slow things down, and keep the students *thinking* for the entire time.

Conclusion (5 minutes) Remind students that there is no lecture on Monday (MLK day) but they will work through Gly3 and Gly4 on Tuesday. They may wish to experiment with the Higgins-Sel'kov Glycolysis Model before then. And they should polish their lab manual up through Gly1, because the Lab Checkin is on Wednesday! Also there is another worksheet due in the next Tuesday section.

Don't forget to collect table tents and take attendance!

Lab 4, Section 3

Lesson summary: Glycolysis!

Key terms: Glycolysis, glucose, pyruvate, ATP, enzyme, period, frequency, nullcline, limit cycle!

Learning objectives: After this lesson students will be able to:

- Identify features in a graph, in which the data exhibits oscillation.
- Match features of a sinusoid (amplitude, period, phase shift) with parameters in a function.
- Identify a term in a change equation which represents a triple interaction.
- Identify terms in a change equation that represent constant change, or first-order decay.
- Write down and solve equations to find the equation of a nullcline.
- Connect a limit-cycle trajectory in state space to a time-series plot.

Before section: Students have completed Osc3 and Osc4, and have probably completed Gly1.

During section: Gly1-4.

After section: Just look back to complete the lab manual (up to and including Gly4) for the Lab 4 Checkin.

Plan

Setup (10 minutes): It is good to get everyone thinking about glycolysis again, after the long weekend. One way is to ask a series of basic questions, in a table-by-table format. For example, "Table 1: What is the input and output of glycolysis?" and "Table 2: Why is glycolysis important for cells?" and "Table 3: If something oscillates with a period of 20 seconds, what is its frequency?" and "Table 4: If something oscillates with a frequency of 5 per hour, what is its period?" etc. Just some warmup questions to get every table talking and responding. Go for 30 seconds or less per table, to get people ready within 5 minutes.

The next 5 minutes can be taken by getting all the groups to open up their Lab Manual to page Gly1, and share their answers within the group (and to the floating TA and ALF) to make sure everyone has basically correct answers.

Collect the worksheets! Take attendance!

Gly2 (20 minutes) With everyone on the same page, it is time to look at Gly2. It might be good to start with Desmos, and the exercise in the middle involving the function

$$S(t) = A\sin(\frac{2\pi(t-\phi)}{p}.$$

They are meant to load up Desmos (1-2 computers per table), and type this into the graphing calculator. Desmos should ask to add sliders for the parameters A, ϕ, and p. Adjusting the sliders should adjust amplitude, period, and phase shift. They may not know the term "phase shift" so this is the chance to tell them! I would like them to write more than "amplitude". Ideally, they should write something explanatory, like "A is the amplitude, the height of a peak relative to the middle of the oscillation."

With Desmos done (10 minutes), the top and bottom exercise asks students to turn back to the graph to look more closely.

The first exercise is a compare/contrast, where I hope students will use the words *period*, *frequency*, and *phase shift* to compare/contrast oscillations. With these words, they should notice things like: the three oscillations have roughly the same frequency; there is a phase shift between the 3rd sinusoid (FBP, or Fructose 1,6 Biphosphate) and the 1st (G6P) and 2nd (F6P) sinusoid. An explanation for phase shift is that the glycolytic reaction steps take a bit of time! The shown graphs capture three steps (steps 2,3,4) of glycolysis,.

These graphs come from the paper "Sustained oscillations in free-energy state and hexose phosphates in yeast." This may be nice to bring in (a sample page or two) or ask students to find online (free, and can google it, but need to click through for library access). In the paper, you can see the equations of a sinusoid, and the authors addressing precisely the questions here. The authors use those key words of period, frequency, and phase shift, and this can be a nice sort of proof to the students that what they are learning is *real*.

Gly3-4 (30 minutes) The page Gly3 begins with technical background, and it is fine to skip to the key pair of change equations with [F6P] and [ADP]. Essentially, there are a bunch of other molecules involved, but the critical dynamics of the system are in the two shown: F6P and ADP. Dissecting change equations is a familiar task for students by now. But some will still struggle to identify the v as a constant influx of F6P. And some will struggle to interpret the term $-k[ADP]$ as meaning that [ADP] naturally undergoes first-order (exponential) decay. Review and remind! The really NEW sort of term is the $[F6P][ADP]^2$ term. They have seen terms like ST in ecology, and $[H][OH]$ in chemical dynamics, when two things interact (a shark and tuna meet, or a hydrogen and hydroxide ion meet). They have also seen terms like P^2 for a self-interaction (one member meets another member from the same population). So it should not be a big stretch to seethe term $[F6P][ADP]^2$ as representing the number of interactions between one F6P molecule and two ADP molecules.

Gly4 is familiar for our students who used the Modeling Synthesizer for other systems. They may have algebraic troubles with finding the A-nullcline, and that is a good chance to remind students what the nullcline means (the A-nullcline means the solution set to the equation $A' = 0$, in state space), and assist with a bit of algebra. Once the modeling synthesizer is all loaded, the students are meant to explore the appearance and disappearance of a limit cycle, and connect the shape of the limit cycle to the oscillations visible in the time series.

Conclusion (5 minutes) Remind students that the Lab 4 Checkin is due tomorrow, and that's the only thing they have to work on. It goes through Gly4, so they need to finish it by then.

Lab 4, Section 4

Lesson summary: Students explore a new predator-prey model (Holling-Tanner), and encounter the *limit cycle* again.

Key terms: Limit cycle, stable/unstable spiral, logistic model, carrying capacity, Lotka-Volterra, .

Learning objectives: After this lesson students will be able to:

- Relate the terms in the Holling-Tanner model to ecological assumptions.
- Relate the biological notion of saturation to a sigmoid shaped graph.
- Identify the Hill function as a function which saturates.
- Identify a limit cycle, and recognize its key features, from seeing trajectories in state space.

Before section: Students have gotten on the same page, by finishing the Lab 4 up to Gly4 and turning it in.

During section: Students work through HT1, HT2, HT3, and perhaps a bit of HT4.

After section: Students complete Osc5, and Osc6 for homework, plus the Lab 4 Worksheet 3.

Plan

HT1 and HT2 (30 minutes): I will have lectured on the Holling-Tanner model the day before this section. I will try to construct and dissect this pair of change equations in lecture. Here is the chance to engage students.

I would carry this out by starting with the basic Lotka-Volterra equations

$$S' = -\delta S + pST \text{ and } T' = \beta T - qST,$$

and ask them what each state variable and parameter means. They should know this... we did it a lot in Math 16A. So they'll know that

- β is the (net) birth rate of tuna and δ the (net) death rate of sharks.
- ST is an interaction term, and p measures the benefit of these interactions for sharks, while q measures the harm of these interaction for tuna.

With that established (5 minutes), I think you shift to the generalized Lotka-Volterra, by writing

$$S' = \alpha S(1 - S/k) + pST \text{ and } T' = \beta T(1 - T/m) - qST.$$

Then $-\delta$ is replaced by the (probably negative) parameter α for the net birth rate of sharks. If α is negative, it means the sharks are dying off. The really new element it k (carrying capacity, or positive equilibrium point) for sharks, and m (carrying capacity) for tuna. They *should* be able to identify k and m as carrying capacities... when one ignores the interactions.

Then you can say that the Holling-Tanner model has two changes. First, we limit tuna predation (qST) by shark appetite. Second, we incorporate the benefit of predation into the carrying capacity of the logistic model. I.e., the more tuna, the more shark population can be sustained.

The purpose of the $A(T)$ exercise on HT1 is to see what happens ("saturation") when shark appetite is limited. Their graph should reflect "saturation" – the number of tuna eaten increases with the tuna population, but only up to the point where sharks have eaten all they can eat! So the graph should be increasing but bounded. Then they can play with Hill functions on Desmos to understand how they fit the graphical understanding.

Page HT2 can be done more interactively, by incorporating tuna into the carrying capacity of sharks (replace k by qT) and then writing the full Holling-Tanner system on the board. If you've work through it enough, students will remember what each term and parameter means, to answer the question on HT2. You may wish for students to write this down within their groups while the TA and ALF are floating, so that one group doesn't just blurt it out and end the thoughtful discussion.

HT3 (20 minutes) Now it is time to refocus with HT3, and loading up the *Holling-Tanner Predator-Prey Model*. This is again a similar task, as they adjust parameters to see what happens. They might begin with the S-nullcline as an algebraic warmup too. If you have time for the algebra practice, each group can solve the equation for the S-nullcline at the board, and compare solutions. I would write the solution as follows: The S-nullcline occurs when $S' = 0$. This occurs when $\alpha S(1 - S/qT) = 0$. This is equivalent to $S = 0$ or $1 - S/qT = 0$. And that is equivalent to $S = 0$ or $S/qT = 1$. And finally, the nullclines are described by $S = 0$ or $S = qT$, a pair of lines.

In the simlulator, students should find the *stable spiral* equilibrium point, and the transition to limit-cycle behavior at a critical value of c. They should examine lots of trajectories, and see how they all get sucked into the single equilibrium point, or into the same limit cycle. The limit cycle yields long-term oscillations that are robust to occasional disturbances.

Starting HT4 (10 minutes) HT4 is a more creative series of exercises, where the students will try to model three species with change equations. I expect this to be an interesting challenge, and I want students to work mostly on their own. But to give them a start, I think you could try to work through a plausible F' equation, using the given property 1. The thought process should be something like this: The filter-feeders might need these bacteria and algae to eat. We could treat them like the "sharks with limited appetite" in Holling-Tanner, and write

$$F' = \alpha F(1 - F/k),$$

where k is the carrying capacity. The carrying capacity would depend on the amount of bacteria and algae available, so a decent model would be

$$F' = \alpha F\left(1 - \frac{F}{pC + qG}\right).$$

There are many variants, worthy of rich discussion. Discussion will help students understand a sophisticated model like Holling-Tanner and others.

Conclusion (5 minutes) Remind students to complete their worksheet for next Tuesday, and to complete Osc5,6 as well as finishing HT1-4 by next Tuesday. Next week we are starting a new source of oscillation... the delay equation!

Lab 4, Section 5

Lesson summary: Students explore time delay as a mechanism for oscillation, in a "racecar" example and in Hutchinson's population growth model.

Key terms: Time delay, reaction time, negative feedback.

Learning objectives: After this lesson students will be able to:

- Fill out a data table based on a change equation with time delay.
- Implement a time delay with a spreadsheet.
- Modify the logistic population growth model with a time delay.

Before section: Osc5 and Osc6

During section: TD1, TD2, TD4.

After section: Complete TD4 as needed. Polish Osc5, Osc6 for lab due date.

Plan

Racecar setup (20 minutes): The page TD1 sets up a familiar situation, in which a car is driving, trying to stay in its lane. To start, I recommend a short lecture/demonstration (10 minutes) followed by 5-10 minutes for every group to add columns to their table and write them on the boards.

When it drifts right, the driver steers left, and when it drifts left, the driver steers right. It should seem a lot like the temperature example from the beginning of the lab (when it's too cold, you turn on the heat, etc.). The point of these situations is that the *time delay* can create an oscillation. If you react a bit slowly, then you will end up overcorrecting. Your room will get too hot, too cold, too hot, too cold. Or the car will swerve left, right, left, right.

The driver's equation is provided as

$$\Delta X/\Delta t = -X(t-0.2) \text{ meters per second.}$$

The "-0.2" is the 0.2 second (or 200 ms) delay. To begin with, students should understand this change equation without the time delay, where it looks like $\Delta X/\Delta t = -X$. It is good to have students work with this a bit, noting that it has a unique equilibrium point at $X = 0$, and this is a stable equilibrium point. As long as $\Delta t < 1$, all long-term behaviors are tendencies towards equilibrium.

For the time delay, I think it is good to start with a little lecture, deriving the table shown on page TD1, to show the technique. Note that the "drift phase" (first 0.2 seconds) must be given to figure out the rest... otherwise you don't have the data necessary to finish the table. Once you show the technique, the students can try getting the next two columns (with $t = 0.9$ and $t = 1.0$. This data table shows the car swinging to the right ($X = 0, 0.1, 0.2$) before reversing direction ($X = 0.2, 0.19, 0.17, 0.15$), and it is not clear whether it will oscillate.

Spreadsheet solution (20 minutes) A spreadsheet is a much more efficient way of looking at this problem, and it exhibits the power of "relative references" in a spreadsheet. Students should load up Google Sheets (or Excel) for this one, with at least two computers open per group.

The most important thing here is that students follow the directions, step-by-step, from the margin. They've done some spreadsheet practice last quarter, but probably not enough. So you might want to run a tight ship, checking that all groups have done each little step (Enter Rows 1 and 2 manually... Copy-paste Cell C2 into C3, C4, etc....) before going onto the next step. They should be able to go far beyond 0.8 seconds, to get a table that lasts for 2 or 3 seconds. Making a time-series plot is a bit trickier, but they can select the two columns A (time, t) and B (location, X), and inserting a chart.

Note that column D is where the change equation is really held, with the formula "$= -B2$" in cell $D4$. Note that this refers from $D4$ to $B2$, because it refers to the X-value two time-units previous (two rows up in the spreadsheet). Once everyone is on the same page with the first two exercises, they should have 10 minutes to explore in the last exercise. The result should be that the presence of oscillation depends on a combination of time delay and negative feedback. If the time delay is small, you need a strong negative feedback to get oscillation. And if the time delay is large, your negative feedback can be weaker and you'll still get oscillation.

Skipping TD3 You may safely skip TD3. Let's see how it goes, but I think it doesn't need to be done. Maybe I'll delete it in Version 3.0 of the Lab Manual.

Population growth with time delay (20 minutes) Now for a biology model! This is the Hutchinson Population Growth model, which is meant to explain some oscillations that are seen within isolated populations (e.g., water-fleas). To get people started, you may ask a group or two to read aloud to the class the first two paragraphs from TD4. Then you can set up the equation:

$$P' = \beta P(t) \cdot (1 - P(t - \tau)/k)\,.$$

Here τ is the time delay. If there is no delay, then this is just the logistic equation

$$P' = \beta P(1 - P/k),$$

with net birth rate β and carrying capacity k. The time-delay idea is that the negative effect of crowding is delayed by τ units of time. Of course, one may debate slightly different ways of implementing this time delay, but the given equation is what Hutchinson studied. The key explanatory sentence is in the lab manual: "if the population exceeds the carrying capacity at some moment, the negative effect on the population will occur τ units of time later."

Before jumping into the simulator, you might draw some time series for the original logistic equation with $k = 100$ as a quick review. And then ask how the time-series would change given the delay. This is similar to the racecar situation, though here there are two equilibria (unstable at $P = 0$, and stable at $P = k$ with time delay). Once students load up the simulator, they should be able to explore parameter values, and for each birth rate they should find the critical time delay at which oscillations emerge.

The cool answer (for TAs!): Stable long-term oscillations occur when $\beta \cdot \tau > \pi/2 \approx 1.57$. So for example, if $\beta = 0.1$, then one should find stable oscillations when $\tau > 15.7$.

Conclusion: (5 minutes) Remember that the last parts of Lab 4 are due Friday! Muscle tremor examples are at the next section on Thursday.

Lab 4, Section 6

Lesson summary: Students study two examples of oscillation in physiology: muscle tremor and ultradian oscillations in glucose. Both have time-delay as an essential component.

Key terms: Sustained oscillations, period and frequency, insulin, glucose, glucagon.

Learning objectives: After this lesson students will be able to:

- Describe oscillating systems in physiology with coherence and precision.
- Relate the abstract notions of time-delay and steep negative feedback to physiological parameters.
- Apply knowledge of oscillations to reason about physiological systems.

Before section: Students have completed TD1, TD2, TD4, to understand the role of time delay in oscillation.

During section: MT1, MT2, and IGO1, IGO2.

After section: Osc7 and Osc8, the synthesis pages, and then complete Lab 4!

Plan

Muscle Tremor (30 minutes): The two pages MT1 and MT2 have a lot of background reading, as they relate to a specific aspect of physiology: muscle tremor. I strongly recommend having students read out loud. It takes just a minute or two to read a paragraph out loud, and it's crucial for context. The exercises on MT1 are meant to be pretty easy – the point is having students get accustomed to this physiological system, and to get them to *use* their newfound knowledge of oscillation to describe systems precisely. Note that the parameter r is the reflex magnitude – that is the "answer" to the last exercise on MT1, and it is essentially written on MT2. But what does that phrase "reflex magnitude" mean? This is a parameter that may really depend on the person! We want our students to understand – particularly in a physiological context – which parameters are intrinsic properties of the person (e.g., kidney function), and which parameters are choices that a physician might make (e.g., dosage). Reflex magnitude is something about the person.

The basic muscle tremor model here is not meant to be perfectly accurate. I've just given a nice and easy model with an equilibrium point at $L = 20$:

$$L' = r(20 - L(t - \tau)),$$

where r is reflex magnitude and τ is the time-delay due to nerve signals. It is similar to the Hutchinson population growth model, but really an easier equation! The questions on MT2 are similar to the Hutchinson questions – looking for critical values of r and τ in order to see sustained oscillations. So in a sense, one lesson is the same as for the Hutchinson model: it is a combination of r and τ that creates sustained oscillations. One needs a combination of a strong reflex response (big r) with a delayed reaction (big τ) to see oscillations.

The deeper point with MT1 and MT2 is that one can potentially use the *frequency* of oscillation to determine whether r is "too high" or τ is "too high", i.e., whether a patient's tremor comes from an over-strong reflex (hyper-reflexia — too big of a reaction, which occurs sometimes after a stroke) or from a slowed reflex loop (as in some cerebellar diseases, where the long-latency reflexes appear to be delayed). Both may exhibit tremor, but the frequency can be different.

A caution if it comes up: parkinsonian rest tremor is *not* a clean example of the time-delay mechanism. The mainstream account is that it is generated centrally, in the basal ganglia and cerebello-thalamo-cortical circuits, rather than by a slow reflex loop. A 2025 modelling paper does argue for a reflex-loop account, so it is genuinely contested — but not settled enough to teach as fact.

Insulin-Glucose: Ultradian Oscillations (30 minutes) Students have spent time with the Insulin-Glucose system in Lab 1, where the system was

$$G' = m - sIG$$

$$I' = qb\frac{G^2}{1+G^2} - \gamma I.$$

Here G is the blood concentration of glucose, and I the blood concentration of insulin. The parameter m reflects an assumption that glucose intake is constant... not realistic, but good for a baseline. The "predation" term $-sIG$ reflects the fact that insulin downregulates glucose in the bloodstream. The term $-\gamma I$ reflects the fact that insulin naturally decays in a "first-order" way. Insulin would naturally all decay away, if it were not actively produced by β-cells in the pancreas. The production of insulin is in the $qbG^2/(1+G^2)$ term; the constants q and b reflect the beta cell mass and efficiency. The $G^2/(1+G^2)$ term reflects the fact that more glucose yields more insulin production... but only up to a point (there is a horizontal asymptote). The β-cells in the pancreas are limited in how much insulin they can produce, so we use this "Hill function" $G^2/(1+G^2)$ which approaches an upper limit of 1 as $G \to \infty$.

That is all in Lab 1, and is worth a review! But here we complicate this model in two realistic ways. First is that there is another source of glucose: when our insulin level is low (e.g., when we're asleep), our liver produces glucose. This yields a positive term in G' which we write as $\alpha/1 + e^{kI-c}$. The first exercise has students explore the shape of this function.

Next, as students go into the simulator, they will see the Insulin-Glucose model with $\alpha = 0$ (no liver production of glucose). They should experiment with this, setting $m = 0$ (no eating) to see what value of α yields survival (glucose in the green safe range). The idea is that the liver production of glucose keeps our body working, even though we often don't eat for 5-10 hours at a time. This only works for so long, because the liver has limited stores of glucose. After the liver glucose stores are exhausted, the body goes into ketosis (getting energy from fat), which people now know about through the "Keto diet." Keto dieting is not built into the simulator :)

The last page, IGO2, asks students to look at real oscillations in insulin and glucose levels. The explanation for these oscillations has been debated in the literature, but the consensus seems to be that time delays are largely responsible. For a while, the alternative hypothesis was that the pancreas had a sort of "pacemaker" – releasing insulin in synchronized pulses. But a time-delay model can also lead to the oscillations we see, and it is plausible given the actual time it takes for hormone signals to have an effect on blood concentrations of insulin and

glucose. It is crucial that students use the given parameter values $m = 2$, $s = 7$, etc., written on IGO1, in order to see the oscillations. They are highly dependent on those parameters as well as on the time delay.

Conclusion (5 minutes) Remind students that their lab is due on Friday! There will be a worksheet due next Tuesday, when Lab 5 will commence!

Chapter 5

Randomness

About Lab 5

Lab 5 is about chaos and randomness... mostly randomness. The modeling that we have done in the class so far has focused on "change equations" – our friendly term for ODEs (ordinary differential equations). We have seen discrete-time and continuous-time change equations, but everything has been deterministic. And throughout, we have seen that the perfectly flowing trajectories and time series of our models do not really resemble the jittery plots one gets from actual field or experimental data. The goal of this lab is to learn how to incorporate a bit of randomness into a model, and how to adjust the whole modeling process accordingly.

The first part of the lab focuses on what randomness is... or rather, what it is not. It is crucial to distinguish *chaos* from *randomness*. Chaos is the word we use to describe *deterministic systems*, when a very slight change in the initial values has drastic effects in the long term. We haven't seen chaos before Lab 5. One reason we haven't seen it is that a smooth system of autonomous (no time-dependence) ODE in 2 variables cannot exhibit chaos. (That's a theorem). We have not studied systems in a 3 variables (3-dimensional state space), so we haven't seen chaos. There are systems in 2-variables with discontinuities that exhibit chaos, e.g.

$$X' = \frac{XY}{X^2+Y^2} - \alpha X,$$

$$Y' = \frac{Y^2}{X^2+Y^2} - \beta Y + \beta - 1.$$

But we haven't seen such systems in this class.

To see chaos, the easiest thing to look at is just a 1-variable system with *discrete time*. The one variable logistic equation exhibits chaotic behavior, in the discrete time setting:

$$\Delta P/\Delta t = \beta P(1-P).$$

Note that this system is deterministic – the initial value of P determines all future values of P. But tiny changes in this initial value will have huge changes for future values of P... at least when β is above a threshold.

In contrast, we use the word "randomness" to describe when there is a "stochastic" element to the change equation. The simplest example would be a purely random "random walk". We might describe such an equation by an equation like

$$\Delta X/\Delta t = \epsilon, \text{ where } \epsilon \text{ is a random variable.}$$

What is a random variable? For this class, it is just a table of numerical outcomes and their probabilities. For example, a random variable ϵ could be: 1 with probability 50% and -1 with probability 50%. If that is the random variable, then the change equation above just means that X either increases or decreases by 1 each unit of time... according to (independent) flips of a fair coin.

After our brief tour of chaos, we build up some language to talk about random variables. Most will come from coin flips, in one form or another, though the coin may be biased, and the outcomes may be unusual. For example, we work with a random walk in which "Heads" means you move to the right, and "Tails" means you go back to zero – a "survival" or "growth and collapse" model. We incorporate these random variables into other growth models, e.g., in good years a tree grows a lot, and in bad years a tree grows a little. The goal is to have students work with these models with randomness baked in, to have an intuition for how they resemble and deviate from deterministic models, and how we can still say meaningful things about the world in the presence of some noise.

Students will struggle with aspects of probability, particularly with fractions and percents, and broader probabilistic reasoning. We are not touching the really counter-intuitive stuff like Monty Hall. But still, many students won't have a good language to describe certainty and uncertainty. That is what they should learn – how to use words like "expectation" in a precise way, and to start to quantify when results are surprisingly high/low. This is a sort of conceptual warmup for statistics, where they will learn all their t-tests, z-values, p-values, etc. We touch on these things in broad terms, from a computational modeling perspective.

Many of the models here are more like games than equations. Students will need to read carefully, because they need to understand the rules of the games precisely rather than plugging in values into a change equation to make predictions. That can be easier in some ways, but the resulting systems have subtle differences. Some, like the Yule model and genetic drift model, and core to population genetics.

We finish this chapter with a study of Brownian motion, which is a purely stochastic process – a random walk . The incredible thing here, is that Einstein realized that by observing the Brownian motion of small (but visible!) particles in a dish of water, and carefully tracking their motion, one could estimate the size of atoms and molecules. In fact, the experiment Einstein suggested, performed a few years later by Perrin, led to a good estimate of Avogadro's number. And moreover, this sort of sealed the deal for scientists... who at the time were not fully convinced that atoms and molecules were real – for a while, they seemed like magic imagined things that made some chemistry formulas work out. Students will perform a simulated version of the Einstein-Perrin experiment to estimate Avogadro's number, effectively counting the number of molecules in a macroscopic sample.

Lab 5, Section 1

Lesson summary: Students will explore the discrete logistic equation, and learn to differentiate chaos and randomness.

Key terms: Chaos, randomness, stochastic, discrete time, bifurcation, butterfly effect, uniformly random.

Learning objectives: After this lesson students will be able to:

- Identify a bifurcation when adjusting a parameter in a system.
- Differentiate chaos from randomness
- Identify key features of chaos.

Before section: Students have completed Lab 4.

During section: RC1, RC2, starting RC3 and RC4.

After section: Students complete their write-ups of RC3 and RC4.

Plan

New groups! (10 minutes): It is the start of a new lab, so time to randomize the groups! Start off by randomly shuffling the name-cards, and assembling groups at the tables. There may be a few students who switched sections, so please take care to identify people without name-cards and name-cards without people, and note them for coordination later. Once students are in their groups, you can give them about 5 minutes for icebreakers and introductions, before setting up the math.

Discrete logistic equation spreadsheet/explorer (30 minutes): You can introduce the discrete logistic equation as a lecture on the board, beginning with

$$\Delta P/\Delta t = \beta P(1-P).$$

Questions for the class include: what is β? What is the carrying capacity here? What are the equilibrium points? Then you might show how this is used to make a table, using $\Delta t = 1$ and a starting population of $P = 0.5$. That should be a straightforward 5-minute demonstration at the board.

Then the big exercise on RC1 is doing this with a spreadsheet. Have two people from each group open up a computer with a spreadsheet (Google Sheets or Excel), and enter the headers: t, P, Dt, and DP. (Use D instead of Δ to avoid having students type in Greek.) The next row should be the initial values, so it looks like the following.

	A	B	C	D
1	t	P	Dt	DP
2	0	0.5	1	

Then you need to ask the students what *formula* goes in cell D2 in order to represent the given equation, if $\beta = 1.5$ to start. The answer should be the formula "= 1.5 * B2 * (1 - B2)" The next spreadsheet steps are the following, making sure to copy and paste through at least 30 rows of the spreadsheet.

- Copy and paste the "1" from cell C2 to all the cells in column C, since $\Delta t = 1$ throughout.
- Enter the formula "= A2 + C2" into cell A3, and then copy and paste that down the column A, to make a counter in column A (1,2,3,4,... for time).
- Enter the formula "= B2 + D2" into cell B3, and then copy and paste that down the column B, to make sure that P changes by ΔP at each step.
- Copy and paste the formula "= 1.5 * B2 * (1 - B2)" from cell D2 into all the cells down the column D, to implement the given change equation.

To change the $\beta = 1.5$ parameter to other values, they can just change the formulas in column D. Qualitatively, they should see typical logistic growth, oscillation, and further bifurcations, even if they don't have the vocabulary for it yet. They can select columns A and B to make a nice chart too.

The page RC2 introduces the *Discrete Logistic Model Explorer*, which does all this quickly. The spreadsheet is the sort of conceptual glue between the raw change equation and the slick explorer and bifurcation plot. The goal of RC2 is to understand the bifurcation diagram as a sort of roadmap to what happens when the parameter β is changed. On this roadmap, they should be able to see bifurcations (forking) as a stable equilibrium becomes an oscillations, and then 2-cycles become 4-cycles, and eventually the emergence of chaos!

Chaos vs Randomness (20 minutes) On RC3 and RC4, students are meant to really understand what chaos is. Chaos is NOT just "stuff goes crazy." or "it looks wild." The two characteristics are essential: chaotic systems are deterministic. We found chaos from a *formula* which *determined* the next term by the same process every time. Secondly, chaotic systems exhibit the butterfly effect – tiny differences in initial conditions yield drastic differences in the long term. Determinism can be seen in the plot at the bottom of RC3. Plotting the next value vs current value in the P-sequence should yield a clear parabolic pattern, because that is what the formula dictates!

In contrast, a random sequence would yield a less coherent pattern, when graphing current value vs next value. Students can begin to understand randomness in RC4, though a deeper study is coming in future sections. Students may struggle a bit with probabilities for uniformly chosen random numbers. For example, if ϵ is chosen uniformly within an interval $[a, b]$, then what this means is that for all c, d satisfying $a < c < d < b$, the probability of landing in the interval $[c, d]$ is the ratio

$$(d - c)/(b - a).$$

In other words, when a random number is chosen uniformly, the probability of landing within a given interval is proportional to the length of the interval. That may require a bit of setup (a practice problem at the board) for the computational exercise of RC4. And then finishing RC4 requires a qualitative comparison of the graph on RC4 with the graph on RC3. Try to push students towards making qualitative statements that are specific and meaningful, with particular reference to the graphs that are shown, and complete sentences.

Conclusion (5 minutes) Remind students that there is a midterm next week! And more randomness this Thursday.

Lab 5, Section 2

Lesson summary: Students study two stochastic growth models, followed by a more formal introduction to random variables and expectation.

Key terms: stochastic model, histogram, random variable, expectation, independence.

Learning objectives: After this lesson students will be able to:

- Use a histogram to describe the results of repeated experiments.
- Incorporate a bit of randomness into classical models of growth.
- Connect the term "random variable" to a table of outcomes and probabilities.
- Compute the expectation of a random variable.
- Compute binomial probabilities, e.g., probabilities of various outcomes with fair and biased coins.

Before section: Students have completed RC1-RC4 to understand the difference between chaos and randomness.

During section: RG1, RG2, Flip1, and starting Flip2.

After section: Students mostly study for the midterm. No worksheet is due next week.

Plan

Tree ring growth (15 minutes): This is a simple sort of stochastic linear growth model, where a tree grows a bit in radius each year. In good years, the growth might be a lot, e.g., 1.5mm, and in bad years perhaps it grows a little, e.g., 0.5mm. One might "expect" an average growth of 1mm under these circumstances. But after 50 years or 100 years, what is your expectation? And what results would be "surprising"? This is meant to help students get accustomed to dealing with expectation and uncertainty, and to describe them in precise ways.

On RG1, groups should independently estimate the minimum and maximum width of a tree ring from the printed figure, and fill out their estimate on their lab manual. Then they can load the *Tree Ring Simulator* and play with it. It is helpful if they can track their data on a table on the board. It should only take about 5 minutes to gather data if they are efficient and have roles – one or two students running the simulator, while one records results on the board.

Then they have choices in creating a *histogram*. It is good here to introduce the sort of anatomy of a histogram – that the heights of bars represent *frequencies* and the bars correspond to a choice of *bins*. There are subjective decisions to make about bins, etc., that one needs to practice. So that's good for students to discuss in groups while the TA and ALF float.

Stochastic exponential growth (20 minutes) The next model, on page RG2, is a stochastic exponential growth model, where bacteria divide at a random rate – there are lucky minutes and unlucky minutes. In lucky minutes, P gets multiplied by 1.04, and in unlucky minutes, P gets multiplied by 1.02. Note that if you track $\log(P)$, then this is the same sort of stochastic linear growth from before! Students may realize this, especially as they complete the semilog plot. They will probably have some trouble giving the formula for $P(x, y)$, but see if you can talk them through it – this should be a sort of review from Lab 2, where they learned that exponential functions come from repeated multiplication. It is safe to leave off the last question as homework.

Random variables and expectations (25 minutes) Flip1 and Flip2 contain some traditional math, and I will introduce the ideas (random variable, expectation) and give examples just the day before in lecture. This will go much smoother if students are up at the board working things out, and have some chances to look at other people's boards as well. Students who struggle with fractions and percents will struggle here, but the first few "easy" exercises can be good ways to get most of the class on the same page. Most students will *not* know about expectations, nor will they know the term "random variable" – so even students who know fractions and percents will need to learn these concepts through practice.

The last exercise on Flip1 illustrates an important point in probability: the "expectation" does not commute with other functions like squaring. (You probably don't want to say it that way to students!) The example there is that the expected area of a circle is not πr^2 where r is the expected radius of a circle. It may be close, but it won't be correct! Expectations are just weird that way. Later in this lab, students will work with an expected "squared-displacement" and similar issues will arise. So that is a good one to get through.

It is advisable to get through most of Flip2 if possible – this brings in two skills: first, finding the expectation from a non-uniform random variable. Second, converting a situation with randomness into a table with coin tosses. This sort of conversion is the key to modeling with random variables.

Conclusion (5 minutes) Remind students that the midterm is coming up, so there is no worksheet due next week! Instead, they can focus on the midterm study guide. The midterm will cover material up to and including Flip2, and that includes some Math 16A stuff too.

Lab 5, Section 3

Lesson summary: Recovering from their midterm exam, students spend about one hour studying systems based on coin flips.

Key terms: Outcome, probability, Independent, expectation, random variable,

Learning objectives: After this lesson students will be able to:

- Compute probabilities of outcomes from multiple coin tosses (fair and biased).
- Group coin-toss outcomes in natural ways (e.g., two heads out of three tosses = HHT, HTH, or THH)
- Model situations with coin toss systems.

Before section: Students have practiced computing expectations for random variables.

During section: Finishing Flip 2, then Flip 3, Flip 4.

After section: Students complete Flip 4 as necessary.

Plan

Gathering with some coin tosses (10 minutes): A classroom of students provides a quick way of seeing probability in action. Almost all of your students will be able to flip a coin, via their smartphone. Just Google "flip a coin" and Google will virtually flip a coin!

With that in mind, you might start with a review/completion of Flip2, where you look at some classic coin-tossing probabilities. To warm up, have every individual in class flip a coin twice, and keep track of the results on paper with an H or T. So after the two coin flips, students should have written HH, HT, TH, or TT. Then, tell them that each H means go up 1 (or win a dollar), and each T means go down 1 (or lose a dollar). Ask a few students what their outcome was (go up 2? go 0? go down 2?) Work with them to tabulate the *outcomes*. Then it is time to tabulate the *probabilities*. Since there are three outcomes, a few students might expect the probabilities to be 1/3 for each. But alas they are wrong, and the chance of "go 0" is 50%, corresponding to the pair of coin-outcomes HT and TH. After you tabulate the outcomes and probabilities, please make sure to call the table a *random variable*. We are trying to get that definition/concept across: a random variable is just a table of outcomes and probabilities!

Expectations, finishing Flip2 (15 minutes) Then it is time to work through mathematical expectations. You might work as a class through the expected location exercise. In the two coin-flip situation, you can also do the expected location (1, -1, or 0) and the class should be able to guess the answer is 0. Someone should even be able to get to the idea of symmetry as a reason – ups and downs are equally likely, so the expectation *must* be zero. That reasoning applies to three, four, or however many coin-flips you try.

The next exercise is far more interesting. There the game is that "heads" means move +1, and "tails" means go back to 0. It is a "growth-and-collapse" model. To warm up the exercise, you can do the two coin-flip example at the board, tabulating

Location (R)	Coin tosses	Probability
0	TT, HT	50 %
1	TH	25 %
2	HH	25 %

Then set them loose on the three-flip version, with possible locations 0,1,2,3.

With both tables on the board, spend the rest of this block on the *computation* of expectation itself, methodically and out loud. Students struggled with this more than with anything else in the lab last year, and the difficulty is almost never the arithmetic – it is not knowing what the procedure is. So give it a fixed shape, and repeat it until it is boring:

1. Write the random variable as a table: one row per outcome, a column of outcomes and a column of probabilities.
2. Check that the probabilities add to 1 (or to 100%). This catches most errors before they can propagate.
3. Add a third column: outcome $\times$ probability, one row at a time.
4. Add that column down. The total is the expectation.

Multiply across, add down. Run it on the growth-and-collapse table the class has just built, so that $E(R) = 0.875$ comes out of a procedure rather than out of the air. Then run it again on something they have already seen – the fair die from Flip1, or the biased coin – so they watch the same four steps produce a number they recognize.

Get students to the board in groups for this. One group's table worked through in public, with the class correcting it, is worth more than any number of individually correct answers, because the mistakes are the point: probabilities that do not sum to 1, a column added across instead of down, an outcome confused with its probability. All three are common, and all three are visible on a blackboard in a way they never are on paper.

Two things are worth saying out loud while you have their attention. First, an expectation need not be a possible outcome – 0.875 is not a location you can land on, and 3.5 is not a face of a die. Second, an expectation is a *weighted* average, not the average of the outcomes: a student who adds the outcomes and divides by how many there are will get the right answer for a fair die and the wrong answer for everything else in this lab.

Lifespan model (25 minutes) This is an important stochastic model for many biologists, as it deals with survival! The key is setting up the random variable as done in the first exercise. I've just tabulated the initial coin tosses. Students may struggle how this works.

The rest Flip3 is conceptually a bit more straightforward. Once the random variable is set up in a table, you just directly compute the expectation (multiply across, add down). The tough thing here is that in this model, the lifespans could be ridiculously long. So the unfortunate students are being asked to evaluate

$$1 \cdot 0.5 + 2 \cdot 0.25 + 3 \cdot 0.125 + \cdots = \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \frac{5}{32} + \cdots = \sum_{n \geq 0} n 2^{-n}.$$

We will be content to allow students to just add the first 10 years to get a decent approximation. There are nice ways to evaluate the infinite sum though :) I recommend leaving the last exercise on Flip3 for homework – it is the same as earlier parts, except with a much higher chance of survival (90% instead of 50%) each year.

Growth and collapse (10 minutes) We go back to the growth and collapse game, where "heads" means go +1, and "tails" means collapse back to zero. You could have the class play this individually with the Google "Flip a Coin" – to play, just ask everyone to start at zero. When you say "Flip!" they all flip their coin. If they get heads they go +1. Otherwise they collapse back to zero. After you say "Flip!" 10 times, they've tracked their number, and you can see what the distribution is. How many zeros are there? How many ones? How many twos? Draw a histogram on the board, and use the word "histogram" a lot so students know what it is.

Conclusion (5 minutes) Time-permitting, they can try the *Growth & Collapse Simulator* to see the same experiment run 500 times over, 50 steps each... and see what the histogram looks like. With enough repetition, growth and collapse models lead to a histogram whose bars decay geometrically – each bar a fixed fraction of the one before it.

Lab 5, Section 4

Lesson summary: Students study two stochastic models in biology, to understand variation due to "neutral" processes.

Key terms: stochastic, tree, natural selection, fitness, drift, neutral, fixation.

Learning objectives: After this lesson students will be able to:

- Use coin tosses to model a situation with randomness.
- Model a system in which individuals are governed by a random process
- Look for "neutral" explanations in biological systems.
- Develop familiarity with neutral sources of variation.

Before section: Students complete Flip4.

During section: Yule1, Yule2, GD1, GD2.

After section: Students complete GD2.

Plan

Exponential population growth (10 minutes): A good way to introduce the Yule process is to review an old exponential model of population growth. Suppose that you begin with 10 bacteria, and each minute exactly 5% of them divide. (Use decimals... don't worry.) Then you expect population growth like 10, 10.5, 11.025, etc. It is a good chance to review this model with students, and will simultaneously complete the bottom two exercises on Yule1. You can do these as a big group. For example, ask one person from each group to get up to the board and write a formula for $P(t)$ according to this model. The correct answer is

$$P(t) = 10 \cdot (1.05)^t.$$

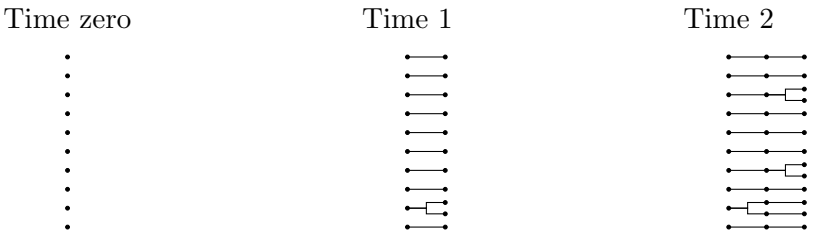
See if they can get there with some hand-holding and questioning. Get to the explanations of why... we studied this in Lab 2 by reasoning: the population starts at 10, and gets repeatedly multiplied by 1.05 for a 5% increase. Then you can complete to the semilog plot question, to have students find that

$$\log_{10}(P(t)) = 1 + t \log_{10}(1.05),$$

and use a calculator to put this in $\log_{10}(P(t)) = mt + b$ form.

It is important that the setup reinforces the expectation of exponential growth, and stays close to the concrete situation of 10 bacteria with 5% dividing each minute!

Yule Process (25 minutes) The Yule Process is the stochastic variant, where instead of saying that exactly 5% divide each minute, you say that each bacterium has a 5% *chance* of dividing each minute. Here is how you do this with the *Rapid Coin Flipper* tool in the Modeling Synthesizer. Have a student load the tool, and change N (number of flips) to 10, and probability of heads to 5%. Draw 10 dots on the board for the starting bacteria at $t = 0$. Then they click "Flip!" and get the results, like TTTTT TTTHT. Ask for the sequence of Ts and Hs, and draw a division if one happens, as below for $t = 1$. Then you might have 11 bacteria. So have the student change it to 11 coin flips if needed. After clicking "Flip!" again, and the result is TTHTT THTTT T, the tree diagram should evolve to look like below-right.



Reasoning with trees is a core skill in biology! Students can then work on the first two exercises on Yule 1 with some floating help. In the second exercise, they will need to draw such a tree themselves. Yule2 is very similar, but assisted by a specialized Yule Process Simulator in the Modeling Synthesizer. The final exercise of Yule2 is conceptually important. A result of a model like the Yule Process is that a cell that divides (by chance) earlier than others will have a far larger tree of descendants. E.g., in the tree you've drawn, that initial bacterium at position 9 divided in the first minute to become two bacteria. Its lineage (set of all descendants) could be twice as large as the others, because of this fact!

Genetic Drift (25 minutes) Genetic drift is a stochastic reason that you can see variation in species even when one isn't "stronger" than another. "Stronger" here means what biologists call a fitness advantage, meaning an advantage that causes the production of more surviving offspring. Evolution predicts that fitness advantages will yield to advantages in population growth, and some species or variants "winning" over others. The alternative "neutral" hypothesis is that a great deal of variation can be explained by the stochastic process of genetic drift.

Here we imagine two variants (e.g., spotted and unspotted banana slugs), living in a sort of equilibrium in the forest. At equilibrium they reproduce and die in a balanced way so the total population stays the same. There is no fitness advantage, so both variants reproduce and die with the same probabilities. But spotted slugs give birth to spotted slugs, and unspotted slugs give birth to unspotted slugs. The way we model this is with a starting population $P = A + B$ of slugs. Here A are the spotted and B are the unspotted. Each produces two surviving children for a total population of $3P = 3A + 3B$ slugs. Then a *random* selection of $2P$ slugs dies, to bring it back down to a population P. Note that A and B are changed, because the deaths are random, while the births simply replicate the A:B ratio.

This *neutral* process sometimes leads to complete extinction of one variant, just by chance. This occurs frequently with smaller populations (bottlenecks), where one variant can completely disappear just by chance (due to randomness in deaths).

Conclusion (5 minutes) Please remind students to turn in their worksheet on Tuesday. And Monday is a holiday... no lecture!

Lab 5, Section 5

Lesson summary: Students explore temperature – the molecular kinetic theory of heat – as a stochastic model.

Key terms: Temperature, Kelvin, Boltzmann's constant, distribution.

Learning objectives: After this lesson students will be able to:

- Interpret temperature as a statement about speed of molecules, depending on mass of molecules.
- Use a graph with a probability distribution to reason about likely and unlikely events.
- Use a stochastic model to predict distributions of outcomes.

Before section: Students complete worksheet, and genetic drift (GD1 and GD2). Students read and attend lecture to learn about the kinetic theory of gases.

During section: Students complete Temp1, Temp2, Temp3.

After section: Students complete Temp4, using the *Atmospheric Molecule Simulator*. They then complete BM1 as best as possible.

Plan

Introduction to temperature (15 minutes): I think it is good to start class (with groups formed) by having students take turns reading the paragraphs on Temp1, with questions to gauge understanding. Also, begin the class by having the big equation (Maxwell-Boltzmann distribution) on the board:

$$P(v) = C\left(\frac{m}{2k_BT}\right)^{3/2} \cdot v^2 \cdot e^{-mv^2/2k_BT}.$$

The left side $P(v)$ represents the probability of finding a molecule moving at velocity v (in a gas). The right side expresses the dependence on temperature and the molecule's mass. The constant k_B is called Boltzmann's constant, and it is a VERY important constant and a bit tricky to measure. Finding k_B is equivalent (through some other science stuff) to finding Avogadro's number, which plays a big role in the next section.

We want students to be able to interpret a continuous probability distribution in a qualitative sense. So you might start with the sorta bell-curve looking curve from Temp1. Label the axes with some made up numbers, like 200 m/s for typical speed, with marks for 100, 200, 300, on the speed axis. Ask students which speeds are more or less likely. The *real* answers involve some integrals, but we are just going for eyeballing the graph to get the concept.

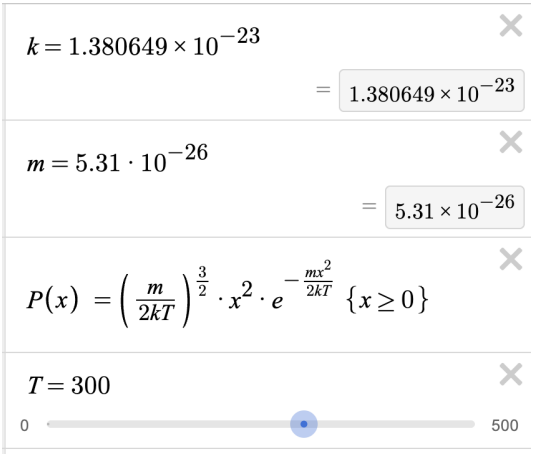
As you move into the student work (on Desmos), you **really need** to provide some guidance so they can see the nice shape of the Maxwell-Boltzmann distribution emerge (like the picture you've hopefully drawn on the board. My advice would be to set up the following. First, give them the constants if you want to accelerate things:

$$k_B = 1.38 \times 10^{-23} \text{ and } m = 5.31 \times 10^{-26}.$$

That's Boltzmann's constant, and the mass of an oxygen molecule (O_2 that is).

Temperature, molecular mass, and typical speed (20 minutes) Load Desmos!

Advise students to make their Desmos input look like the screenshot here. Notice I've used Desmos to enter the constants before the formula. I've used $P(x)$ instead of $P(v)$, because Desmos recognizes x as the independent variable. I've put T on a slider. The student will **NEED** to set the horizontal range (velocity) between about 0 and 1500, and the vertical range (probability) between 0 and 0.001! Using Desmos, the exercise on Temp1 should be straightforward. Look for the peak of the distribution to find the typical speed of an oxygen molecule at room temperature.



On Temp2, you might want to skip the first data-gathering exercise for a moment, and have students do the quicker exercises about ozone, and the broader relationship between, temperature, molecular mass, and typical speed for gas molecules. These are primarily thinking problems. The problem about Ozone can be done quickly with Desmos, by changing the mass parameter m by multiplying it by 3/2.

The first problem on Temp2 may require a bit more time. They can choose various temperatures, adjust the slider in Desmos, and find the typical speed of an oxygen molecule. In this way, they can (1) gather the data on Desmos, (2) record it in their lab manual, and **then** enter it into the *Linear Regression with Log Scaling* tool. That should show a relationship of the form

$$\log(\text{Typical speed}) = m\log(\text{Temperature}) + b.$$

The slope m indicates the exponent in the power law.

A brief intermission to collect worksheets!

Temperature and equilibria (20 minutes) The idea of this page Temp3 is the following. Students have already met *bistable* systems: systems with one state variable (X) having two *stable* equilibrium points. An example is a (real physical) light switch. It is stable in the off position, and stable in the on position, but there is an unstable equilibrium point in the middle.

Temperature affects stability; in a "hot" system, the state can jitter and jump from a neighborhood of one equilibrium point to a neighborhood of another equilibrium point. That's the whole concept!

Conclusion (10 minutes) You might introduce the *Atmospheric Molecule Simulator* before sending them off to complete Temp4 by the next section. This involves the study of a random walk (up/down) with two new elements. The first is a "ground" where the random walk cannot go below zero. The second is "gravity" which pulls the random walk a bit more down than up. This gives a Markov process, where the stable distribution is exponential... just like the atmosphere! The simulator model is reasonable in the "exosphere" where the density is very low and air molecules don't collide with each other very much. But in the usual atmosphere (where planes fly and stuff) the collisions among air molecules play the bigger role. Remind students to complete Temp4 AND BM1 by Thursday.

Lab 5, Section 6

Lesson summary: Students re-enact the Perrin-Einstein experiment, to estimate Boltzmann's constant and Avogadro's number.

Key terms: Brownian motion, diffusion, squared-displacement, expectation, viscosity,

Learning objectives: After this lesson students will be able to:

- Determine whether parameters/constants can be measured, or need to be deduced from experiment.
- Understand the rate of diffusion, and the linear growth of squared-displacement.
- Solve algebraic equations and handle units (familiar and unfamiliar) to estimate a scientific quantity.
- Appreciate the significance of Avogadro's number, and the insights necessary to estimate the number.

Before section: Students have completed Temp4, and grasp the temperature/speed connection. Students have worked through BM1 before class.

During section: Students check their work on BM1, and continue through BM2, BM3, BM4.

After section: Students complete Lab 5 and submit it on Friday!

Plan

Brownian Motion Review (10 minutes): Students should have the basics of Brownian motion from lecture, and from reading and trying BM1 on their own. But 10 minutes should be good to get people on the same page with questions like:

- What is Brownian motion? [Answer: random visible wigglings of small particles in water.]
- What causes this motion? [Answer: temperature. Lots of random motion of molecules yields bumping of bigger particles... even visible ones!]
- How big is a molecule? How do the paragraphs on BM1 answer this question?

Then you can ask the class how many molecules of water are contained in a human cell. See how many answers you get? Any consensus? They should bring this answer into class!

Then it is time to go high-tech and load up the *Brownian Motion Simulator* — at least two laptops open per group, so everyone can see a screen. Complete BM1 efficiently to move onto this. They might reason about the starred exercise at the end of BM1 – from what they learned about gases on the previous few pages, they can work out how much hotter the water would have to be for a particle to wander twice as far in the same time.

Data gathering and processing (25 minutes) The core "experiment" here consists of running the Brownian Motion Simulator repeatedly. In the new version, you just need to click the "Diffuse" button. The data is gathered automatically, and can be copied and pasted into a spreadsheet. This *can* be done with careful copy-paste work, though it may be better to have one student run the simulator, and another student enter the data into a spreadsheet (on two different computers). What they are tracking is the *squared-displacement* $d^2 = x^2 + y^2$ as a function of time. In a single run of the simulator, this d^2 will bounce around randomly. But if you *average* d^2 over many experiments, it should grow linearly!

Excel can create a best-fit line, or students can put the time vs. average d^2 data into the linear regression tool. It is good to require the line to go through the origin, and a spreadsheet might allow this. But it shouldn't matter too much. The *remarkable* thing is that d^2 grows roughly linearly with time. This is a core fact about random walks and Brownian motion. An implication is that d (the actual distance from the origin!) grows proportionally to $\sqrt{t}$. In other words, diffusion spreads according to a square-root law.

Estimating Boltzmann's constant (20 minutes) Once students have done their experiment, they should have an empirical law of the form

$$\mathbf{E}(x^2 + y^2) = mt + b, \text{ with } b \approx 0.$$

In his PhD thesis, Einstein used prior ideas – the molecular-kinetic theory of heat and work of Stokes on movement of particles in a liquid with viscosity – to argue that

$$\mathbf{E}(x^2 + y^2) = \frac{2k_B T}{3\pi r \eta} \cdot t,$$

where k_B was a then unknown constant, r the radius of the particle, and η the viscosity (stickiness) of the liquid. On BM3, students will wrap their heads around this, and then they will do just a bit of algebra. The point is that they empirically found a slope m (just as Perrin did in a 1910 experiment!), and Einstein relates that slope to other numbers, and so

$$\frac{2k_B T}{3\pi r \eta} \approx m.$$

Everything on the left side may be measured except for k_B. Thus one can *solve for* k_B! That gives Boltzmann's constant, by observing Brownian motion!

Avogadro wrap-up (5 minutes) Most of our students are in chemistry, and all of our students have (briefly) seen the "mole" concept from Math 16A if not in chemistry. The core concept is that there is a convenient number which links two questions: (1) how many (invisible!) molecules of a substance there are, and (2) the easily measurable mass of a substance. For most of its history the "mole" was defined as the number for which a mole of carbon atoms weighs 12 grams – Carbon-12, the pure isotope, which gets its name from this fact. (Since 2019 the definition runs the other way: the mole is defined as exactly $6.02214076 \times 10^{23}$ things, and the 12 grams became something measured rather than declared.) A mole of atoms is about 6.02×10^{23} atoms. That is Avogadro's number. Knowing that number allows us to convert mass (grams) to number of atoms! A mole of hydrogen atoms weighs 1 gram. A mole of iron-56 atoms weighs 56 grams. Etc.

Students finish this lab by estimating Avogadro's number, essentially counting the number of molecules in a sample. The key is the ideal gas law $PV = k_B NT$, which relates three

measurable quantities (pressure P, volume V, temperature T) to the unmeasurable number of molecules N. Thus if one knows (or can estimate) k_B as students have just done, they can figure out N from measurable quantities. The rest of BM4 is just reading, and some typical algebra and mucking about with units. Their final estimate should land near 6×10^{23} – typically within 20 or 30 percent of it, with almost all of the remaining error coming from the slope they measured. More experiments narrow it, since the scatter falls off like $1/\sqrt{n}$. Getting the order of magnitude right is the headline, but do not undersell a close result: this is a real measurement of a fundamental constant, made by watching specks jiggle.

Conclusion (5 minutes) Remind students about the lab due tomorrow! Lab 6 coming next week!

Chapter 6

Order

About Lab 6

Lab 6 is called *order*, and the theme is how order can emerge from randomness at scale. More precisely, if a large number of individuals (cells, molecules, etc.) behave according to the same probabilitistic rules, then order can arise at the large scale. As always, examples are taken from across biology and chemistry, from osmosis at the physical/molecular scale, to cell-state transitions, to epidemiology and population modeling. The material from Lab 6 builds on all the previous labs, and hopefully reinforces the concepts along the way.

The new mathematics of Lab 6 is linear algebra. As in Lab 1, we will see vectors as state vectors – tracking a few (two or three) populations. But rather than state vectors changing according to "change equations" (ordinary differential equations), they will change according to "transition matrices". This could be framed using a discrete-time (linear!) change equation, but the matrix model gives us more tools for analysis. By the end of this lab, equilibria can be found by solving linear equations (or systems), and long-term behavior can be predicted from eigenvalues and eigenvectors.

Linear algebra should be familiar to the teaching assistants for this course. But the framing might be new. Here are the important conventions and language we should consistently use.

- All vectors are column vectors, and we use capital letters to denote them. E.g., $X = \begin{pmatrix} 100 \\ 200 \end{pmatrix}$. All vectors will be *state vectors* for a system with *compartments*. The entries of a vector will always denote the *number of individuals* within each compartment. For example, a two-compartment model might have individuals that are "on" or "off". If there are 500 individuals that are on, and 800 that are off, then the corresponding state vector is $X = \begin{pmatrix} 500 \\ 800 \end{pmatrix}$.
- All matrices are square matrices. We use capital letters for these, usually M. E.g. $M = \begin{pmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{pmatrix}$. All matrices will be used to represent transitions from one state to the next, via matrix multiplication. So if the current state is $X = \begin{pmatrix} 500 \\ 800 \end{pmatrix}$, then the next state would be
$$M \cdot X = \begin{pmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{pmatrix} \cdot \begin{pmatrix} 500 \\ 800 \end{pmatrix} = \begin{pmatrix} 450 + 160 \\ 50 + 640 \end{pmatrix} = \begin{pmatrix} 610 \\ 690 \end{pmatrix}.$$

Observe in this last example that the matrix M had column sums $0.9 + 0.1$ and $0.2 + 0.8$ equal to 1. And all values of the matrix were between 0 and 1. This is a *Markov* matrix, and the entries have

a very clear interpretation. Namely, 0.9 means that 90% of on-individuals stay on; while 0.1 means that 10% of on-individuals turn off. And 20% of off-individuals turn on, and 80% of off-individuals stay off. Note that the total number of individuals stays constant: 500+800 = 610+690.

The first half of this lab, roughly, is about modeling with such Markov matrices. These are incredibly useful for systems where individuals can switch between two or more states. If individuals switch with *probabilities* given by a Markov matrix, then one can use a bit of algebra and computer simulation to find equilibrium, and to see how things bounce around near the equilibria. We look at two systems in detail: one is the ion channel system (channels open and close), and the other is a 3-compartment model for breast cancer cells.

The second half of this lab returns, in a thematic sense, to the sort of population modeling we began with in Lab 1 – population ecology. Here we look at age-stratified population growth using "Leslie matrices". The idea is that we use different ages (or developmental stages) as our compartments. Aging moves the individuals from a young compartment to an old compartment. Reproduction creates new individuals in the youngest compartment. Death removes individuals from the compartments. Here the matrices are not Markov matrices, because the total population is not fixed. But we can use some analysis of eigenvalues/eigenvectors to find interesting long-term behaviors, including exponential growth, stabilizing age-ratios, and oscillation.

After a brief matrix model used in epidemiology, the end of this lab is an extended synthesis and reflection to conclude the course. Indeed, while the linear algebra is a significant new thing for students, part of the goal is to really cement the prior knowledge by hitting themes from across Labs 1-5. We leave students, hopefully, with a fond reflection on their mathematical growth.

Lab 6, Section 1

Lesson summary: Students explore a 2-compartment Markov model through computational simulation, before learning a bit of matrix/vector mechanics.

Key terms: Compartment, osmosis, expectation, transition, transition probability/matrix, vector, matrix, dimensional.

Learning objectives: After this lesson students will be able to:

- Use a simulation to find the outcome from a 2-compartment Markov model.
- Add vectors, and multiply vectors by square matrices.
- Connect vectors to states of a compartmental system.
- Connect matrices to transitions from one state to the next.

Before section: Students attend a lecture about osmosis, and are warmed up for Lab 6.

During section: Osm1, Osm2. Then LA1, LA2.

After section: Complete LA3, LA4.

Plan

Osmosis ice-breaker (15 minutes): If you would like an all-class ice-breaker to set the mood, you can try having them perform osmosis. Split the room and students into two halves, with a center line of your choosing. Label the two halves A and B, and call them "compartments" A and B! They can get out their phones and ask them to Google "roll a die". This will bring up a Google page which virtually rolls a die, with a random outcome 1,2,3,4,5,6. The rule is this: if they are on side A, and they roll a 1 or a 2, they switch to side B. And if they are on side B, and they roll a 6, they switch to side A. Once they are on their side, ask them to repeat (click the roll button). Then repeat again, and again. Remind them that their rule changes when they change sides! Note that there should be more A-to-B transitions than B-to-A transitions. Use the words "transition" and "compartment" to get used to the vocabulary! Let the process go for 2-3 minutes. They should be able to get faster with rolling and walking back and forth across the class.

Quietly, while students are playing this game, put their name cards into new random groups at tables.

Leading questions for students:

- Which transitions are more common? From A to B or from B to A? How much more common?
- What do you think will be the long-term outcome? Students will probably reason that there will be more in compartment B. But will *all* the students end up in compartment B? What proportion? What if there were 1000 students?

It is a new lab, so time for new groups! Please assemble the students into their new randomly determined groups of 4-6 students per group. Given them 2 minutes to introduce themselves to their group.

Osmosis simulation (25 minutes) The mathematics of osmosis (Osm1, Osm2) in this class is the same as the ice-breaker. There are molecules that have a higher probability of one transition A-to-B than for the reverse transition B-to-A. The first two exercises, on Osm1, are exercises in careful tracking. If you have 1000 molecules in compartment A, and each has a 20% chance of moving to compartment B, then you would expect 800 in A and 200 in B after that transition. But the students have to keep track of both compartments with their differing probabilities, over a 3-second period. You may wish to demonstrate the first second, where the transition is

$$(1000, 1000) \text{ transitions to } (800 + 100, 200 + 900) = (900, 1100).$$

Note how I've broken it up; the 800+100 represents the A-to-A transitions and the B-to-A transitions. The 200+900 represents the A-to-B transitions and the B-to-B transitions.

Once students have gotten through Osm1, they can work in groups for about 15 minutes on Osm2. The first exercise in Osm2 is probably the most difficult. Why does equilibrium occur when $0.2A = 0.1B$? You might ask one or two groups to share their answer, once you determine they have a good answer by floating around. What I *hope* is that some groups figure out that equilibrium occurs when the A-to-B transitions ($0.2A$) are perfectly balanced by B-to-A transitions ($0.1B$). Then students may struggle to solve the equation $0.2A = 0.1B$, even with the hint that we begin with 2000 water molecules (so $A + B = 2000$). Students do not like systems of linear equations any more. But here is a friendly intermediate step: the equation $0.2A = 0.1B$ implies that $A = 0.5B$ (by dividing both sides by 0.2. So you are looking for two numbers A and B which sum to 2000, and for which A is half of B.

The explorer should be visually compelling, and show how this happens in practice. Even though the individuals behave randomly, they follow those 20% and 10% rules for transition probabilities. The system should stabilize (roughly) near the equilibrium they have found. The last question on Osm2 is a bit tricky; the equilibrium won't change, but the system will approach equilibrium faster.

Vectors and Matrices (20 minutes) The page LA1 is meant to give students a way to package the information of compartments and transitions into vectors and matrices. You might do the first exercise on the page on the board, with student assistance. Show them how to package their "state vectors" into a column vector. And then you can lecture on the following paragraph to show how the probabilities are packaged into a transition matrix. I find it helpful to level the matrix with the From and To words; it is important that they consider "staying in the same compartment" as a transition, and find the complementary probabilities.

Note that LA1's transition matrix uses the same convention as Osm1 and Osm2: a 20% chance of the A-to-B transition, and a 10% chance of B-to-A. It is worth saying out loud that a transition matrix is read *by columns*. The first column describes the individuals who are currently in compartment A, so its two entries are the A-to-A and A-to-B probabilities, and they sum to 1. Students who read the matrix by rows instead will get every later exercise backwards, and that is the most common mistake on this page.

To finish the day, groups can work through the vector and matrix mechanics on LA2. This requires some careful reading, and arithmetic. The definition of multiplication here is really weird the first time you see it. But here in this class, we can give a justification – we define multiplication $M \cdot X$ so that it represents the "next state" of the system. Multiplication *implements* the transition, from the state X to the new state $M \cdot X$.

Conclusion (5 minutes) Remind students to complete the lab manual through LA3 and LA4 by Thursday. It's basically a lot of arithmetic to get used to vector and matrix mechanics. There will be a further worksheet due next Tuesday.

Lab 6, Section 2

Lesson summary: Students will use a 2-compartment matrix model to understand the behavior of ion channels.

Key terms: Ion channels, gated, patch clamp, current, agonist, transition, equilibrium, ratio.

Learning objectives: After this lesson students will be able to:

- Interpret a patch-clamp readout in terms of open/closed states of an ion channel.
- Relate a patch-clamp readout to transition probabilities.
- Work with scientific units, including milliseconds and picoamps.
- Find the equilibrium state in a 2-compartment matrix model.
- Relate the probability-ratio to the equilibrium state in a 2-compartment matrix model.

Before section: Students complete LA3 and LA4, to have a basic grasp of vectors and matrices.

During section: Students explore the ion channel model through IC1, IC2, IC3, IC4.

After section: Students complete a worksheet for Tuesday.

Plan

Introduction to the ion channel system (20 minutes): Before class begins, draw a patch-clamp readout on the board, to roughly mimic the readout on page IC1. Make sure to label your axes. Note that time is in milliseconds here, and the vertical axis is electrical current in picoAmps. Pico is one trillionth; we are talking about trillionths of an ampere of electrical current going through a single ion channel in a single cell. I will have lectured on this extensively the day before!

Take the first 10 minutes to get people in their groups again, and have representatives read out loud the paragraphs on IC1, including the text of the first exercise. That is when you can ask students to do the exercises on IC1 in their groups. If they get stuck, you can try to estimate the open/closed time on your board drawing of a patch-clamp readout. That will show the technique. Their estimate does not need to be great! The important thing is to be able to make reasonable estimates from a graph, even in the presence of background noise (the small squiggles that are prevalent throughout the patch-clamp readout).

Ion channel simulator (15 minutes): Once students complete the paper-and-pen analysis of a patch-clamp readout, they can use the simulator. The simulator does a pretty good job of showing what a real patch-clamp readout looks like, with various transition probabilities. Here the channel is randomly making a transition from open-to-closed, or from closed-to-open, with different probabilities. The readout just captures a single ion channel.

Unlike in the case of osmosis, where we started by considering lots of molecules, we are just looking at a single ion channel. But we are looking at it over a longish period of time. So we look at *how much time* an ion channel is open/closed, rather than *how many* ion channels

are open/closed... at least in the beginning. Students will explore how the patch-clamp readout depends on the transition probabilities. If open-to-closed occurs more frequently than closed-to-open, we expect the channel will be closed more often! The very end of IC2 asks about multiple channels; this is important for real cells, which are covered with thousands of channels!

Equilibrium and ratio (30 minutes): To study equilibrium, we leave the patch-clamp readout, and consider a large number of ion channels undergoing these transitions. The first goal is to model this with state vector (how many open, how many closed) and transition matrix (a Markov matrix). You might help walk students through this, including the computation of $X_1 = M \cdot X_0$. And then let the groups compute $X_2 = M \cdot M \cdot X_0$. The rest of IC3 is really classical algebra... and requires students to have completed LA3 and LA4 to get things right.

We finish with a Theorem, which describes the equilibrium in any 2-compartment Markov matrix system. READ this page before teaching it. Here is how I think of the theorem:

The equilibrium ratios mirror the probability ratios.

For example (as done in IC4), suppose that closed-to-open has a probability 10%, and open-to-closed has a probability 1%. This is a 10:1 ratio of probabilities. We therefore expect a 10:1 ratio in the equilibrium, i.e., 10 times as many open as closed. (Double-check to avoid getting things backwards: there are more closed-to-open transitions, so there will be more open in equilibrium.)

Ratios have befuddled students for a few thousand years. (One can find textbook problems about making sandals in ancient Egypt which befuddled students back then.) Our students will take genetics someday, where they will again have to face ratios. So IC4 also provides some practice. Maybe if students do a bit with ratios every quarter, it won't seem so bad by the time they need it.

To keep students working and thinking, through the somewhat dry material of IC3 and particularly IC4, it is best to get them at the board writing their solutions. You might take a break and have one group visit another group's board, assign presenter and scribe roles within groups, etc.

Conclusion (5 minutes) Remind students that there is a worksheet due next Tuesday! And Wednesday's Lab 6 Checkpoint will go through BC2, so we will be doing BC1 and BC2 in section next week.

Lab 6, Section 3

Lesson summary: Students finish their study of Markov processes with a 3-compartment model related to breast cancer cell states.

Key terms: Compartment, transition matrix, state vector, cell state.

Learning objectives: After this lesson students will be able to:

- Use a 3x3 matrix to organize transition probabilities in a 3-compartment model.
- Multiply a 3x3 matrix by a 3-dimensional vector to compute the "next state".
- Hypotheses about long-term behavior based on looking at a transition matrix.
- Interpret a Markov model to strategically change a system.

Before section: Students have completed a worksheet, and are comfortable with basic vector/matrix mechanics.

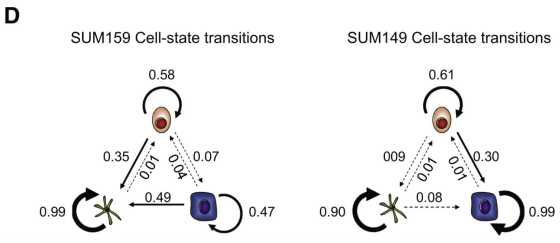
During section: Students explore the 3-compartment breast cancer model, BC1 and BC2, to completion.

After section: Students submit their lab check-in.

Plan

Reading and diagramming (20 minutes): There is a bit of scientific background here, and everyone needs to be on the same page to get started. I advise starting with reading out loud from BC1. As they get into the three states (S, B, L), you can tell them that these "cell states" are the three compartments for this model, and the cells will "change states" according to certain transition probabilities. This sort of modeling is used in cancer biology, where in a tumor environment cells tend to change states – this makes cancer difficult to treat. One treatment might seem to kill all the cells in a "bad" state, but then more cells transition into the bad state, and in the long term the cancer is not cured. Another approach is to look for treatments that affect the transition probabilities.

Here is Figure 2D from the cited paper, "Stochastic State Transitions Give Rise to Phenotypic Equilibrium in Populations of Cancer Cells" (Gupta et al., Cell, 2011).



That is where I got the transition probabilities, and the general structure of BC1 and BC2. I think a starting exercise would be to get everyone to produce a diagram like the one above, using the given transition probabilities to create a compartment-and-arrow diagram. Then they will be set up for the exercises on BC1 and BC2.

Matrix practice (15 minutes) Now it is a good chance for students to work at the board, multiplying transition matrix by state vector to find the next state. For the exercises on BC1, students should also be able to figure out the answers *without* matrix mechanics. This can be a good thing, so they can check their answer.

Simulation and interpretation (30 minutes) The page BC2 uses an online tool to explore the long-term behavior from a 3-by-3 transition matrix. Once students load the tool, make sure they change the name of the compartments (A,B,C) to Basal, Luminal, and Stem-Like, so they don't get mixed up! Then they can *carefully* enter the transition probabilities given on BC1. It's good if each group has two computers up and running. Note that students will have to work out a few transition probabilities (the B-to-B, L-to-L, S-to-S) on their own, to make the columns sum to 1.

Then they should be good to go, figuring out that the transition matrix gives a predictable outcome – no matter what starting values they begin with, the final state should have the same B:L:S ratio. Starting from 1000 cells in each state, the equilibrium is about 2920 basal, 9 luminal, and 70 stem-like. Basal dominates because it is by far the stickiest state – 99% of basal cells stay basal – and because both of the other states send a large share of their cells to basal. Note that stem-like settles well *above* luminal here, which tends to surprise people. Students are then asked to explore a different system of transition probabilities. The point here is that there are a lot of different kinds of tumors, different gene mutations, etc., which give rise to different transition probabilities. This can drastically change the course of the disease, and the difference in cell state proportions can be visible to a pathologist.

The final exercise asks students to really think about the implications. After getting used to transition matrices during three discussion sections, students can think about what this means for *interventions*. Note that if one changes starting values, the end-result is roughly the same in these tumors. So a treatment which simply kills all the cells in a given state will have a temporary effect as the equilibrium proportions will be restored. But if you can fundamentally change the transition probabilities... you may be able to effectively treat the cancer. That is a strategy which people try across various kinds of cancers, e.g. melanoma. Biologists are trying to exert control over cell state transitions for broader purposes too... see, e.g., "Control of cell state transitions" `https://www.nature.com/articles/s41586-022-05194-y` by Rukhlenko et al., in Nature (2022).

Conclusion (10 minutes) There is a little extra time today (I hope), to let students finish up Lab 6 through BC2, for their Lab 6 Check-in due tomorrow. On Thursday, students will continue working on matrix models... but without the Markov assumption.

Lab 6, Section 4

Lesson summary: Students use a matrix model (the Leslie matrix) to study age-stratified population growth.

Key terms: Age-stratified, Leslie matrix, eigenvector, eigenvalue, equilibrium vector.

Learning objectives: After this lesson students will be able to:

- Use a matrix to capture rates of birth, death, and aging in an age-stratified population.
- Identify when a vector is an eigenvector of a (2x2 or 3x3) matrix, and find the associated eigenvalue.
- Interpret eigenvectors in terms of states with stable ratios.
- Plot the trajectory in state space associated to a 2-compartment matrix model.
- Visualize real eigenvalues, using directions of proportional change.

Before section: Students have completed their Lab Checkin, and are familiar with transition matrices and state vectors.

During section: Students work through Les1, Les2, a bit of Les3, and start Les4, to work with Leslie matrices and age-stratified matrix models.

After section: Students complete Les1-Les4, and then complete Epi1, Epi2 as homework.

Plan

Black bears (15 minutes): We begin with a somewhat complicated *age-stratified* model of a bear population. Age-stratified means that we place the bears into compartments based on age. Here we use just two compartments: juvenile (non-reproducing) and adults (reproducing). There are various changes that happen, including birth and death and aging!

A good first goal is to construct, at the board, the complicated diagram at the right of page Les1, and ask someone from each group to explain what each arrow in the diagram *means*. This is different from our Markov compartment models, which individuals just migrate among compartments. A matrix model can be used, even though the entries are no longer probabilities, as Patrick (George) Leslie described in the 1940s.

As students work through the exercises in Les1, there are a few tricky points. Students may figure them out if you ask them, but groups may need to talk to each other.

Tricky point 1: if 50% of the adults are females, and each adult female gives birth to one juvenile each year, then A adults in one year will produce $0.5A$ juveniles. That is where the 0.5 comes from in the top-right of the Leslie matrix.

Tricky point 2: The fact that 25% of juveniles become adults each year is not reflecting a stochastic process, where a juvenile has a 25% chance of suddenly become an adult. Rather, this is a way of modeling the fact that the juvenile stage lasts for four years. In practice, one can create more accurate models by adding more age-strata (like one per year), but this 25% is a easy simplification.

Equilibrium and eigenvectors (20 minutes) Les2 introduces the dreaded topics of eigenvalues and eigenvectors. It may be best to hit the exercises first, and the jargon second. What I mean is that students may first try to find an equilibrium vector for the Leslie matrix $M = \begin{pmatrix} 0.65 & 0.5 \\ 0.25 & 0.9 \end{pmatrix}$. Can they find a number J of juveniles and A of adults, which stay fixed over time? There are various ways to approach this, but you can set this up as a system of equations:

$$0.65J + 0.5A = J \text{ and } 0.25J + 0.9A = A.$$

The first equation rearranges to $0.5A = 0.35J$, so $A = 0.7J$. The second rearranges to $0.25J = 0.1A$, so $A = 2.5J$. Both can hold at once only if $0.7J = 2.5J$, which forces $J = 0$, and then $A = 0$ as well. Note that neither equation on its own rules anything out – it takes the two together. Students may find other explanations. You can explain this graphically too... both equations describe lines through the origin, and they have different slopes. So they only intersect at the origin.

After students solve this exercise, you can point out what it means: there is NO equilibrium in this bear system, except when there are zero bears.

The next exercise asks students to examine the state vector with 100 juveniles and 100 adults. The next state of the system is 115 juveniles and 115 adults. Students are meant to discover that iterating the transition will simply *scale* the state vector by 1.15. There will always be the same number of juveniles and adults! Once they discover that this state vector is just scalled by the matrix, you can introduce the term eigenvector (the state is an eigenvector) and eigenvalue (1.15, the scaling factor). Basically, the eigenvalue becomes the growth rate of the state vector, and *proportion* remains constant. So we look at eigenvectors as a sort of modified equilibrium – instead of the system staying exactly the same, the proportionality stays the same while the total population may grow or shrink.

Visualizing the change (25 minutes) Les3 harkens all the way back to Lab 1, where students first learned about vector fields and trajectories in state space. By connecting vectors X to $M \cdot X$, they should see the vector field along which the populations change. The only equilibrium is at $(0, 0)$, but the trajectories should flow towards the diagonal eigenvector-growth trajectory. The long-term expectation is that a nonzero bear population will increase, and the ratio of adults:juveniles should approach 1:1. This is a sort of dynamical interpretation of "eigenvector".

Try to avoid having students plot all the points and arrows in Les3 during section. Move it along, because they can do that on their own. It is probably better to get them started on Les4, which they visualize eigenvalues and eigenvectors, and experiment with a slight change in the Leslie matrix and its implications for the survival of black bears. There are some important lessons for our ecologists in there towards the end of Les4.

Conclusion (5 minutes) Remind students that next week is the last week of classes! We'll have study materials for a final exam, etc. They must complete the two epidemiology pages Epi1, Epi2 on their own for homework. Next Tuesday, they'll finish the study of Leslie matrices with a species having 3 age-strata.

Lab 6, Section 5

Lesson summary: Students complete their study of Leslie matrix models, including a bit on eigenvectors and oscillation.

Key terms: Leslie matrix, oscillation, eigenvector, eigenvalue.

Learning objectives: After this lesson students will be able to:

- Provide a Leslie matrix that models a system of births, deaths, and aging.
- Draw time-series plots to capture population changes with multiple age strata.
- Identify when oscillations may arise from a Leslie matrix model.
- Connect eigenvalues to the long-term behavior in a Leslie matrix model.

Before section: Students have completed most of Les4, and Epi1, Epi2.

During section: Students complete Les5 and Les6, and go back to discuss Epi1, Epi2 if time permits.

After section: Students write *rough drafts* for Syn1 and Syn2, to bring into the last discussion section.

Plan

Interpreting a Leslie Matrix (15 minutes): A good place to begin would be to put two Leslie matrices on the board:

$$M = \begin{pmatrix} 0.65 & 0.5 \\ 0.25 & 0.9 \end{pmatrix} \text{ and } M = \begin{pmatrix} 0.1 & 1.4 \\ 0.4 & 0.2 \end{pmatrix}.$$

The first is from black bears (Les4), and the second from (Les5). This is a chance to see if the class can remember (closed book?) what all the matrix entries in M mean. For example, we know from the first column that 25% of the bears become adults each year, while 65% stay juveniles, and 10% die. So the 0.65 can be interpreted as the fact that bears take a while to reach adulthood... not that the juveniles have an average of 0.65 offspring. The full interpretation is in Les1, and it is good to review this since it is complicated!

Next, ask the students to think of an organism whose Leslie matrix would be described by the second matrix. Have them try to write about it in Les5, but focus on good sentences written on scratch paper or the board to start. For example $0.1 + 0.4 = 0.5$, so 50% of the juveniles may die. And among the adults... about 20% of the adults remain adults, while 80% die. But as an intro activity, see if students can choose an organism for their table, and really describe what the Leslie matrix would *mean* in their context.

Les5 (20 minutes) Once the students have interpreted their Leslie Matrix, it is time for them to work out future states, starting from 30 juveniles and 100 adults. Try to have them use matrix multiplication, for at least 5 steps in each group. They can tabulate their data in a table on the board, with columns: Time, Juveniles, Adults. Once they have 5 time steps on the board, ask the groups to create a *really neat* time series for their lab manual. Use a straightedge for the axes. Label both axes carefully. Choose the axis range *after* the data is computed. Make a legend to distinguish the juveniles and adults. This is a chance for graphical excellence.

The last exercise on Les5 is meant for students to play with visualizing a matrix, as a linear transformation. They should first enter the matrix entries at the top-right of the 2-dimensiona Matrix Visualizer tool. Then they can click the Show Eigenvectors to see the eigenvectors, which are labeled by the corresponding eigenvalues. They can select the *unit square* shape. And perhaps even better, they can click on the plot to add individual points. Then clicking the "Next state" button will move all the state vectors according to the transition matrix M.

This will take some fiddling and questioning to get students to make key observations. Here are some prompts.

- With the eigenvectors turned on, click a whole bunch of points in the first quadrant, and then click "Next state" a bunch of times. What do you notice about the points? (Answer: they sort of line up along the $\lambda = 0.9$ eigenvector. Why? that is the "dominant" eigenvector. Also, as they line up, they seem drawn towards the origin at (0,0). Why? The eigenvalue $\lambda = 0.9$ is less than 1. So the population is decreasing by 10% each step.)
- What do you notice about the unit square? (Answer: it seems to flip-flop and contract to lie on the $\lambda = 0.9$ eigenvector. The flip-flopping is a form of oscillation, and it happens due to the negative eigenvalue $\mu = -0.6$.

Oscillation from a Leslie matrix (25 minutes) The last Leslie matrix example is an insect model which exhibits oscillation. Students should be able to find their Leslie matrix with their groups with little assistance by this time. The matrix should be

$$M = \begin{pmatrix} 0 & 0 & 1000 \\ 0.02 & 0 & 0 \\ 0 & 0.05 & 0 \end{pmatrix}.$$

Basically, eggs become hoppers, hoppers become adults, and adults all die after they produce eggs. So each year, there will only be one of three age-strata present. And indeed this happens with some insects.

Note that $0.02 \cdot 0.05 \cdot 1000 = 0.02 \cdot 50 = 1.0$. Since this is perfectly equal to 1, $M^3 = 1$ (the identity matrix), and we find perfect oscillation. It is certainly a contrived model, and slight variations in those numbers will make the species go extinct or expand exponentially.

As students explore this one, they can use the *Three-Compartment Matrix Model* as suggested in the Lab Manual. Or they can use the brand new *Leslie Matrix Population Model* tool instead... that will show the same information... perhaps even a bit better.

Conclusion (5 minutes) Class is wrapping up with just one more discussion section! I suggest assigning Syn1 and Syn2 for homework, as rough drafts (on separate paper), since it can make for a good discussion and chance for sharing in the last section. Or it may be necessary to return to Epi1 and Epi2 if students struggled with that one.

Please remind students to (1) Complete their SETS, and (2) Complete the end-of-quarter survey on Canvas. These are separate tasks, and (2) is graded (worth points on Canvas).

Lab 6, Section 6

Lesson summary: Students review and reflect on what they have learned in Math 16A and 16B.

Key terms: Randomness. Order. Exponential model. Change equation. Matrix model.

Learning objectives: After this lesson students will be able to:

- Identify situations where order emerges from a stochastic (random) process.
- Differentiate a nonlinear change equation from a matrix model.
- Identify situations where exponential growth and decay naturally occur.
- Explore the scope of mathematical modeling in biology.

Before section: Students have drafted Syn1 and Syn2 and completed the earlier sections of Lab 6.

During section: Students discuss in their groups, and then as a class, their answers to Syn1, Syn2, Syn3.

After section: Students will draw a beautiful picture for Syn4, then complete their lab, surveys, and study for the final!

Plan

This is really a flexible day, where the goal is getting the right combination of synthesis, reflection, and completion towards upcoming deadlines. Upcoming deadlines: the Lab 6 final draft is due tomorrow! So students may wish to spend some time going through difficult parts. But Syn1 and Syn2 also require some discussion and are worth sharing. Note that Syn3 and Syn4 are more personal reflections and can wait until after discussion section.

My request is that you really *make a plan*. Finish the class strong. Set some goals like the following:

- Students should realize that they really have learned a lot in two quarters!
- Students should connect what they have learned to their desired field of study.
- Students should feel like they can get everything done that they need to get done.
- Students should know that the teaching team (TA, ALF, ACE, etc.) is there to support them as they study for the final and complete their lab manual.

Make your plan on the back of this page.

Plan Write your plan here!

Conclusion (5 minutes) Remind students that the Lab 6 Final Draft is due tomorrow night. And the final is next week! Remind students of how to submit things on Gradescope, and review sessions, office hours, etc. Please thank them for their efforts, and thank your ALF for their support too!

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